MR Ch. 1-2

Paul Milgrom, John Roberts - Economics, Organization, and Ma

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DOES ORGANIZATION MATTER?

The principles of organization got more attention among us than they did then in universities. If what follows seems academic, I assure you that we did not think it so.

Alfred Sloan[1]

BUSINESS ORGANIZATION

Crisis and Change at General Motors

When Pierre du Pont appointed Alfred Sloan to head General Motors in 1921, the company was in crisis. The demand for cars had fallen in the 1920 recession, and although the firm already had huge inventories of unsold cars, the factory managers continued to produce with abandon. In response to this falling demand, Ford Motor Company had cut the price of its Model T by about 25 percent—a reduction that GM with its higher costs could not match. Trying to hold the line on prices, GM saw its sales fall by 75 percent between the summer and fall of 1920. By 1921, Ford Motor Company's Model T held a 55 percent share of the U.S. automobile market, compared to just 4 percent for Chevrolet and 11 percent for all of General Motors' brands combined. Taking relentless advantage of the huge cost advantage that came from producing a single product at very high volume, Ford was expanding its production capacity in order to increase its already dominant position in the car market.

Even apart from the immediate difficulties caused by the recession, General

[1] Alfred Sloan, *My Years With General Motors* (Garden City, N.Y.: Doubleday, 1964), p. 50.

Motors faced a fundamental long-term problem. It simply could not produce a car that would offer more value at a lower price than the Model T, and it was squandering the resources and capabilities it did have, as divisions like Cadillac, Buick, Oakland, Olds, and Chevrolet competed mostly with one another. What was needed first was a new, more coherent marketing strategy focused squarely on competing with Ford. Sloan's plan was as follows: GM would design different cars for different segments of the market. The Cadillac division would make luxury cars for the highest-income buyers, and the other divisions each would serve successively lower-income segments, with Chevrolet making a model that would be sold for even less than the Model T. The Model T would still appeal to some buyers, but Sloan believed that most customers would choose to buy cars that were either more luxurious or less expensive than Ford's product.

PRODUCT DIVERSITY AND DESIGN COORDINATION There was, however, one important hitch: Carrying out this plan involved a combination of diversity of products and close coordination in design that exceeded anything that had ever been attempted before. There would need to be a variety of new car designs, new dealerships, market information about the customers in each new market segment, separate factories to manufacture each type of car, and different supplies for each factory—a huge amount of variety. At the same time, the many parts of the organization would need to be coordinated in various ways. They would have to cover the different segments of the market without competing too much with each other. They would need to share ideas about how to improve products and reduce manufacturing costs, coordinate their research and development efforts, cooperate with the supply divisions that produced major components like bearings, radiators, and spark plugs, and standardize designs enough to achieve economies of scale in parts production. Compared to Ford's one-product strategy, Sloan's segmented-market strategy required that many more decisions be made and much more information be continuously gathered and evaluated. The organization used by Ford Motor Company would be no model for the new General Motors.

The former organization of General Motors as essentially a collection of car companies and suppliers operating without any central direction was no model either. The car divisions had failed to coordinate their parts designs, raising costs for all of them. The accounting system, which allocated costs among the producing divisions, was unable to keep accurate track of which decisions by which units raised costs and therefore failed to guide divisional managers to economize. For example, the individual divisions continued production even in the face of huge accumulated inventories of unsold cars during the recession because the system failed to assign inventory holding costs to the divisions.

GM'S MULTIDIVISIONAL STRUCTURE Sloan studied GM's organization and decided a radical change was needed. The new organization would be a multidivisional structure with a strong, professional staff in the central office. There would be no infringement on the basic autonomy of the divisions in making operational decisions. Each separate division would make and sell a car targeted for an assigned market segment. Each would have its own managerial team with authority to make its own operating decisions. Unlike other business organizations, GM's central office would not be responsible for day-to-day operations. Instead, its primary roles would be to audit and evaluate each division's performance and to plan and coordinate overall strategy. The central office would also be responsible for the research, legal, and financial functions of the corporation. It would survey market prices to make sure that the prices used for internal accounting purposes were a good reflection of actual costs. This would

enable the company to evaluate internal supply divisions on the basis of their profitability, as if each were a separate firm.[2]

Henry Ford, accustomed to being well informed about every important decision in his company, was skeptical about GM's reorganization and especially about how far its top management would be removed from its operations. He commented:

> To my mind there is no bent of mind more dangerous than that which sometimes is described as the "genius for organization." This usually results in the birth of a great big chart showing, after the fashion of a family tree, how authority ramifies. The tree is heavy with nice round berries, each of which bears the name of a man or an office. . . . It takes about 6 weeks for a message from a man living in one berry at the lower left-hand corner of the chart to reach the president or chairman of the board.[3]

However, Sloan's organization, which looked so cumbersome to Henry Ford, quickly transformed GM into a fearsome competitor. From 1927 to 1937, Ford lost $200 million, whereas GM earned over $2 billion. GM's market share grew to 45 percent in 1940, whereas Ford's once commanding share shrunk to a mere 16 percent.

The creation of a multidivisional structure not only enabled General Motors to compete successfully with its new strategy, it also set the stage for a continuing expansion of the company's product line. In the years that followed, General Motors added products ranging from trucks to kitchen appliances to its product portfolio. Such an expansion would not have been possible using the older forms of business organization. The multidivisional form that GM helped to pioneer has become a standard organizational feature of the corporate world, enabling many companies to produce a wide array of products. General Motors' new organization was well suited to its needs, but GM was hardly the last automobile company to discover the advantages of organizational innovation.

Toyota

In the early 1950s, Toyota was a small automobile manufacturer serving the Japanese market. Compared to its giant U.S. competitors, Toyota suffered from a drastic lack of capital and a tiny scale that made it impossible to match its competitors' low production costs. Although Toyota enjoyed much lower labor costs than the U.S. firms at the time, many other countries had even lower labor costs. However, none had been able to parlay these low costs into a substantial competitive advantage in what was then a high-technology, capital-intensive industry.

"JUST-IN-TIME" MANUFACTURING Like other automobile companies in Europe, Japan, and the Eastern Bloc, Toyota tried for a time to mimic the advanced mass-production techniques of its U.S. competitors. Soon, however, under the leadership of Eiji Toyoda and Taiichi Ohno, it began to develop a distinctive approach that was better suited to the scale and nature of its operations. One of the most famous of Toyota's innovations was the development of the "kanban" or "just-in-time" (JIT)

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[2] Similar organizations were independently developed in the same era by the Du Pont Company, Standard Oil of New Jersey, and Sears. The fascinating history of the emergence of the multidivisional firm is told by Alfred Chandler in *Strategy and Structure: Chapters in the History of the American Industrial Enterprise* (Cambridge, MA: MIT Press, 1962).

[3] As quoted in Alfred Chandler, "The Historian and the Enterprise," Ninth Annual David R. Calhoun, Jr. Memorial Lecture delivered at Washington University (March 29, 1988); and Alfred Chandler, *Strategy and Structure*.

manufacturing system. This system was ideally intended to eliminate all inventories from the production process. In traditional manufacturing industries, goods processed on one machine would be held in a buffer inventory until the next machine in the sequence was ready for its operation. Inventories separating the successive stages protect each machine's operations from delays or disruptions at adjacent stages of production. However, inventory systems are subject to very large economies of scale, so Toyota could never achieve cost parity (equality) with its larger competitors if it relied on such a system.

In place of inventories, Toyota established a system of closer communication and tighter coordination between successive stages of the production process, so that each stage would be informed "just in time" when it had to deliver its product to the next stage. Without inventories to buffer the disruptions caused by defective products and broken machines, Toyota engineers had to work to improve the reliability of every step of the process. The same changes that reduced the number of interruptions in the production process often reduced the number of defects in Toyota's cars as well, as flaws were caught immediately rather than piling up in the in-process inventory. The absence of inventories also meant that Toyota had to be linked more tightly to its suppliers than were U.S. firms, communicating with them about day-to-day needs and helping them to improve the reliability of their own systems. At the same time, the need to repair broken equipment quickly caused Toyota to train its equipment operators to carry out maintenance and repairs themselves. In contrast, maintaining and repairing machines was a separate specialty with a separate job classification in the United States, and when a machine broke down, its operator stood around waiting until a repair specialist turned up to fix it.

Under Alfred Sloan's leadership, General Motors moved to take full advantage of the large scale of its operations by including the same parts in many of its different cars. By using the same chassis or engine or brakes in several models, GM could afford to develop specialized manufacturing equipment for these components, substantially reducing the production costs. In contrast, Toyota did not enjoy such scale economies and instead emphasized improving the flexibility of the equipment it did use, so that the same equipment could be quickly reset to produce different models. Given this emphasis, it should come as no surprise that Toyota had become a world leader in the use of industrial robots as early as the 1960s.

Because General Motors' specialized equipment was not easily adapted to the production of radically new designs, GM had major redesigns of its models only about once every 12 years in the 1950s and 1960s, whereas Toyota redesigned its vehicles twice as often, introducing improvements with each new design. By the early 1970s, Toyota's technological prowess in the design and manufacture of small cars had earned it a considerable share of the world market. As its sales grew, Toyota built new manufacturing plants that were much larger than those built earlier by the U.S. car companies, allowing the company to enjoy many of the scale economies that remained in the production process.

COORDINATION WITH OUTSIDE SUPPLIERS Another feature that distinguished the Toyota organization from its North American competitors was Toyota's great reliance on outside suppliers. In contrast, GM was very highly vertically integrated. GM's huge volumes and use of the same components for many models allowed the company to utilize fully the output of efficient-sized parts factories. Furthermore, ready access to capital allowed it to produce these parts and components itself. In its early days, Toyota could not achieve efficient scale in the building of components, nor could it afford to own its own parts makers. Therefore, unlike GM, Toyota chose to rely on outside suppliers not just for its basic inputs like sheet steel, screws, and fabric for

seats, but also for the more complex components and systems, such as headlamps, brake systems, and fuel-injection systems. This also opened the possibility that its suppliers then could achieve larger scale by producing for other auto manufacturers as well.

The JIT system necessitated close coordination between Toyota and its suppliers. The need was reinforced by the frequent redesign of the vehicles and the fact that the suppliers were providing high-level components that had to fit together, rather than simple standardized commodities. As a result, simple market arrangements with the suppliers became problematic. Instead of seeking numerous suppliers for each part or component and shifting business among them to induce price competition, as GM did, Toyota built long-term relations with a much smaller number of suppliers. These long-term relations facilitated communication and made the suppliers willing to face the risks of investing heavily in both skills and machinery to meet Toyota's specialized needs.

The Hudson's Bay Company

The history of the automobile industry provides a clear example of the importance of a coherent organization that is well suited to the firm's size, capabilities, and market strategy. But the importance of the details of business organization can also be seen from centuries past.

On May 2, 1670, the Governor & Company of Adventurers of England Trading into Hudson's Bay was formed as a joint stock company by a royal charter of King Charles II of England. The charter gave the company a monopoly over trade in all the lands draining into Hudson's Bay.[4] This is an immense area of 1.5 million square miles, covering much of Quebec, most of Ontario, all of Manitoba, much of Saskatchewan and Alberta, the eastern part of the Northwest Territories, parts of Minnesota, North Dakota, and Montana, and a bit of South Dakota. The area is larger than 10 Japans, 15 United Kingdoms, or 30 states of New York. The company's total legal monopoly nominally covered all trading of goods in the region, but in fact the firm was in the business of trading European manufactures to the native peoples for animal furs, and especially beaver pelts.

Now known as the Hudson's Bay Company (HBC), the firm is still in existence. It is the world's oldest commercial entity that continues its original line of business. In the late eighteenth and early nineteenth centuries, however, it was in desperate shape, being thoroughly beaten by a rival, the North West Company (NWC), that operated under a minor legal deficiency (it was violating the royal monopoly) and what should have been an absolutely debilitating technological inferiority. However, the NWC used a much more effective organizational structure and strategy that put it closer to its customers than the HBC, encouraged more flexible, effective responses to changing conditions than the older company could manage, and gave its employees stronger incentives for initiative and effort. This strategy and structure more than offset the older firm's huge cost disadvantage. Only when the HBC mimicked key aspects of the NWC's strategy and structure was it able to compete effectively.

HBC's Organization The stock in the Hudson's Bay Company was owned by a group of wealthy aristocrats in the United Kingdom. Management was provided from London by a committee of the owners under the leadership of one of their number,

[4] A well-researched and extremely readable source on these matters is Newman's two-volume history of the Hudson's Bay Company. This is the source of most of the details reported here. See Peter C. Newman, *Company of Adventurers* and *Caesars of the Wilderness* (London, New York, and Toronto: Penguin Books, 1988).

the Governor. None of these worthies ever set foot in the area of the company's actual operations during the seventeenth and eighteenth centuries, and they were often monumentally ignorant of the conditions that prevailed in the field. They recruited employees to carry out the trade in North America and paid them a flat salary. Both the general strategy that these employees were to implement and many of the finest particulars about operating decisions were made in London (along with detailed rules regarding the employees' conduct and behavior). Of course, communication between London and Hudson's Bay was extremely slow in the days of sail, especially because the bay is frozen solid for much of the year. Therefore, the lag between the reporting of information and the receipt of a response from management was frequently as long as 15 months.

The company's strategy was to build a few trading posts along the shores of Hudson's Bay and to trade only there. Native tribes near the bay were the initial customers, but they rapidly became independent middlemen, obtaining furs from more distant peoples in return for goods obtained from the company. This pattern had the usual inefficiencies of "double monopoly": The intermediaries took a significant markup for their services. However, the strategy fit well with the HBC's personnel and pay policy and with its management's preference for control. In addition, the employees in the field for the most part had no interest in leaving the relative safety and comfort of their dismal "forts" and "factories" to risk the barren wilderness in hopes of increased company profits but with no extra reward for themselves.

The North West Company

Until the surrender of French claims in Canada to the United Kingdom in 1763, the HBC faced intermittent competition from French Canadian *coureurs de bois*, independent traders from the St. Lawrence who traveled by canoe to trade with the native peoples directly on their home trapping grounds. However, the colonial authorities in New France frowned on such trade, fearing it would lure men away from farming and foster too much independence, and they taxed the returns heavily, quite often simply confiscating the furs. Indeed, the HBC itself had been established at the instigation of two such entrepreneur-explorer-traders, Radisson and Groseilliers, who tired of losing their revenues to the French government.

Once British government and commercial law came to Canada, Scottish and other English-speaking immigrants to Montreal established an effective trade based on the *coureurs de bois* model which competed with the Hudson's Bay Company. By 1779 these traders had formed a partnership called the North West Company, but even before then they had established permanent fur trade posts as far away as the Athabasca River delta, 3,000 miles northwest of Montreal.

THE NWC'S PROBLEM The North West Company (NWC) suffered under an immense technological disadvantage, but they suffered heroically. Montreal and the Hudson's Bay forts of the HBC were approximately the same sailing distance from the source of trade goods and the market for furs in England. Montreal, however, was separated from the prime fur country by an extra 1,500 miles of trackless swamp, bare rock, and impenetrable bush. The Nor'Westers' (North West Company traders') solution was to move the trade goods to the north and bring the furs back in fragile birch-bark canoes powered by French Canadian *voyageur* paddlers. This solution was possible because the interlocking system of Canadian rivers and lakes permits travel from the Atlantic to the Arctic and Pacific Oceans with very few, reasonably short land bridges. Nevertheless, using this system presented monumental challenges.

North West Company canoe brigades would leave the fur-trading posts in the north as soon as the ice left the rivers, traveling across half the North American

continent, through absolute wilderness, along fast, rocky rivers and across storm-swept lakes, often having to lug their canoes and all the goods they carried around raging rapids. Their destination was the company's inland headquarters, first located at Grand Portage on the northwest corner of Lake Superior, just south of the current border between Canada and the United States, and later moved 40 miles north to Fort William. There they would meet canoe brigades that had come up from Montreal, loaded with trade goods, via the Ottawa River and Lake Huron. The trade goods and furs were exchanged, then the Nor'Westers would retrace their paths back to the fur country, where they would spend the winter trading with the native peoples. Meanwhile, the furs would be carried back to Montreal, where they would (if the river was still open) be loaded on ships for England. As a result, there was a minimum lag of 15 to 18 months between the NWC's purchase of trade goods in England and its sale of the furs it traded for them, with 24 months a more common length of time. In contrast, trading at its Hudson's Bay forts allowed the HBC to sell furs within four to six months of buying the goods they traded for them. The difference in working capital needs, combined with that in transportation costs, should have made the HBC dominant, for its costs for importing goods were half those of its rival.

THE SUCCESS OF THE NORTH WEST COMPANY In fact, by the end of the first decade of the nineteenth century, the NWC had seized nearly 80 percent of the trade and was immensely profitable, whereas the HBC was losing money at such a rate that its officers considered getting out of the fur business altogether. A key factor in the Nor'Westers' success was their strategy of building trading posts in the fur lands, which put them close to their customers and gave them a market advantage. But at least as important was their organizational structure, which embodied systems of incentives and decision making that encouraged the effort, imagination, flexibility, and innovation that were crucial to making the market-oriented strategy work.

The HBC was rigidly hierarchical. Rules and controls from distant London circumscribed every action and decision of its employees in the field, leaving little possibility of flexibly responding to emerging conditions. Innovation was discouraged or even punished, and performance was rewarded only by the possibility of someday gaining a promotion to the next rung of the bureaucratic ladder. Employees were chosen for their ability to withstand the excruciating boredom of their indentured terms of service by the frozen bay and for their willingness to work cheaply and follow orders. They were disciplined by floggings for breaking the company's myriad regulations.

Organized and managed this way, the HBC was no match for the NWC. The NWC was a partnership, with two classes of partners who shared in the profits of the enterprise. The senior, Montreal-based partners were responsible for acquiring trade goods and financing and for marketing the furs at the London auctions. The "wintering partners" ran the trade in the field. The two groups met annually at Grand Portage or Fort William to exchange information, set policy, and divide the profits that arose from their efforts. Operating decisions in the field were largely left to the individual wintering partners on site, who could respond quickly and imaginatively and who were well motivated by their ownership shares. Other employees were chosen for their fit with the aggressive, entrepreneurial style that characterized the North West Company and its partners. These people were given real responsibility, performance-related pay, and a serious chance to become a partner.

THE REORGANIZATION OF HUDSON'S BAY COMPANY The HBC ultimately responded to the NWC's challenge, beginning in 1809 when new owners gained effective control of the company after its share price had fallen from £250 to £60. The response was

simple: mimic their rival, build trading posts inland to compete directly with the Nor'Westers, institute a profit-sharing scheme that allocated half the profits to the officers in the field based on performance, give other employees more incentives and more freedom of action, and recruit a new class of employees who would respond to this new organizational strategy.

The Nor'Westers immediately saw the danger that the HBC's new strategy presented, for they always were painfully aware of the cost disadvantage they faced. Their immediate response was an attempted hostile takeover. They sought to buy a controlling block of shares in their rival so as to gain access for themselves to the short transportation route through Hudson's Bay. For once, being based in Canada rather than London was a disadvantage, for before they could implement their strategy, the shares in question had been purchased by the new, aggressive owner-managers and their allies.

The HBC managed to transform itself remarkably quickly from a feudal royal monopoly to an effective commercial competitor, and its inherent cost advantage became decisive. The competition continued fiercely and even bloodily for a decade, but by 1820 the NWC essentially was beaten. The competition ended in a merger that nominally treated the two rivals as equals but in fact gave control to the victorious HBC. However, the NWC's aggressive spirit survived in the merged company, which expanded its range of successful operations in the next half century across the Rockies to the Pacific and even to the Hawaiian Islands and Asia. Today, the HBC has left the fur business, but it is a major real estate firm and the owner of the largest chain of department stores in Canada.

Organizational Strategies of Modern Firms

Providing incentives through compensation and ownership was an important element in the NWC's challenge and in the HBC's successful response. Design of the compensation and ownership structure continues to be an important feature of the organizational strategies of modern firms. A current example from the financial-services industry involves some interesting twists.

Salomon Brothers and the Investment Banking Industry

Salomon Brothers is a major investment bank headquartered in New York City. Like many other investment banks, Salomon was originally organized as a partnership, but it became a publicly traded corporation after the partners sold their ownership claims in 1981 to Phillips Brothers, a firm in the commodity-dealing business. In 1984, John Gutfreund, the head of the Salomon subsidiary, became CEO of the parent company, which had been renamed Phibro. The corporate name was subsequently changed to the current Salomon Inc. During the mid-1980s Salomon was reputed to be the most profitable of all the Wall Street investment banks and, on a per employee basis, the most profitable corporation in the world. Its stock sold in the neighborhood of $60 per share.

Investment banking emerged as a separate industry in the United States after 1934, when the Glass-Steagall Act prohibited commercial banks, which accept deposits and make loans, from underwriting securities as they had previously. Investment banks arose to take over the underwriting function.[5] Modern investment banks, including

[5] Underwriting the issue of a new security first involves the underwriter's agreeing on a price for the security with the issuing entity. The underwriter pays this price for the security, then markets it to investors. The underwriter thus bears the risk that the security will not be saleable at the agreed price.

Salomon, are involved in two main activities: corporate finance, and sales and trading. The former includes underwriting new securities issued by corporations, governments, and not-for-profits; advising on, helping with, and organizing financing for mergers, acquisitions, divestures, and financial restructurings; and advising on corporate financial policy. Sales and trading involve the trading of financial instruments (stocks, bonds, options, warrants, mortgage-backed securities, and so on) both for resale and for the firm's own investment account, and the sale of these investments to (large) individual investors and institutions.

BOND TRADING AND THE CULTURE OF SALOMON BROTHERS Salomon Brothers always has been particularly strong in "fixed income" sales and trading. The general public tends to think of Wall Street in terms of buying and selling stocks. However, in the 1980s the real sales and trading action was in bonds, spurred by the volatility of interest rates that marked the period combined with the huge growth of the U.S. government debt and the debts of U.S. corporations and individuals. Because the price of bonds and other securities that pay a fixed rate of interest varies inversely with market interest rates,[6] volatile rates meant that bond prices could swing wildly, creating opportunities for speculation (buying or selling bonds in the hopes of realizing large profits when prices change, but risking comparably huge losses) and arbitrage (the process of buying and selling to make a nominally riskless profit by taking advantage of price disparities between markets or equivalent assets). The growth of indebtedness meant that there were a great many bonds to trade, especially after Salomon traders developed new securities that packaged together home mortgages to form tradable financial instruments. Thus, even before the explosion of "junk bond" activity centered on Michael Milken of the rival investment bank of Drexel Burnham Lambert, huge profits were being made in bond sales and trading. Salomon traders were the leaders in this business.

Bond trading is a frenzied business in which immense sums of money are at risk. Traders will buy hundreds of millions of dollars worth of bonds in a day, acting on very limited information and under extreme time pressures from competition. Each trader has a pair of telephones and spends much of the day with one in each ear, screaming orders. Tiny movements in interest rates and bond prices result in gigantic gains and losses. The risks are monumental, but the profits that come from successful trading are commensurate. The sort of people who are attracted to this work are an especially individualistic, risk-taking, competitive lot. The leaders of the firm, including Gutfreund, are former bond traders, and they have set the style of the corporate culture at Salomon Brothers. In this culture, performance as measured by profits generated is the sole source of power, prestige, status, and respect.[7]

THE PERFORMANCE-PAY SYSTEM Salomon's compensation system for its employees, from the clerical staff through the approximately 150 managing directors and the more senior executives, involves a base salary plus an annual bonus. The bonus is determined in a fashion that approaches an individual piece-rate system. Almost every transaction is separately priced, with charges for credit risk and overhead, and the resulting profit

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[6] Consider a bond with a maturity value of $100 that pays $5 in interest per year—a 5 percent rate of interest. Now suppose that interest rates rise to 10 percent, so that newly issued bonds with a maturity value and price of $100 must pay $10 per year in interest. Then investors will be unwilling to pay as much as $100 for the old bond, and its price will fall to a level where it again becomes an attractive investment. For a ten-year bond with a maturity value of $100 to yield 10 percent, its price must fall to $74.28.

[7] This culture is described very entertainingly in the best seller by Michael Lewis, *Liar's Poker* (New York: W. W. Norton & Co., 1989).

is credited to the individuals or departments involved. The individual bonuses then are determined on the basis of performance evaluations in which these calculated profit contributions play a key role. However, the bonus payments actually are negotiated; the bonus is not simply computed by formula from the profit contribution.

Because the bonus payments can be extremely large—million-dollar bonuses are not uncommon and the bonus is commonly two thirds of aggregate earnings—the bonus system is a key incentive device in the firm. This performance-pay system is quite elaborately defined and extensively applied by the standards of most businesses. It has very effectively encouraged Salomon's people to work extremely hard and to take great risks to increase both their own and their departments' profits. By doing so, it has reinforced the firm's fiercely competitive corporate culture.

The system has not, however, encouraged cooperation between departments. For example, if corporate bond traders obtained information that might be valuable to Corporate Finance, they might not bother to pass it on because there was nothing in it for them. It also has caused managing directors to focus excessively on their individual departments and accounts, rather than developing the firm as a whole and building its long-term success. Finally, the performance-pay system has resulted sometimes in highly dysfunctional attempts to "steal" other departments' profits.

STOCK OWNERSHIP In May 1990 Salomon Brothers attempted to respond to these drawbacks by reforming its bonus scheme. Myron Scholes, a finance professor on leave at the firm from Stanford University, was instrumental in designing the solution.

Under the new scheme, once the individual bonus has been determined, a fixed percentage is withheld and used to buy stock in the firm. This stock is purchased in the open market, so that the capitalization of the firm will be unchanged. The stock then is held in a trust for the employee, who will not be able to withdraw it for five years. In effect, the value of each employee's current bonus is tied to the overall market value of the firm five years in the future. The plan covers all employees, although there are different percentages applied to those at the managing director level and above. It is estimated that in five years the employees will own 20 percent of the firm.

The plan won acceptance because of the incentive effects it embodies. The explicit aims were to change the culture, to encourage a long-run perspective and cooperation, and to align the employees' interests with one another and with those of the stockholder owners. It was also expected that this scheme would influence favorably the type of people who would be attracted to work for Salomon. In the long run, this "self-selection" or "clientele" effect on recruitment and turnover could be very important. Of course, the plan puts a major new element of risk into the employees' personal incomes because they no longer receive cash that they can invest as they choose in a safely diversified portfolio. Instead, they are locked into a single stock whose price might move up or down. One element in the acceptance of the plan was that the company was selling essentially for the book value of its assets at the time the program was introduced; therefore, an upward movement in its stock price was relatively likely. Nevertheless, the firm also attempted to offset the increased risk to some extent by specifying that it would buy 15 percent more stock for the employees' trust than the bonus pool would have generated.

Many companies attempt to provide incentives to employees, and especially to senior executives, through stock ownership. Employees may be allowed to buy stock at a reduced price, and executive bonuses are often paid in stock or in options that give the right to buy the company's stock in the future at a fixed price. Salomon's technique of using the trust helps ensure that the employees will actually hold on to the stock rather than sell it soon after receiving it. The intent is to ensure that

employees will be concerned with the firm's long-run performance.[8] The trust also has tax advantages, because any appreciation of the price of the stock that occurs while it is being held in the trust is free of capital gains taxation.[9]

Several years' experience will be needed before we fully can judge the effectiveness of this innovation in compensation. However, the system did pass a key market test: As of 1990, announcement of the plan appeared to have raised the price of the 100 million shares of outstanding Salomon stock by between $1.50 and $2 per share from a base of $21. Thus, the plan raised the market value of the firm by between 7 and 10 percent, and the present value of the firm's profits was estimated to have risen by at least $150 million as a consequence of adopting the plan. Because the plan's direct cost to the firm is, if anything, slightly higher than the original cash-only bonus (because of the promise to buy the extra 15 percent more stock for the trust), investors in the market estimate that the new incentives' effect on revenues will be even greater than the direct profit effect.

The Changing Economies of Eastern Europe

Salomon Brothers' pay reforms are an example of experimentation with ownership structure and incentive plans at the level of the individual enterprise. A much more dramatic experiment at a macro level was being carried out in the late 1980s and early 1990s in Eastern Europe, where whole economies were being redesigned.

Recent History

The world was shocked and the Western capitalist democracies were stunned in 1957 when the Union of Soviet Socialist Republics launched Sputnik, the first space satellite, and then quickly followed this triumph by putting the first man in space. Barely a dozen years before, the Soviet Union had emerged from the Second World War having suffered the deaths of 14 million of its citizens, both in battle and through deprivation, and the occupation and destruction of large portions of its territory. Before the war the Soviet economy had been rapidly industrializing under Stalin's Five Year Plans, but the progress was uneven and the Soviet economy was still relatively backward. During the war, the Soviets were forced to depend heavily on food and munitions sent by their allies, especially the United States.

The immediate postwar years saw the outbreak of the Cold War between the Soviet Union and its former allies. This conflict initially went very well for the communist side, with the installation of Soviet-dominated communist regimes in the Eastern European countries occupied by the Red Army and the adoption of communism in China, North Korea, and North Vietnam. Suddenly, the Soviet Union had demonstrated scientific and engineering prowess that seemed to exceed that of the West, and soon the Soviet leader, Nikita Khrushchev, was boasting that the economic might of the Soviet Union would bury the West. Communism, as an alternative to capitalism, not only seemed to enjoy its claims of ethical superiority but also seemed capable of producing remarkable economic progress.

By 1990, however, communism had been rejected as a total failure throughout

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[8] It is possible, especially for sophisticated investors such as these people, to offset the lock-in effects to some degree by adjusting their portfolios. A simple way would be to sell Salomon stock short, contracting to deliver the stock at a fixed price five years hence. This would lock in a price that the employee would be able to get for the stock in the trust; therefore, the future price of the stock would no longer be of concern. The firm no doubt would be very displeased with an employee who shorted its stock.

[9] If the employee had received the stock outright at the time the bonus was earned, held it for five years, then sold it, the increased value over the period would have been a capital gain and been taxed by the federal (and perhaps the state and local) government.

most of Eastern Europe, and even the Soviet Union itself was struggling to create a market economy after 70 years of communist organization. The system simply had not produced. Living standards had been flat or falling for a decade in the Soviet Union and in most of the members of the Soviet bloc. There were constant shortages of food, housing, and all other sorts of consumer products. As a result, workers, with the implicit consent of their managers, took several hours off each day to wait in line at shops. Luck, influence, and patience were the means by which the available goods were rationed; the money that had to be paid for them was at most a secondary consideration: "We pretend to work, and they pretend to pay us." The goods that were available were relatively expensive, shoddy, and often unsafe. For example, more than 2,000 television sets a year exploded in Moscow alone, causing injuries and fires. Agricultural output, which had grown immensely throughout the rest of the world, had stagnated under state control and collectivization, and the Soviet Union had become dependent on grain imports from the West to feed its people. Meanwhile, Soviet agricultural officials annually blamed the repeated crop failures on yet another year of unusually bad weather.

Communist industry was both technologically backward and monumentally inefficient; measured total factor productivity in Soviet industry actually fell in many major sectors in the decades following the triumph of Sputnik, and Eastern Europe faced the most serious industrial pollution problems in the world. Maintaining the Cold War military competition with the United States and NATO was devouring an immense percentage of the already low Soviet gross national product (GNP), nearly bankrupting the country. Instead of the "workers' paradise" and equality that were promised, communism had delivered a repressive, totalitarian government and institutionalized privileges for the politically favored.

Building Socialism

Soviet communism found its intellectual origin in the writings of Karl Marx, a nineteenth-century classical economist. However, Marx provided no blueprint for organizing an economy along communist lines. For Marx, the triumph of communism was simply foreordained by the inherent contradictions in the capitalist system. The actual task of designing the first communist system fell initially to Nikolai Lenin, the first leader of the Soviet state that emerged after the Revolution of 1917, and then to Joseph Stalin, who ultimately succeeded Lenin and led the Soviet Union from the late 1920s until after the Second World War. It was Stalin's vision that defined the basic form for the Soviet economy: state socialism and central planning under the direction of the Communist Party.

The system that Stalin developed replaced private ownership of the means of production by collective—or, rather, state—ownership of land, buildings, machines, and other capital. Factories belonged to the communist state as representing the proletariat, and individual farms were forcibly socialized into collective farming enterprises. Private business essentially ceased to exist, except on the smallest scale. Workers were guaranteed jobs and had latitude in their career choices, but their mobility was limited and being unemployed became a criminal offense.

Prices were set by the government and left unchanged for extended periods. They therefore bore little or no connection to costs and could not serve to ration demand when shortages appeared. Prices also were not allowed to direct resources to their highest-value uses, increasing supplies of goods where there was shortage and favoring purchases from low-cost producers. Instead, a system of detailed central planning and extensive vertical information flows was instituted to replace the decentralized decision making, horizontal communications, and price-guided coordination that are characteristic of market systems. Central planners in Moscow decided

how much of what goods were to be produced in each period by which factories. The plans directed where inputs were to be obtained, where outputs were to go, and what prices were to be paid. Similar systems were instituted in Eastern Europe after the communists gained control there. In Poland at one point even the production of pickled cucumbers and the number of hares that would be shot by hunters were included in the plan, although later the planning was less pervasive. In all cases, investment decisions were centralized, with capital allocated by the state. As circumstances changed and unforeseen contingencies arose that rendered the original plan infeasible, information would be passed up to the central planners who would attempt to redefine the plan and coordinate activities.

The initial intent of the socialist system was to replace economic incentives with political and moral appeals to the workers' patriotism and socialist consciousness. Pay was divorced from supply and demand—"From each according to his abilities, to each according to his needs"—and the aggregate amount of GNP to be devoted to consumer goods was determined by the planners so as to leave enough extra resources to finance the planned investments. However, whether the means of realizing the plan were administrative orders or designed economic incentives, individual self-interest intervened. This led to the sort of difficulties epitomized by the familiar (if possibly apocryphal) story of a factory meeting its target of 10,000 kilograms of nails by producing a single nail weighing 10,000 kilos. Later, when the planning and incentive systems were refined, incentive problems still arose, especially with regard to quality. A worker in a Baltic television manufacturing facility told of the rush at the end of each month to meet production targets and earn bonuses:

> We never use a screwdriver in the last week. We hammer the screws in. We slam solder on the connections, cannibalize parts from other televisions if we run out of the right ones, use glue or hammers to fix switches that were never meant for that model. And all the time the management is pressing us to work faster, to make the target so we all get our bonuses.[10]

The 2,000 exploding televisions a year in Moscow are now understandable.

THE RATCHET EFFECT A particular incentive problem that proved to be of fundamental importance was the *ratchet effect*. The central planners were never as well informed about the productive capability of any particular factory as were the factory managers. The planners would set targets for input usage and output levels and then use the information gathered from experience to judge what would be possible in future planning periods. Those factories making their quotas were rewarded in various ways; those failing were punished. From the point of view of the factory manager, the incentives inherent in this system were especially perverse. By exceeding the quota this period, the manager could expect to be "rewarded" with a higher quota for the indefinite future, as performance expectations are ratcheted up or raised to a higher level. As a result, there was no reason ever to exceed quota. Instead, the incentives were to meet the quota barely, or even—provided a good excuse were available—deliberately miss it somewhat so that the next period's quota would be lower and easier to meet. In particular, there was an inexorable tendency to hoard resources, to hold back on effort and output, and to underreport capabilities.

These tendencies were intensified by the experience of factories and managers who responded to experiments with stronger and better incentives by increasing production, only to be accused of having defrauded the state before. These perverse

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[10] Clive Cook, "A Survey of Perestroika," *The Economist*, April 28, 1990, 6.

incentives led to low productivity in the economy. They also burdened the planners with systematically distorted information. The task of planning even the key resource flows in the economy would have been formidable if the information being conveyed was accurate. Huge amounts of information needed to be transferred. Even employing strong simplifying assumptions about the structure of technology, the computational problems were extreme, and the problems of responding to unforeseen, emergent events were overwhelming. The factory managers' misrepresentation of productivity information and their secret hoarding of resources made the task impossible.

Beginning in the 1960s there were many attempts to reform the communist economies. Moves were made towards decentralization designed to give more decision-making authority to local planners and individual factories and to provide better incentives. None of these attempts were truly successful, although Hungary's liberalization moves following 1968 had positive effects.

The Collapse of Communism

The collapse of communism in Eastern Europe was dramatically rapid. There had been a history of worker uprisings and attempted breaks with Moscow—in East Germany and Hungary in the 1950s, in Czechoslovakia in the 1960s, and in Poland in the late 1970s and early 1980s. All had been more or less successfully squashed, often with the use of Soviet troops. Then in 1989 and 1990, within a single year following signals from Soviet leader Mikhail Gorbachev that the Soviet Union would no longer intervene in the internal affairs of the Eastern European countries, noncommunist governments appeared, the Berlin Wall came down and travel restrictions were removed, independent presses evolved, multiparty elections were held and capitalist-oriented parties were elected, communist parties were disbanded or re-formed as socialist parties, East and West Germany were united under the West German model, and experiments in moving to market economies began in the other Eastern European countries. These changes were most radical in Poland, which quite simply decided to restructure itself as a free-enterprise economy as soon as possible.

In the Soviet Union Gorbachev's policies of *Glasnost* (openness) and *Perestroika* (restructuring) were originally meant simply to reform communism and make it work better.[11] Even with these intentions, the policies met strong resistance from the entrenched party functionaries and bureaucrats. On the other side, "radicals" and noncommunists won election victories and began exerting pressure for the elimination of communism and its replacement by a market system.

THE CHALLENGES OF ADOPTING A MARKET SYSTEM These economies face great problems in moving to more market-oriented systems. They must determine property rights and decide who will be allowed to own the currently state-owned enterprises, how title will be transferred, and what prices will be charged. They must set up capital markets and create banking, financial, and monetary systems. They have to design meaningful accounting systems so that firms can be valued and their performances judged. They need to redraft their laws to allow for new forms of economic organizations, new patterns of ownership, and new sorts of transactions. They have to find managers who can operate in a market system and compete in a world market.

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[11] George Shultz, the former U.S. Secretary of State, is reported to have been told by his Soviet counterpart, Edvard Schevardnadze, that the Chernobyl nuclear reactor accident made the Soviet leadership more fully appreciate the need for Glasnost. The Kremlin leaders were unable to get reliable, timely information on the crisis through their bureaucratic channels. Instead they found that the Cable News Network (CNN) broadcasting by satellite from the United States was their best source of information on a major event in their own country. This drove home to top government leaders that their closed system had to be opened up for the benefit of the country.

They have to educate their populations to the new rules of the game and gain acceptance for these rules. They must decide on competition and regulatory policies and find a way to deal with the fact that simply privatizing the giant, inefficient state firms will yield a system of giant, inefficient private monopolies. They must decide how much to wean their industries from state subsidies and develop tax systems to finance government activities. They have to decide whether and when uncompetitive firms will be allowed to fail and create social-service and support systems to handle the human costs of the dislocations that their economies are sure to face, both during and after the transition.

The great difficulty in all this is that these tasks are all interdependent and in need of coordination. Private enterprise will not work without the discipline of potential failure. Output markets are of no use without input markets, including ones for capital, through which producers can obtain the resources they need. Neither of these are of much value without well-defined property rights and mechanisms for contract enforcement. All the parts have to come together and fit reasonably well for the system to work. The worst outcome might be some halfway compromise. Consider the joke told in response to criticisms that Poland's "cold-turkey" foreswearing of communism and adoption of free enterprise is too much, too soon: It was decided in a certain country to change the side of the road on which traffic drove from the left to the right, but there was concern about making too radical and rapid a switch; so as an experiment only the trucks changed sides for the first year.

Patterns of Organizational Success and Failure

Henry Ford's account of organization missed the mark. The study of organization is not about how berries are arranged on a tree of authority but about how people are coordinated and motivated to get things done.

In these few, brief historical accounts, several patterns have begun to emerge. First, and most fundamentally, organization and business strategy can be as important as technology, cost, and demand in determining a firm's success. Despite its superior technology, greater resources, and scale advantages, Ford Motor Company under Henry Ford's management lost its battle with Alfred Sloan's General Motors. General Motors, in turn, lost market share to a smaller and technologically weaker Toyota, which labored under the same kinds of disadvantages. The Hudson's Bay Company suffered many defeats in its competition with the fledgling North West Company. The successful competitors in these stories gained advantage partly from the strategies they adopted in their markets, but a large part of their advantage also came from their innovative organizational structures and policies and especially from the match of their strategies and their structures.

Which aspects of organization matter? In these examples, incentives are one important element. The Hudson's Bay Company employees were not much inclined to show initiative and judgment when they could be flogged for their errors but got no share of any extra profits they earned for the company. At General Motors in 1920, the failure to charge divisions for the cost of the inventories they accumulated was responsible for a huge inventory build-up leading to a financial crisis. Salomon Brothers, with its many independently operating traders, originally had the incentives partly right. However, its emphasis on individual performance evaluation discouraged employees from cooperating and from sharing information with one another. Finally, the communist countries, with their ideological commitment to economic equality, ran afoul of incentives at every turn.

Another important shared feature of successful organizations in these examples

is the tendency to place authority for decisions in the hands of those with information. Salomon Brothers, with two telephones and enormous discretion in the hands of its traders, is an extreme example. General Motors, with its multidivisional organization, placed product and marketing decisions in the hands of divisional managers. Toyota's decision to give responsibility and authority for machine repair and maintenance to those who operate the machines led to improved reliability. And the North West Company gained advantage over its larger competitor by operating as a partnership with the wintering partners in the field making timely decisions based on up-to-date information about local market conditions.

Although delegating authority to those with the information needed to make good decisions is an important part of good organization design, it is of little use unless the decision makers share the organization's objectives. We have already mentioned incentives as a way to align individual and organizational objectives. The additional point we want to make here is that incentives are especially important when more initiative is expected from employees. The same point, viewed from a different angle, is that delegation of authority is much more valuable when those being empowered have also been given incentives to work for the organization's objectives. It is no accident that the North West Company both relied on the judgments made by traders in the wilderness and made them partners who shared in the profits. Nor is it an accident that the traders at Salomon Brothers whose decisions can have a multimillion-dollar effect on the company's holdings are provided with both information and the incentive to use it well. Similarly, in Eastern Europe, giving more discretion to factory managers without also providing for decentralized ownership could not resolve the problems.

In the language of economics, incentives and delegated authority are *complements*: each makes the other more valuable. Evaluating complementarities—how the pieces of a successful organization fit together and how they fit with the company's strategy—is one of the most challenging and rewarding parts of organizational analysis. In our General Motors example, the reason for the multidivisional structure was to carry out Sloan's new market-segmentation strategy. The delegation of decisions to divisional managers was also combined with improved accounting information to help evaluate those decisions and with coordination from the central office to ensure that the parts of the organization were not working at cross purposes. At Toyota, an organization based on very low inventories was naturally vulnerable to disruptions at any stage in the production process and to simple changes in plans. If low inventories were to be achieved, the rest of the organization had to emphasize reliability, quick response to machine breakdowns, quick and close communications with suppliers, stable production plans, and other features that substitute for the buffer function of inventories. Toyota's emphasis on reliability fit with an emphasis on quality in its marketing, and its use of flexible equipment, necessitated by its initially small scale, fit well with its more frequent redesign of its several models.

In this book, our economic analysis of organizations is based on elaborating the several ideas that these brief histories suggest. We study **coordination**: what needs to be coordinated, how coordination is achieved in markets and inside firms, what the alternatives are to close coordination between units, and how the pieces of the system fit together. We also study **incentives** and **motivation**: what needs to be motivated, why incentives are needed and how they are provided in markets and firms, what alternative kinds of incentive systems are possible and what needs to be done to make incentive systems effective. In the last chapters, we see in more detail why these aspects of organization do matter, when we apply the principles to make a detailed study of a few important functions of the business enterprise.

EXERCISES

Food for Thought

1. In fast-food chains, some decisions about standards are made centrally and others are left to the individual outlet managers. Who typically makes which kinds of decisions? Why? Can you think successfully about the fast-food business by dividing the issues between coordination and motivation?

2. Armies in battle have especially severe organization problems. What kinds of decisions are made centrally and which are left to commanders in the field? What principles dictate the division? Can you think successfully about the problems of military organization by dividing the issues between coordination and motivation?

3. In late summer of 1991, it was revealed that beginning in February of that year, the Salomon Brothers' managing director in charge of its dealings in the markets for U.S. Treasury securities had secretly violated government rules designed to prevent any one buyer from purchasing more than a specified share of the new government debt obligations being auctioned at any one time. Moreover, senior executives at the firm had learned of these violations of the law but had failed to act on them. In the resulting scandal, four of Salomon's top executives (including Gutfreund) were asked to resign, and the future of the firm was imperiled.

One theory was that the employee who violated the rules was motivated by his competitive, aggressive nature: He bitterly resented the government's attempt to limit his actions and was determined to show them that he was smarter and tougher than they. Another was that he was motivated by greed: He wanted to monopolize the market in these government bonds and reap the rewards. In any case, his actions, and those of the senior executives who failed to stop him and report his wrong-doing, were very costly to the firm: The stock price, which had risen by almost a half in the preceding year, fell back to where it had been before, and important customers ceased dealing with the firm.

Does this episode mean that the incentive scheme described in the text was fundamentally flawed?

It turns out that one simplifying assumption, that is, the condition of no wealth effects discussed in the next section, takes care of this problem completely.[6] When this condition is satisfied, only one pattern of behavior is consistent with efficiency, and that is the pattern that maximizes the total value created in the transaction.

Wealth Effects, Value Maximization, and the Coase Theorem

In many economic decisions, the choice actually made depends on the decision maker's wealth. A poor person (or a poor country) may not have the resources to pursue some courses of action that a richer one could. Even when the same alternatives are affordable, a poorer person might still make different trade-offs than a richer one. For example, the poor person might be reluctant to take financial risks that a rich person would welcome. The changes in choices resulting from increased wealth are known as **wealth effects**.

The Value Maximization Principle

Although wealth effects can sometimes be significant, this is not always the case. In fact, the formal analysis of problems in the economics of organizations is greatly simplified when wealth effects can be ignored entirely. Moreover, it is precisely in ignoring wealth effects that such key management concepts as "creating value" become unambiguously defined.

No Wealth Effects We say that there are *no wealth effects* for a certain decision maker with respect to a set of possible decisions when three conditions hold. First, given any two alternative decisions y_1 and y_2, there is a definite amount of money $\$x$ that would be sufficient to compensate the decision maker for switching from y_1 to y_2 (or from y_2 to y_1). Second, if the decision maker were first given an additional amount of wealth, then the amount needed to compensate the decision maker for the switch from y_1 to y_2 would be unaffected. Third, the decision maker must have enough money to be able to absorb any wealth reduction necessary to pay for a switch from the less preferred to the more preferred option.

None of these conditions can be expected always to hold. For some people, for example, there may be no amount of money that they would accept as compensation for a serious risk of loss of life or limb, or for being forced to live far from family and their childhood homes, or to live in a culture where they cannot exercise their religious beliefs. Nevertheless, the condition that there is some monetary amount that would compensate for a change of circumstances holds widely for many of the most common kinds of business decisions.

To examine the implications and applicability of the second condition, suppose that a corporation suddenly finds itself richer on account of an unexpected increase in the value of its assets. If there are no wealth effects, then the price the corporation would demand for its goods and the returns it would demand from its planned investments would remain unchanged. In this example, the absence of wealth effects

[6] Like other modelers, economists use the terms *assumption* and *assume* in a different sense than they are often used in ordinary discourse. In everyday conversation, assuming something connotes believing it is true. Making an assumption in an economic model carries no such connotation. An assumption is merely a working hypothesis used to abstract from the complexity of the real economic world. The purpose of making an assumption may be to derive a good approximate prediction or to highlight a single force or effect for closer study and better understanding. Assumptions are used in both ways in this text.

seems likely to hold, at least over a broad range of wealth levels. As a second example, however, suppose a worker wins the jackpot in the state lottery. If there are no wealth effects with respect to current consumption choices, then the winner would not buy a new house, or quit his or her job, or do any of the things lottery winners normally do. Here the assumption of no wealth effects seems particularly inappropriate.

The third condition connects the decision maker's initial wealth and the changes in the nonfinancial situation that are being considered. The amount by which a worker's pay could be cut to offset his or her psychic gain in being allowed to shift working hours to miss rush-hour traffic is probably small relative to income, and the condition is likely to hold in this case. On the other hand, holding a worker in a nuclear reactor facility financially responsible for the effects of his or her mistakes is apt to run into wealth constraints, and so the assumption of no wealth effects will be inappropriate.

In general, these examples suggest that the assumption of no wealth effects is most restrictive—least likely to be valid and most likely to lead to incorrect conclusions—when the decision makers are individuals and when large cash transfers or significant changes in personal living conditions are involved. When the sizes of the cash transfers are small relative to the decision maker's financial resources, assuming that there are no wealth effects (or that they are small enough that they can safely be ignored) is more likely to be a good approximation to reality.

THE EQUIVALENT VALUE INDEX The utility function of a decision maker who shows no wealth effects with respect to a set of decisions can be represented very simply.

Let x represent the decision maker's monetary wealth and let y be a list of all the other influences or characteristics associated with decisions that affect his or her preferences: social approval, job assignment, effort exerted on the job, and so on. An important example involves uncertain income and expenses, where x is interpreted as the certain, unconditional amount of money that will be received (or the average, expected amount) and y reflects a risky component of income. In general, the utility function takes the form $u(x,y)$, where x and y may interact in complex ways. If there are no wealth effects, however, then there is always a cash equivalent value $v(y)$ that can be assigned to the list y and the decision maker's utility function can be written in the form $u(x,y) = x + v(y)$. In other words, by adding the cash equivalent value $v(y)$ to the decision maker's wealth x, we obtain an index of personal welfare, which we may call his or her *value index*.[7] The importance of the value index is that, when it is valid, a related index—the total value of the affected parties—is an appropriate measure of welfare changes for group decision making.[8] We state the matter as follows.

> The Value Maximization Principle: An allocation among a group of people whose preferences display no wealth effects is efficient only if it maximizes the total value of the affected parties. Moreover, for any inefficient allocation, there exists another (total value maximizing) allocation that *all* of the parties *strictly* prefer.

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[7] To see how the three preceding conditions relate to this formula, note first that the change in x necessary to compensate for a change from y_1 to y_2 is easily computed to be $\Delta x = v(y_2) - v(y_1)$, because for any given initial wealth M, $M + \Delta x + v(y_2) = M + v(y_1)$. The change in x to compensate for switching from y_2 to y_1 is then just $-\Delta x$. The calculated amount Δx is independent of the initial wealth, M, as the second condition requires. With a general utility function $u(x,y)$, there might be no amount Δ that would make the equality $u(M+\Delta, y_2) = u(M, y_1)$ hold, as the first condition requires, and even if there were such an amount, its magnitude would generally depend on M. Finally, as long as the initial wealth M is larger than the largest possible difference in the $v(y)$ values between two alternative ys, so that making the transfer Δx cannot require more money than the individual has, then the third condition holds.

[8] The total value is also sometimes called the total (consumer and/or producer) surplus.

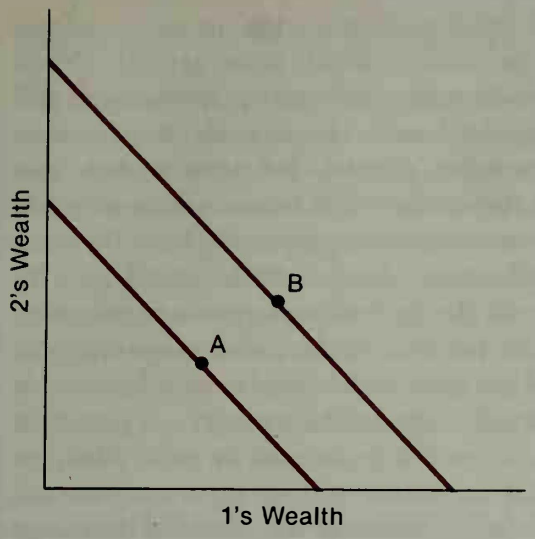

Figure 2.1: Any point like A on a line of lower total wealth is Pareto dominated by some point B on the line of highest total wealth. Points like B on the line of highest total wealth are undominated.

The Logic of Value Maximization

To establish the principle for a particular example, consider an investment decision involving two people with utility functions that satisfy the condition of no wealth effects: $U_i(x,y) = x + v_i(y)$, $i=1,2$, where y represents inputs to be provided by the parties.[9] The investment generates total cash income of $P(y)$. We may think of $v_i(y)$ as representing the personal cost investor i bears in supplying the agreed inputs. In that case, $v_i(y)$ is a negative number for positive levels of y. The income $P(y)$ will be divided between the individuals, with x_1 being paid to individual 1 and x_2 being paid to individual 2, so that $x_1 + x_2 = P(y)$. For any particular allocation (x_1,x_2,y), *the total utility or value of the two parties is* $[x_1 + v_1(y)] + [x_2 + v_2(y)]$, which (because $x_1 + x_2 = P(y)$) is equal to $P(y) + v_1(y) + v_2(y)$. *The total value depends on y alone and not on the profit shares x.* As we vary the profit shares x_1 and x_2, the individual utilities of the two parties change, but the total utility remains fixed.

Figure 2.1 illustrates the situation. Each line depicts the possible levels of value for the two parties for any fixed investment decision y, as the profit shares x are varied. The fact that the lines are straight and run at 45° to the axes reflects the fact that total value is independent of its distribution between the parties; it is possible to move utility or value from one party to the other (via the xs) without affecting the totals. It is clear from the diagram that any point like A on a line that corresponds to lower total value is Pareto dominated by some point like B on the line of highest total value. It is also clear from the figure that no point on the line of highest total value is Pareto dominated by any point on that line or any other line. Consequently, *an allocation* (x_1,x_2,y) *is efficient if and only if y maximizes the total value:* $P(y) + v_1(y) + v_2(y)$.

The intuition of this result is simple: With more total it is always possible to distribute it in a way that makes everyone better off. A full mathematical proof of the proposition is developed in the exercises at the end of the chapter.[10]

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[9] We can think of y as being the pair (y_1,y_2), where y_i is i's contribution. Then writing $v_i(y)$ rather than $v_i(y_i)$ allows the possibility that each individual's costs depend on both parties' contributions. At the same time, writing $v_i(y)$ allows that v_i depends only on y_i.

[10] In the presence of wealth effects, an allocation that maximizes the sum of utilities is still efficient, but there may also be efficient allocations that do not maximize the sum of utilities.

Applying Value Maximization Although this discussion is set in the context of two people making an investment, the principle itself is much more general. When preferences take the form we have described, then any decision (x,y) is efficient if and only if y is chosen to maximize the total value of the parties. Importantly, the efficiency of the choice (x,y) does not depend on the selection of x, which determines only how the fruits of the joint enterprise are shared. When the value maximization principle applies, the issue of the distribution of value is completely separable from the issue of how value is created. While this separation is not always realistic (see Chapter 8), it is often reasonable and it does always simplify the analysis of economic organization problems. For that reason, it is a good starting point for thinking about organizations.

The abstract model we have described has wide application because the variable y can be given so many different interpretations. The model can apply to people in a community who must decide on the resources y to be devoted to parks, libraries, or some other public good. It can also apply to overtime assignments (y is then the identity of the person assigned to work overtime), to water rights (y identifies the owner of the rights), to the brand of computer to be used (y names the brand), or to the location of the new office building (y specifies the location).

The Coase Theorem

In practical business situations, the way the benefits of an agreement are divided among the parties will depend, of course, on what assets each brings to the bargaining table, how patient each party is, what outside opportunities are available, and so on. Nevertheless, if the parties engage in efficient bargaining, that is, if they reach an agreement from which there is no possibility of further mutual gain, and if the value maximization principle applies, then regardless of what cash changes hands at the time of agreement, y will be chosen to maximize the total value of the parties to the agreement. Only the distribution of the costs and benefits will be affected by the strength of their relative bargaining positions, and this distribution will show up in the xs. This conclusion is summarized in a proposition that is also due to Coase:

> The Coase Theorem: If the parties bargain to an efficient agreement (for themselves) and if their preferences display no wealth effects, then the value-creating activities (y) that they will agree upon do not depend on the bargaining power of the parties or on what assets each owned when the bargaining began. Rather, efficiency alone determines the activity choice. The other factors can affect only decisions about how the costs and benefits are to be shared (x).

This celebrated proposition is the foundation of the transaction costs approach to the theory of the firm and other economic organizations. With the assumption of no wealth effects, the Coase theorem and the efficiency principle mean that all real activities are determined to maximize the total value of the parties, taking into account the costs of organization (transaction costs) along with all other kinds of costs. For any given plan of production specifying who is to make what with which resources (and thus the aggregate production costs that will be incurred), if we think of transaction costs as the costs of managing the transactions (including the costs of writing contracts, supervising workers, enforcing contracts, and resolving disputes), then the efficient organization for that plan is the one that minimizes transaction costs.

To understand the significance of this perspective, it is helpful to contrast it with other perspectives that have been vigorously advocated.

The Transaction Costs Approach versus Alternative Views

The transaction costs approach differs sharply from the Marxian approach. According to Marxian theorists, organizational arrangements are a reflection of underlying power relationships and class interests. In contrast, according to the transaction costs perspective, the choice of a firm's organization (y) does not depend on the a priori distribution of power between the owners of capital and the laborers. For example, in Yugoslavia, where employee control of firms has been the norm, labor hired managers who organized the factories in similar ways and instituted similar kinds of controls (even over labor) as those seen in capitalist firms.

Applied to relationships among firms and between firms and their customers, the transaction costs approach suggests that business arrangements should be understood as attempts to increase the total wealth available for sharing among the parties. Scholars trained in the Harvard school of industrial organization and antitrust would try instead to explain the arrangements as attempts by the firms to increase their ability to manipulate prices for the products they sell or the inputs or labor they buy. For example, resale price maintenance—the practice by manufacturers of limiting contractually the rights of distributors and retailers to set prices—has been attacked by members of the Harvard school as anticompetitive. The transaction costs approach suggests that it in fact may be efficiency promoting. Adherents of this approach might argue that, without resale price maintenance, discounters would free-ride off the service and customer education provided by other dealers, relying on them to provide product support but then grabbing the sales for themselves. This would drive the amount of such support below efficient levels.

The correct way to decide between competing hypotheses is to confront them systematically with detailed evidence. The observation concerning management methods in the Yugoslav firm represents an instance of this, but it is by itself insufficient. We report more of the relevant evidence later in this book.

In any case, in application, it is important to remember that the Coase theorem and its various implications depend on restrictive hypotheses regarding preferences and, perhaps more importantly, on the ability to make unlimited transfer payments between the parties. The implications do not hold when some of the parties have very limited capital with which to make payments. Thus, although it would be reasonable to apply this analysis to study the terms of a contract between General Motors and Toyota, it would be a mistake to apply it uncritically, for example, to study land tenure in a developing country or the institution of slavery in the pre-Civil War American South.

Organizational Objectives

It is traditional in economics and management texts to assume that firms seek to maximize profits and, more generally, to ascribe well-defined goals to organizations and to presume that the organization acts in pursuit of these objectives. Occasionally we employ such a hypothesis in this text, when it is convenient to treat the organization as a purposeful entity. More often, however, we do *not* presume that organizations per se have goals that they seek to realize. Rather, we are careful to treat organizational decisions and actions as the outcomes either of strategic interplay among self-interested people responding to incentives designed to influence their behavior, or of collective or managerial attempts to compromise the interests of the parties affected by the decisions. Only when the value maximization principle applies is there an objective that we can ascribe to the firm that is implied by considerations of efficiency alone.

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EXERCISES

Food for Thought

1. One of the main tenets of economic analysis is that people act in their own narrow interests. Why, then, do people leave tips in restaurants? If a study were to compare the size of the tips earned by servers in roadside restaurants with those frequented mostly by locals, what would you expect to find? Why?

2. For most large Japanese firms, a majority of the voting shares are owned by the company's major lenders (banks and insurance companies), its customers and suppliers, and related firms with whom it has longstanding relationships. How would this ownership structure affect the objectives that these firms seem to pursue?

3. Would you expect the organization of agriculture in developing countries to be arranged in a way that maximizes the total wealth of the farmers, workers, and lenders? If arrangements do not maximize total wealth, what kind of variations would be most likely? Explain your answer.

4. In California's fruit farms, farm workers who pick fruit are commonly organized into teams that are paid according to the number of trees that are cleanly picked. The teams themselves decide how to divide the pay among their members. What attributes of this transaction account for this arrangement?

5. Cable television companies lay cables to individual households in the communities they serve to carry the television signal. How specific is this investment? What kind of arrangements would you expect the cable companies to make with local communities about the pricing and taxation of cable services?

Mathematical Exercises

1. Suppose four families share a common stretch of beach and they are considering a program of improvements, including a stairway and a play structure for children. The value of spending y in total for the improvements is $5y - \frac{1}{2}y^2$ for families #1 and #2, $7y - \frac{1}{2}y^2$ for family #3, and $4y - y^2$ for family #4. What is the efficient level of expenditure on beach improvements?

2. Continuing with the situation hypothesized in problem 1, show that if the cost of the improvements is to be shared equally, then family #4 will be unwilling to bear its share of the cost. What is the largest improvement that all the families would agree to if the cost of improvement must be shared equally? Demonstrate that the resulting expenditure is inefficiently low by finding an alternative level of expenditure and pattern of cost sharing that the families unanimously prefer.

3. (Mathematical proof of the value maximization principle.) Suppose that there are N individuals. Person n's utility when outcome y occurs and he or she receives cash compensation of x_n is given by the utility function $x_n + v_n(y)$. Suppose that decision y generates net profits $P(y)$, which may be a positive number if we think of y as representing an investment or a negative number if we think of y as representing some public good, such as parks or roads, at a total cost of $-P(y)$. The profits are divided among the individuals with individual n receiving x_n (or paying $-x_n$). The payments must add up to the amount available, that is, $P(y) = x_1 + \ldots + x_N$. Prove that an allocation $(y,x_1,\ldots,x_N)$ is efficient if and only if y maximizes the total value $P(y) + v_1(y) + \ldots + v_N(y)$. [*Hint:* There are two things to prove. First, you must show that if an allocation does maximize the total value, then there cannot be an allocation that Pareto dominates it. Second, if the allocation does not maximize the sum, then you must show that there is another allocation that dominates it. To show the latter, take any y with a higher total value and show that the x_ns can be chosen so that the gain in total wealth is divided equally among the participants.]

4. (Characterizing a utility function when there are no wealth effects.) Suppose that a decision maker's preferences are such that for any two decisions y and y', there is an amount of cash compensation or a cash payment such that y combined with cash compensation of $C(y,y')$ would be just as good, from the decision maker's perspective, as y' with a zero-cash compensation. Further suppose that this amount $C(y,y')$ does not depend on the level of other cash payments made to or by the decision maker. Finally, suppose that the decision maker prefers more money to less. Fix any possible decision $\bar{y}$ and define $v(y) = C(\bar{y},y)$. Show that with this definition, the utility function $x + v(y)$ represents the decision maker's preferences; that is, the decision maker will prefer an allocation (x,y) to another allocation (x',y') if and only if $x + v(y) > x' + v(y')$. [*Hint:* Argue first that the decision maker is indifferent about having (x,y) or $(x+v(y),\bar{y})$. Therefore, for any utility function that represents the decision maker's preferences, $U(x,y) = U(x+v(y),\bar{y})$ and, similarly, $U(x',y') = U(x'+v(y'),\bar{y})$.]

MR Ch. 3-6

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- Design attributes: Ch. 4, "The Variety of Coordination Problems and Solutions" (p. 90–93)
- Scale, scope, core competencies; complementarities; coherent strategies: Ch. 4, "Coordination and Busine
- FFT 4.1–4.3, Math 4.2: Ch. 4, Exercises (p. 119–120)
- Formal model of design decisions (for Math 4.2): Ch. 4, Mathematical Appendix (p. 121–124)
- Investments and specific assets; achieving commitment: Ch. 5, "Bounded Rationality and Contractual Inco
- Bargaining over a sale; incentive efficiency; bargaining costs; bargaining position (De Beers): Ch. 5, "Priva
- FFT 5.1–5.2; Q 5.4–5.5: Ch. 5, Exercises (p. 163–165)

market transactions. For example, they may work cooperatively with their suppliers to develop together the specifications for the inputs they seek, and they negotiate complex requirements contracts under which one firm has the power to direct how much the other will supply to it. Less formally, they may share information on plans and estimated requirements. They set up joint ventures and enter other sorts of alliances, they design complex royalty agreements and franchise contracts, and they put law firms on retainer, paying them even if they do not use their services. Within these relationships, they may exchange large amounts of information and often formulate joint plans.

In Chapter 3 we explored reasons why a price system would not yield an efficient outcome and so might be replaced: increasing returns, externalities, missing insurance and futures markets, excessive search costs, and the possibility of unemployment equilibria. In this chapter we focus mostly on situations where, in principle, the price system could be used and the fundamental theorem of welfare economics would hold, and yet other mechanisms for coordination are actually employed.

Planning and coordinating economic activity never come for free. It takes real resources to plan—people with offices, files, data banks, and the computing and communication equipment to support them. In addition to the planners' time, planning demands time from production people who must fill out forms, complete reports, and answer the planners' queries. At the end of the planning process, errors inevitably still occur, both because the prices or plans are based partly on guesses and partly on erroneous, incomplete, or misleading information and because miscalculations and mistakes occur.

In actual economies, a loose mix of systems is used to coordinate and manage the various kinds of activities. What determines which system is or ought to be used in any particular set of circumstances? How can we account for Weitzman's observation that prices are often ignored in internal decision making, so that "the price system as an allocator of resources does not pass the market test"? To answer these questions, we need to study coordination problems and systems in more detail.

The Variety of Coordination Problems and Solutions

Robinson Crusoe, living alone before meeting Friday, took care of all his own needs. Gathering and preparing his own food and securing his own shelter, Crusoe spent no effort coordinating his activities with those of anyone else. The need for coordination comes from specialization, in which various tasks are divided among a group of people, each of whom relies on the others for part of the job. As noted in Chapters 2 and 3, specialization creates the opportunity for enormous increases in productivity. People who specialize in a job can prepare specialized tools, gain specialized training, develop specialized methods, and exploit their accumulated experience to get more done, more quickly and with fewer resources.

The kind of coordination that is most effective depends on the nature of the task. It is helpful to distinguish among several kinds of problems to understand the kinds of solutions that are used. The most general kind of coordination problem we consider is called a **resource allocation problem**. This is a problem of allocating a fixed set of resources among various possible uses. The term resource can be interpreted broadly enough to classify virtually every kind of important economic or business decision as a resource allocation problem. In this chapter, however, it is useful to distinguish particular attributes of resource allocation problems that make one system of coordination or another especially effective.

Design Attributes

We are especially interested in problems with **design attributes**. These are problems in which (1) there is a great deal of *a priori* information about the form of the optimal solution, that is, about how the variables should be related, and (2) failing to achieve the right relationship among the variables is generally more costly than are other kinds of errors, including especially slight misspecifications of the overall pattern, as long as the individual pieces fit. In this discussion, the word design is a general term describing a system in which the pieces must fit together in a predictable way, thus narrowing the search for efficient decisions. Two common kinds of problems with design attributes are synchronization problems and assignment problems.

SYNCHRONIZATION PROBLEMS An extreme example of a **synchronization problem** arises in the sport of crew, in which it is crucially important that each rower make his or her stroke at precisely the same moment. The coxswain solves the synchronization problem by determining a rhythm for the crew and calling out the signal for each stroke. Like most centrally directed solutions, synchronization has the disadvantage that the centrally made decision cannot be fully responsive to information of the others in the system. In this case, the coxswain can only guess how tired the individual crew members are. This could be an important disadvantage if the coxswain pushes the crew too hard early in the race, leaving them too weak for a strong finishing sprint. However, the great advantage of the system is it synchronizes the actions of the crew, making their individual efforts much more effective. The costs of not setting quite the right pace are very small compared to those of failing to have everyone pulling in unison.

Though it seems ridiculous to contemplate using prices in this context, it is illuminating to see what a price solution for this problem would be and why it would not work well. In this application, a price system would entail the coxswain telling each rower the "price" of effort, that is, how valuable a unit of extra effort at this moment would be to the team. Then, each rower would choose his or her own action, taking full account of his or her own physical condition and the summary information supplied in the form of prices by the coxswain. In principle, if the coxswain could determine the right prices and could costlessly communicate them to the crew, and if the crew could make the right decisions based on that information, the resulting level of effort would be just right. In practice, a system of prices would fare badly for various reasons. First, it would be too difficult or costly for the coxswain to obtain the relevant information from the rowers and then to determine the prices. Second, communicating the prices back to the rowers would be too difficult and too slow. Third, the crew might respond inaccurately to the prices, failing to achieve coordination even if the prices were set correctly. Finally, small errors that disturbed the synchronized rhythm of the crew would be very costly. The price system has the same advantages in this application as in others with multiple producers. It takes full account of information about the individual condition of each producer, but that advantage comes at too high a cost in this synchronization problem.

ASSIGNMENT PROBLEMS Similar difficulties arise in **assignment problems**, in which there are one or more tasks to accomplish and there is a need for just one person or unit to do each. The coordination problem is to ensure that each task is done and that there is no wasteful duplication of effort. For example, if someone is seriously injured in an automobile accident, there typically is a need for one ambulance at the site of the accident as soon as possible. In practice, someone calls for an ambulance and then a central dispatcher assigns a particular ambulance to drive to the site. Even

if the dispatchers are in constant communication with the units, they may not be fully aware of all the relevant circumstances about the location and condition of the ambulance, of surrounding traffic, or of the training and experience of the crew. In principle, the dispatchers could try to determine a system of prices to determine which ambulance could provide the best service in the most timely fashion, but in practice they just select one and send it. If the prices were set incorrectly, more than one ambulance might rush to the scene, or no ambulances might respond. A system of prices performs poorly in assignment problems because it often leads to unnecessary duplication or costly delay.

It is interesting to contrast the way decisions about the uses of ambulances are made with decisions about how many ambulances to keep available, what equipment they should have, and how the ambulance staff should be trained. These latter decisions, made in relatively unhurried circumstances, take account of people's wages and salaries as well as the prices of equipment, vehicles, and training; they also utilize estimates of the benefits of additional ambulance capacity. At higher levels of decision making, people may also take account of the opportunity value of funds used for ambulances, which might otherwise be used to hire additional police officers or to build up an emergency fund or to buy additional park land.

DESIGN PROBLEMS AND ORGANIZATIONAL ROUTINES Our crew and ambulance examples combine several features: a sense of urgency about the decision, the extreme dependence of the optimal course of action on particular circumstances (Are the opponents ahead or behind? Is there an accident, and if so, where?), and the coxswain's and central dispatcher's substantial knowledge about the form of an optimal decision. Together these make central control an attractive alternative to a decentralized approach, as through a system coordinated by prices. However, when design problems arise repeatedly and call for largely the same solution each time, it may be unnecessarily expensive to solve each of them anew by centralized direction. Instead, established organizations set up routines that guide decentralized solutions to the recurrent design problems. For the most common kinds of demands made on the organization, no upper-management discretion needs to be exercised because those who first become aware of the demand know what to do and who to notify, without ever involving higher-level personnel. If someone notices that paper for the copy machine is getting low, then (depending on the established routine) he or she either simply orders more and leaves a sign telling others that paper has been ordered, or else he or she notifies the designated individual in the office whose job is to order more paper. With well-established routines, each part of the organization can rely on the others to do their parts. If the environment of the organization changes, however, then the same routines that were effective may become counterproductive in the new environment, and new routines will need to be devised if the organization is to continue to achieve its goals.

Innovation Attributes

Highly decentralized decision making, whether guided by prices or by organizational routines, will perform poorly whenever the optimal resource allocation depends on information that is not available to any of the people at the operating levels of the organization. Coordination problems with this **innovation attribute** are most commonly present when the organization is trying to do something that is outside its experience, such as introducing a new kind of product, entering a new market, or adopting a new approach to manufacturing. If a firm wants to consider replacing its system of buffer inventories of work-in-process at each work station with a just-in-time system (in which production is synchronized so that each station receives the necessary inputs just in time to process them), then the current production and

inventory managers may not have the experience to give a well-informed assessment of the costs and benefits. The decision cannot be properly made using just the information already available in the system.

When innovation attributes are present, effectively solving the coordination problem commonly involves someone gathering or developing the needed information and then communicating it to decision makers in the organization. This task might be taken on at his or her own initiative by someone at an operating level who has recognized that the problem exists. In other cases, the task will require more resources than lower-level personnel are able to muster on their own, and higher-level decision makers will have to become involved, allocating resources and assigning people to get the needed information.

In any case, the fact that a decision requires gathering new information from outside the organization does not imply that the decision ought to be centralized. People in operating positions may be best equipped to combine new knowledge with local knowledge, or they may be better motivated to make the new system successful if they are also the ones who designed it. An effective way to handle this problem may be to educate members of the organization about the proposed alternatives and to involve them in the final decision. What is certain, however, is that a simple price system relying only on the responses of individuals using their local knowledge cannot reliably achieve an optimal plan in these circumstances.

Comparing Coordination Schemes

The sample coordination problems we have described differ from one another, and the solutions that people actually adopt vary widely. Some decisions may be extremely urgent, with little time to process information. Others may require extremely close synchronization or coordination of assignments, with little tolerance for faults. Some require only making effective use of information that is already in the organization, whereas others require the infusion of new information.

Coordination systems also vary. Certain centralized command systems demand little upward communication of local knowledge and yet they still arrive quickly at reasonable plans and communicate clearly what the operational personnel are expected to do. Others require much more upward communication, but are correspondingly more responsive to local knowledge. Decentralized systems emphasize communicating information to support local decisions, with the required volume of communication being between the two centralized extremes. When these various systems fail, they fail in predictably different ways. The price system may work too slowly or lead to duplication, or it may require too much interpretation of information by decision makers. The coxswain's call and the ambulance dispatcher's instructions are quick and their meanings are clear, but they are insensitive to local knowledge.

There are more aspects to the coordination problem than just what the means of coordination will be. The manufacturing example provides an illustration of how a system could be designed to reduce the value of close coordination while still capturing gains to specialization: The system of buffer inventories eliminates the need to synchronize production but requires costly holdings of inventories.[4]

CRITERIA FOR COMPARING SYSTEMS We use three criteria to compare how well different systems perform in solving various kinds of economic problems. First, if all the information required by the system were reported, if all reports were made honestly

[4] Inventory costs include the interest on funds used to finance inventories, losses due to spoilage, theft, obsolescence, the cost of space in the storage area, and certain other costs that come from the tendency of inventories to conceal the effects of production performance that varies over time.

coordination be achieved? The answer is that the design variables themselves should be communicated. Each rower in the crew must know the intended stroke rate and the timing needs to be communicated to the rower. Each member of the product introduction team must know the introduction date and the weight and other key design parameters of the new car. The ambulance driver must be told which crisis to attend, when, and where. This form of control minimizes communication by the Hurwicz criterion and reduces the cost of error associated with more indirect methods. The appendix to this chapter contains a more formal treatment of design decisions, establishing that communicating the design variables is informationally efficient for the class of design decisions, just as communicating prices is informationally efficient for the class of resource allocation problems treated in the informational efficiency theorem.

Coordination and Business Strategy

Strategic business decisions present complicated problems. Good strategic decision making virtually always requires that effective use be made of line managers' knowledge about how the operations actually work and what capabilities the business has, but they may also involve using knowledge about new technologies, new markets, new business partners, or new forms of organization about which the line managers' knowledge may be limited. In our lexicon, strategic decisions often have innovation attributes. In addition, especially in manufacturing industries, but in some service industries as well, there are important economies of scale that mitigate against completely decentralized decision making. Finally, as we argue later, business strategy decisions commonly have important design attributes. There are predictable elements of fit in any good strategy that make it important to coordinate the actions of various parts of the organization closely. All these factors work against using prices or other very decentralized means of coordination and favor direct communication and other more systematic, centralized control systems.

Scale, Scope, and Core Competencies of the Firm

As we have already seen, when there are scale economies, the efficient level of output in a firm cannot be determined by prices alone. *Operational scale* itself is a design variable. Depending on the volume of sales that a firm anticipates, it will adjust its production capacity and the size of its sales force and secure supplies and distribution equipment and facilities (such as trucks and warehouses)—all tailored to the expected scale of its operations. If the actions taken by the marketing, production, personnel, distribution, and procurement managers are to be coherent, then all these people need a shared vision of the intended scale of operations.

Scale and Structure The anticipated scale of a firm's operations predictably affects more than just the scale of each part. As the GM and Toyota examples from Chapter 1 illustrate, it also determines the degree of specialization the firm should adopt. With larger operations, a firm may be able to afford more specialized equipment, more distribution outlets located nearer to customers, a larger number of plants, training programs for its employees tailored to particular circumstances, and so on. A smaller firm, operating without specialized equipment, may be more likely to rely on suppliers for many more of its components because the suppliers may be better positioned to enjoy economies of scale of their own by serving many firms. Thus, a larger firm with more specialized capital equipment may find it profitable to be more vertically integrated than would its smaller competitors. By definition, economies of scale in production allow a firm to reduce its costs compared to small-scale production, and

these costs are an important element in determining the prices to be charged. Lower marginal costs, other things equal, lead the firm to charge lower prices, which increases the potential product demand, which in turn supports an increased scale of production.

Operational scale is a design variable because it meets the two conditions. It has predictable implications for the various parts of the organization, and many of the mistakes associated with incorrect perceptions of scale by parts of the organization—for example, having too few raw materials or components to keep an expensive factory operating at full capacity—can be very high. Thus, firms that are large enough to assign different management functions to different decision makers take special pains to forecast market growth, competitors' plans, technical changes, input availabilities, and so on, all so that they can use those forecasts to plan the growth (scale) of their own operations and coordinate based on these plans. By making sure that its managers share common expectations about what it is trying to do, the firm takes an important step toward coordinating their plans and behavior.

ECONOMIES OF SCOPE Even when a firm operates at too small a scale in any individual product market to enjoy significant economies of scale, it may still enjoy them in producing components that are used in each of several products. For example, a firm like General Electric may enjoy economies of scale in producing small electric motors, using those motors to make food processors, hair dryers, fans, vacuum cleaners, and various other products. A firm like Casio may enjoy economies of scale in the manufacture of liquid crystal displays (LCDs), using them to produce calculators, wristwatches, electronic address books, and other products.

In these sorts of circumstances, the firms are said to enjoy *economies of scope*; that is, they can produce their several products together at less cost than could a group of single-product firms. Unsurprisingly, economies of scope entail all the same needs for coordination that economies of scale do. The problems are often harder, however, because coordination in planning is required among the managers responsible for different products. As Casio grew, for example, a forecast of large sales in the market for calculators led to falling costs for LCDs, making it more profitable to enter the market for wristwatches that use LCDs.

CORE COMPETENCIES When a firm introduces new products relatively frequently, one very important kind of scale economy that it may enjoy is at the level of product development. That is, a firm may acquire generalized expertise in the important skills that are required to design and market new products in a set of related markets or in using a set of related technologies. For example, a computer maker may develop expertise in microprocessor design, display technologies, memory chips, operating systems, computer manufacturing, data communications, networking, and so on—skills that it expects to be able to apply over and over again as it continues to introduce new products. Scale economies at this level are so important in modern management theory that a new name has been coined for them: **core competencies of the firm**. In a dynamic environment, a firm's capacity to introduce new products and to manufacture them efficiently can be even more important than are the economies of scale it achieves in making its existing product line. In that setting, a strategy of developing scale economies translates into one of building the core competencies of a firm.

In an abstract sense, core competencies are just another kind of shared component, but there is an important practical difference in that the cost of building the competency is shared with a series of products that does not yet exist. Investments in new manufacturing technologies today may actually raise the costs of today's products if the old manufacturing system is well understood and well implemented. The gains to be enjoyed on account of the new system will come over a longer period,

as the firm learns to use the new system, refines its methods, and adjusts the rest of its operations to take full advantage of the capabilities of the new technology. In such a case, managers need to plan for the demands of generations of products not yet even imagined. They also need to keep the price of the current products low enough to sustain a strong volume of sales, even if the company appears to be losing money on each sale, recognizing that these losses are actually an investment in the capabilities needed to produce profitable products in the future. This is just another illustration that when there are economies of scale, individual production and pricing decisions cannot be evaluated in isolation.

More controversially, the same ideas may be applied at the level of national industrial planning—a process about which many economists are skeptical but which has been used successfully in Japan and South Korea. The key first step in successful planning is to identify (groups of) industries to promote, which "fit" together with each other and with the country's existing competencies and advantages. For example, in Japan, the push to develop high-definition television (HDTV) in the 1990s is firmly grounded in that nation's strong positions in semiconductors, consumer electronics, and display technologies. To coordinate Japan's drive into HDTV, the Japanese national broadcasting company, NHK, has announced a set of technological standards that serves both to focus cooperative development efforts within groups of Japanese firms and to intensify competition among groups.

Complementarities and Design Decisions

Complementarities among a set of activities are an important source of design attributes. The standard definition of complementarity in economics is market oriented. Two inputs to a production process are said to be *complements* if a decrease in the price of one causes an increase in the demand for the other. In order to be able to employ the concept of complementarity usefully to study choices of levels of various internal activities as well as levels of input purchases, we introduce an alternative, more inclusive, definition: Several activities are mutually **complementary** if doing more of any one activity increases (or at least does not decrease) the marginal profitability of each other activity in the group.[8]

For example, where there are declining marginal costs due to learning or other kinds of economies of scale in producing a component, then the activities of producing various products using those components are complements. If General Electric's marginal cost of producing small electric motors declines with increasing volume, then the activities of producing electric fans and food processors are complementary because producing more fans makes it cheaper and therefore more profitable to produce more food processors as well.

Complementarities lead to predictable relationships among activities. A decision to increase the level of one activity will raise the profitability of any contemplated increases in levels of any complementary activities. Thus, high levels for all the elements of a group of complementary activities go together. This predictability is one

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[8] In mathematical terms, the complementarity relationship among a group of activities can be characterized as follows. Let $x = (x_1,\ldots,x_n)$ be the levels at which the activities are conducted and let $\pi(x)$ be the resulting profits. If π is a smooth function, then the activities are mutual complements if for all $i \neq j$, $\partial^2\pi/\partial x_i \partial x_j \geq 0$; an increase in the j^{th} activity raises the marginal return to the i^{th} activity.

When a change occurs that makes any one of the complementary activities more profitable or less costly and encourages the firm to do more of that activity, then the marginal returns to the other activities are also increased, leading to more of those activities as well. The increased levels of the complementary activities further increase the marginal returns to the first activity, possibly leading to another round of increases in it and all the related activities.

of the two defining features of design attributes. The second feature is the high cost of failure to match or fit the parts together. This feature is a separate aspect of the problem that is not logically implied by the conditions of complementarity alone. However, we say that a group of activities is *strongly complementary* when raising the levels of a subset of activities in the group greatly increases the returns to raising the levels of the other activities. When a group of activities is strongly complementary, design attributes are always present.

The box entitled "Modern Manufacturing Strategy" describes one particular business strategy that illustrates the complementarities between particular aspects of product strategy, manufacturing policy, equipment choice, personnel and compensation policies, supplier relations, accounting methods, and more. What is most important to notice about the strategy is the coherence of its parts. The parts of the strategy that are connected are mutually supporting because they call for complementary activities. Part of the success of the modern manufacturing strategy in the 1980s and 1990s grows out of the way it takes advantage of the new technologies of the time, such as rapid, low-cost communications, highly flexible equipment, and computer-aided design, which make many of the elements of the strategy more effective and less costly.

Other manufacturing companies in other times have succeeded using other strategies, also well fitted to the available technology. In the first part of the twentieth century, for example, Ford Motor Company adopted an entirely different but equally coherent strategy that was appropriate for its time, based on producing a single product (the Model T Ford) in high-volume factories using specialized equipment and rigidly disciplined labor paid according to the job being done. This strategy reduced costs and increased the quality of the automobile so much that Ford was eventually able to capture more than 50 percent of the automobile market.

The continuing changes in technology and factory prices in the modern era, including the falling costs and growing reliability of highly automated equipment that works with no direct labor involvement, are likely to lead to further revisions in manufacturing firms' strategies.

Modern Manufacturing Strategy

One kind of strategy that has taken on a new importance in the late 1980s and 1990s is the "modern manufacturing" strategy, which is actually a whole group of similar strategies. The particular variant we describe involves producing a wide range of related products for customers with specialized needs. The firm strives to remain the quality leader in its markets with frequent new product introductions and frequent improvements of existing products.

Market behavior of this kind has implications for a wide range of activities in the organization. At the level of manufacturing, the small quantities demanded of each product prevent the company from setting up specialized production lines for each, forcing it to maintain a high level of flexibility. Because the demand for each product is small and because there are important economies of scale in inventory systems, the company will find it profitable to organize in a way that avoids holding inventories of finished products. It will eschew keeping local product warehouses and instead may

even ship directly from the factory to the customers, perhaps using air freight if speed is important. In order to serve customers quickly, communications between the sales staff and the factory need to be closer than for a company that relies on finished goods inventories. Strong communications are always a key variable when products are to be completed "just in time" to meet customer demands. Since the average inventory level for a product is directly proportional to the production run, the company will want to use more frequent, smaller production runs to make its products. The use of frequent, small production runs forces the company to reset its production line more frequently to change the product being made. This makes it more profitable to use flexible production equipment, which can be quickly and easily changed over from task to task.

Frequently introducing new products adds another dimension to the strategy. Obviously, it increases the need for product designers and design engineers. It also means that equipment originally purchased to make one product will surely need to be switched to producing other products, further increasing the importance of flexibility. And it increases the importance of communications between the design team and the manufacturing department, since the product must be designed to be manufacturable with existing equipment.

The modern manufacturing strategy also affects personnel and compensation policies, supplier relations, and accounting systems in predictable ways. Short production runs mean that factory workers will frequently be changing tasks, moving from making one product to another. This tendency is compounded by the frequency of product redesigns and new product introductions. Flexible manufacturing thus requires multiskilled people who can do more than one task, and companies that adopt this strategy will often compensate workers on the basis of the skills they acquire, rather than on the basis of the particular job that they are presently assigned to do. In a system that is based on just-in-time manufacturing and low levels of buffer inventories, the rate of work at each step of the process must be in constant balance; if the molding equipment runs faster than final assembly even temporarily, inventories will accumulate. Consequently, systems tend to avoid piece-rate compensation, which would encourage workers to work at their own varying paces.

Frequent product redesign makes it important that the firm has accurate information about the cost of producing various alternative designs, both in order to make well-informed design choices and to price the product correctly. Since several products are being made on the same machine, the correct allocation of machine costs becomes an important focus of the accounting system. The variable that limits the use of machines is generally time, and so the increasing tendency among firms that use the modern manufacturing strategy is to assign machine costs to the individual products on the basis of the machine time that the product requires.

To the extent that the firm's strategy allows it to use general-purpose manufacturing equipment, it has a greater ability to rely on subcontractors to make its product in times of high demand and to invest in equipment that it cannot keep busy. Another alternative for the firm is to sell its excess capacity to other firms, that is, to become a subcontractor itself.

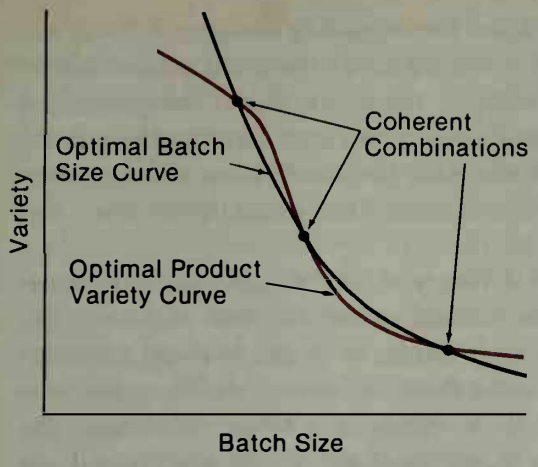

Figure 4.3: There may be several coherent combinations of batch size and product variety, but only one will generally be optimal.

Complementarities, Innovation Attributes, and Coordination Failure

When complementarities are present, the various aspects of a firm's strategy must be aligned properly for the firm's strategy to be optimal. Alignment alone is not enough for optimality, however. The parts of the strategy that Henry Ford adopted in introducing the Model T fit together well, and for a period of time the strategy worked effectively. However, as described in Chapter 1, Ford's strategy was overtaken by Alfred Sloan's strategy at General Motors, which was overtaken in turn by a variant of the modern manufacturing strategy introduced by Eiji Toyoda and Taiichi Ohno at Toyota.

Even in the absence of explicit strategic planning, the various managers in a firm may be able to adapt their choices to each other's, learning how best to pursue the firm's objectives in their own narrow domains. Figure 4.3 illustrates the possibilities for, and a potential problem with, an adaptation of this kind.

The figure portrays the decisions of two managers—a manufacturing manager and a marketing manager—in the same firm. The marketing manager chooses the number of product varieties to offer to customers, and the manager of manufacturing decides how large the batch sizes should be, that is, how many of one item to make before switching production over to another item in the product line. Suppose that each seeks to make the decision that maximizes aggregate firm profits.

Offering more products allows the firm to tailor its offerings more closely to customer needs. Then it can choose to receive a higher price for its products or to increase the number of units it sells. This benefit of increased variety must be weighed against certain costs. Holding the batch size fixed, increasing the number of varieties, raises the level of inventories proportionately. The larger the batch size, the costlier are increases in product variety and the smaller is the optimal number of products. This relationship is depicted by the optimal variety choice curve in Figure 4.3.

Similarly, given the number of different products being produced, there is an optimal batch size. Lower batch sizes increase the frequency with which the firm must incur the set-up costs involved in switching production from one product to another, but they also reduce average inventories and their costs. When the number of varieties is increased (holding total unit sales constant), the marginal inventory cost of increasing batch size rises. For this reason, increases in product variety lead to smaller optimal batch sizes. This relationship is depicted by the optimal batch size curve in the figure.

DECENTRALIZED DECISIONS AND COORDINATION FAILURE Suppose that each manager sets the variable he or she individually controls to maximize the firm's profits, given

what the other manager is expected to do. Then the marketing manager's choice will lie along the optimal variety choice curve in the figure at the point determined by the batch size this manager expects the other to select. Similarly, the production manager will select the point on the optimal batch size curve corresponding to the product variety he or she anticipates. Points where the two curves cross are ones where each manager's choice is the right one to maximize firm profits, given the other manager's choice.

As the figure shows, when the optimal variety choice is a decreasing function of the batch size and the optimal batch size is a decreasing function of the number of varieties, there can be several different combinations of product variety and batch size at which each manager's choice maximizes the firm's overall profits, given what the other is doing. Each of these combinations represents a coherent strategy. The lower right-hand intersection point can be likened to Henry Ford's strategy with the Model T—low variety and large production runs. The upper left-hand point can be likened to a "niche market" version of the modern manufacturing strategy—many varieties produced in small batches. Of course, the actual strategies involve many more choices than are illustrated in the figure, making the choice problem much more difficult.

Still, it is possible that the managers would find a coherent combination on their own, even without any explicit attempt to coordinate their decisions. For example, each might observe what the other is currently doing and adjust optimally to this choice. Such a decentralized process might converge without any communication between the managers (although it is not guaranteed to do so). In any case, it is sure to take time to reach coherence in such an uncoordinated fashion, during which the firm is operating inefficiently. Having the two managers meet and share their information would, on the other hand, seem very likely to achieve coordination on one of the coherent patterns.

There is, however, no reason to suppose that the particular coherent combination the managers arrive at will be the best one. In general, only one combination is actually profit maximizing, although this diagram does not contain enough information to determine which it is.[9] Ford's strategy was best for the auto industry in the early part of the century, while many industries now find that the modern manufacturing strategy of a broad product line combined with small batch sizes is optimal. Could the operating managers, using only the information they might normally be expected to have gained from their own experience, somehow settle on the best one? The answer is quite likely "No." Even if the managers pooled their information and attempted to make a coordinated decision, they may still not have enough information to determine which of the coherent strategy combinations is best.

INNOVATION ATTRIBUTES AND STRATEGIC COORDINATION The key difficulty is that as the environment with its demand conditions and the costs and productivity of different technologies changes, the two curves may change more or less smoothly, and, with them, the precise description of each of the coherent strategies. Yet a small change in the curves may hide the fact that the relative profitability of the different coherent patterns has changed and so the actual profit-maximizing strategy has shifted radically.

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[9] Determining which is best requires looking at a third factor, the total profits accruing to any pair of choices. The two curves represent "ridge lines," traces of the highest points on the profit surface in the relevant east-west or north-south direction. Depicting the height (the total profits) along the ridge lines graphically requires a third dimension. If this third dimension were shown, it would reveal that the profit surface here is like a mountain with three peaks, one at each of the intersections of the curves. Any of these might be the highest, so that the intersection point under it is the strategy pattern that actually maximizes profits.

For example, gradual changes over the years in production technologies and customer needs have not radically altered the general form of the coherent pattern corresponding to mass production. Yet in many industries these cumulative changes have made the modern manufacturing strategy much more profitable than the mass production strategy. This means that profit maximization would require a radical shift in strategy.

The operating managers may be able to use their familiarity with demand and cost characteristics to keep up with local movements in the coherent patterns, marginally adjusting their decisions to track the changes in what is best. But recognizing that a radical strategic shift is desirable is likely to require information that neither manager has, and the local adjustments they are likely to make are never going to track the global shifts in which strategic configuration is best. A shift to much smaller variety and larger batch sizes might enable the firm to use highly specialized equipment that is better tailored to the particular products being made, leading to much lower costs. Estimating the benefits of such a change would require knowledge about technologies and equipment that may be completely unfamiliar. Similarly, a shift to a much wider product line and smaller batch sizes may require much more flexible equipment and completely revised relations with suppliers, which might again be well beyond the range of the production manager's experience. Also, the marketing manager may not know how much customers would be willing to pay for more highly tailored products, or by how much sales would fall if a narrower but cheaper product line were offered.

As in this example, the overall strategic choice in the presence of complementarities is frequently a design decision with innovation attributes. The pieces of the strategy have to fit together, and the information needed to identify and choose between alternative patterns is unlikely to be available freely within the organization. In these circumstances, information must be acquired from outside the organization and assessed, a coordinated decision must be made on the design variables, and then this decision must be communicated to the affected parties. All this presents a role for central coordination and, in particular, for top management. We discuss this in the next section.

We emphasize again that the example we have been discussing involves only two choice variables, and yet complementarities still lead to potential failures of coordination. In general, the problem of selecting among multiple coherent policies becomes more difficult as the number of variables to be adapted increases and as the strength of the complementarity relations between pairs of decisions grows. Understanding the complementarities in the system makes it easier to identify potentially profitable changes of strategy and to anticipate the scope of the changes needed to implement a new strategy.

Management, Decentralization, and the Means of Coordination

We have seen a variety of situations in this chapter in which the price system, the quintessential decentralized coordination mechanism, is less than ideal because it is too brittle and so does not deal well with imperfect information or because it requires too much communication. When the price system fails, the search for efficiency means that other mechanisms must be adopted. Sometimes these supplement the market's operations, but often they involve the creation of formal organizations that largely supplant the market over a range of activities.

The coordination problem is not solved by merely putting a nonmarket form of organization in place, however. Instead, it is transformed into a problem of management. The ideas in this chapter are useful for thinking about the tasks of

nomics (surveyed by Russell Cooper and Andrew John), but this theme has not before been systematically applied to problems in the theory of organization.

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Exercises

Food for Thought

1. During World Wars I and II, even the major market economies switched to centralized forms of planning to allocate many scarce resources. Are there reasons based on efficiency to prefer such centralized planning in wartime? Are there other reasons?

2. For many years, Swiss craftsmen made the finest watches, which were powered by winding a spring by hand. The Swiss lost ground to competitors in Japan and elsewhere as more accurate, less expensive battery-powered electronic watches were perfected. What were the core competencies of watchmaking in the era of Swiss dominance and how were they changed by new technologies? What other products are made by the new watchmakers?

3. In 1987, Sony Corporation, a leader in consumer electronics (televisions, compact disk and audio cassette players, home video recording and playback machines), purchased the CBS Record Company. In 1989, Sony purchased Columbia Pictures Entertainment, one of the giants of the movie and TV production industry. In 1990, Matsushita, the world leader in consumer electronics with its Panasonic, National, and Quasar brands, purchased MCA Corporation, which makes movies, television shows, and music recordings. Why would these companies want to expand into these particular industries?

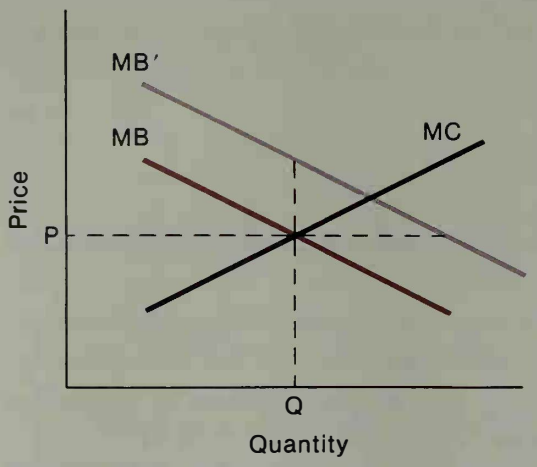

Figure 4.4: An error scenario in which the planner is mistaken about marginal benefits.

Mathematical Exercises

1. Consider the problem of supplying an input to a division of a firm or to a firm in a planned economy. Suppose the executive or planner knows the marginal cost function for the input. This is represented by the fixed supply curve in Figure 4.4. However, the planner is uncertain of the benefit. Her best estimate of the marginal benefit curve is represented by the curve MB in the figure, and on that basis the price should be P and the quantity supplied Q. Consider an error scenario in which the actual marginal benefit is higher, as indicated by the curve MB'. Mark on the graph the quantity that would be produced using a price control in the error scenario and draw a vertical line at that quantity level. Identify the triangles with areas that correspond to (1) the loss suffered under the error scenario when Q is announced under a quantity planning system and (2) the loss suffered under the error scenario when the price P is announced and the division chooses its quantity to maximize profits. Show that the ratio of these losses with an uncertain benefit curve is just the inverse of the ratio with an uncertain cost curve as derived in Equation 4.1.

2. A new product is being introduced in competition with another firm. Having the product ready for production in t months will result in net profits after the introduction of $144 - t^2$ if $0 \leq t < 12$ but will generate no profits at all if $t \geq 12$. Three departments must be ready before production can begin, and getting ready quickly is a costly matter. Department 1 can be ready in t months for a cost of $3(12 - t)$, department 2 for a cost of $4(12 - t)$, and department 3 for $5(12 - t)$. Until all three departments are ready, production cannot begin. (1) What is the optimal date t^* at which to introduce the new product, that is, the time that maximizes the net profit after product introduction minus the cost of readying the three departments? (2) What will the firm's profit be if the project is mistakenly hurried and all three departments are asked to deliver one month earlier than the optimal date? One month later? (3) What will the firm's profit be if department 1 and department 3 both are asked to be ready at the optimal time, but department 2 is asked to be ready one month earlier than the others? One month later? (4) Is the timing of product introduction in this example a *design decision*?

3. A small community must decide how much land to set aside and improve for public parks, to be used for playgrounds, ball fields, and so on. Suppose there are one hundred families in the community and that there are no wealth effects on the families' preferences. That is, family n's utility for x_n dollars of personal savings and y acres of park land is at $x_n + v_n(y)$. If a plan to set aside and improve y acres of land at cost $c \cdot y$ is proposed with family n paying t_n in taxes to finance the project, what conditions must be checked to test the efficiency of the plan? What is the minimal amount of communication required to check whether the proposal is efficient?

Mathematical Appendix: A Formal Model of Design Decisions

In this appendix we study a mathematical model of the problem of introducing a new product, for example, a new model car. Our principle purpose is to show that this problem indeed has the characteristics of a design decision and to establish that the informationally efficient way to handle such a problem is to announce the design attributes (rather than prices).

There are N system components that go into making the product, each of which is produced in a single, separate facility or department. In the case of a car, these would include systems like headlamps, engines, brakes, and so on, each of which is itself a complex product with many parts and many steps involved in its manufacture. We establish units so that exactly one unit of each component goes into making the final product. For example, a car requires two headlamps, so we regard a pair of headlamps as being one single unit in our formulas.

Suppose the corporation makes available a set of corporate resources for the project, and that there are k different kinds of corporate resources. There are $\bar{x}_1$ units of the first type of resource, $\bar{x}_2$ of the second, . . ., and $\bar{x}_k$ of the last. The vector $\bar{x} = (\bar{x}_1, \ldots, \bar{x}_k)$ describes the corporate resources available for the project.

The allocation of these resources among the N units is denoted by the list $(x^1, \ldots, x^N)$, where $x^n = (x_1^n, \ldots, x_k^n)$ denotes the vector of resources allocated to plan and produce the nth component. Of course, the total amount of each resource allocated to the various departments cannot exceed the amount available. The factory that makes component n is ready to begin production at date t^n and with capacity y^n. In addition to the corporate resources, the factory may hire outside workers, suppliers, and contractors to do parts of the job. The total amount of outside expenditures required to design and prepare to make system n by date t^n at capacity y^n using corporate resources x^n is $C_n(x^n, y^n, t^n, z^n)$. This function C_n is a *cost function* and z^n is a parameter of the cost function that is known only by the local manager. (This means that only the local manager knows how costly it is to meet the capacity and timing objectives with a given allocation of centrally allocated resources.) We assume that it costs more to get ready earlier and to build additional capacity; that is, C_n is decreasing in t^n and increasing in y^n.

Because manufacturing a unit of the product requires one of each of the system components, the rate of sales cannot exceed the minimum of the production capacities, and sales cannot begin until the last of the system components is ready for production. Let us therefore write the revenues as:

$$\text{Gross Revenue} = R[Min\,(y^1, \ldots, y^N), Max\,(t^1, \ldots, t^N)] \qquad (4.3)$$

Revenues depend on two numbers: the amount that is produced and the date at which it is available. The first of these is equal to the minimum of the outputs of the various components and the second is the date at which the last component is ready. Naturally, R is increasing in the total available capacity and decreasing in the time to introduction of the product.

An organization seeking to maximize profits (gross revenues less the total costs incurred in all units) would want to select the allocation of the corporate resources (x), the unit capacities (y) and the readiness dates (t) to solve:

$$\text{Maximize } R[Min(y^1, \ldots, y^N), Max\,(t^1, \ldots, t^N)] - \sum_{n=1}^{N} C_n(x^n, y^n, t^n, z^n) \qquad (4.4)$$

$$\text{subject to } \sum_{n=1}^{N} x^n \leq \bar{x}$$

where the inequality constraint reflects the limited availability of corporate resources.

Let us suppose that the profit expression in Equation 4.4 is concave and that the optimum involves positive amounts of each resource being used by each unit. Then, a plan (x,y,t) is optimal if and only if there exists a vector of prices p^*—one price for each resource—and numbers t^* and y^* such that the equality constraints

$$\left.\begin{aligned} y^n &= y^* \\ t^n &= t^* \\ \partial C_n/\partial x^n &= \mathrm{p}^* \end{aligned}\right\} \quad \text{for all } n. \tag{4.5}$$

and adding-up conditions

$$\begin{aligned} \textstyle\sum_{n=1}^{N} x^n &= \bar{x} \\ \textstyle\sum_{n=1}^{N} \partial\, C_n/\partial y^n &= \partial R/\partial y \\ \textstyle\sum_{n=1}^{N} \partial\, C_n/\partial t^n &= \partial R/\partial t \end{aligned} \tag{4.6}$$

are all satisfied. The equality constraints say that each unit's capacity and readiness date are the same, and that the impact on costs of changing the amount of any of the corporate resources allocated to a unit is the same as at any other unit. The first adding-up condition says that the corporate resources are fully utilized. The second says that the marginal cost of adding more capacity in each unit just balances the marginal revenue from the added capacity. Finally, the last says that the marginal costs of having the output from every unit available a little sooner just balance the additional revenue that results.

Prior Information and Efficient Planning

In this formulation, it is clear that the choice of the timing and capacity variables has one of the characteristics of a design decision. Even without knowing the actual cost and revenue functions, we know in advance that at an optimum all the times t^n will be equal and similarly all the capacities y^n will be identical. There is a priori information about how these variables fit, and one consequence of this is that Hurwicz's theorem does not apply. A price system is not guaranteed to be informationally efficient.

Nevertheless, Hurwicz's basic framework still applies. We can still proceed by identifying the minimum amount of communication necessary to verify the optimality of a proposed plan (x,t,y) and checking whether a proposed coordination system achieves that minimum. For this case, the relevant extension has been derived by Fumikata Sato, who has established that, in addition to communicating the $Nk+2$ numbers that comprise the plan itself (x,t,y), any system that can verify the plan's efficiency must communicate at least $k+2N$ additional numbers.[11] Here is a system that achieves that bound. The central coordinator announces prices, an introduction date, and a capacity ($k+2$ numbers). Each production department n responds by reporting what resources it wishes to purchase (k numbers) and its marginal costs for additional capacity and earlier factory readiness (2 numbers). This information allows one to check Equations 4.5 and 4.6 and so to verify whether the plan is efficient. It requires the communication of $N(k+2)$ numbers by the N departments plus $k+2$ numbers by the coordinator, so this system verifies the efficiency of an allocation with a minimum of communication.

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[11] Fumikata Sato, "On the Informational Size of Message Spaces for Resource Allocation Processes in Economics with Public Goods," *Journal of Economic Theory*, 24 (February 1981), 48–69.

By comparison, any attempt to guide capacity choices and factory readiness dates using a price system would fare quite badly. For example, suppose there were $T \geq 2$ possible delivery dates. Describing a full plan would involve the Nk numbers that give the allocation of the k different corporate resources to the N plants, plus a description of the ready capacity for each of the N plants for each of the T dates, the further NT numbers. Then there would need to be a price for each of the k resources, and a price for capacity for each of the plants at each possible date. In sum, an augmented plan under a price system would involve $Nk+NT+k+NT = k(N+1)+2NT$ numbers. This exceeds the minimum required. Indeed, if the number of possible introduction dates T is large, the amount of additional communication is similarly large, being equal to $2NT-2(N+1)$.

The High Relative Cost of Failures of Fit

The other major attribute that defines a design problem is that failures of fit are more costly than are failures to find the right design. To see how this condition arises in the present model, let us first consider the cost of an error in which the parts fit but the design is not optimal. Specifically, let t^* be the optimal introduction time and suppose that all the t_ns are set equal to some different but coordinated date $t^*+\epsilon$. To estimate the cost of a small error, we substitute this expression for each t_n in the objective function and then take the derivative with respect to ϵ. Using the third adding up condition in Equation 4.6, we find that this derivative is zero at $\epsilon = 0$. That is, the marginal cost of an error of this kind is zero. This kind of small design error is not at all costly. Intuitively, the reason is that the product introduction date is optimally set so that the marginal cost of seeking a slightly earlier introduction is just equal to the marginal benefit. A small change from t^* to t should have a zero effect on profits, to a first-order approximation. Similarly, setting all the y^ns equal to $y^*+\epsilon$, the derivative with respect to ϵ is zero. A small error in setting y leads to a cost that is zero to a first-order approximation.

By comparison, small *errors of fit* among the variables are much more costly. Suppose, for example, that all the variables but one, say y^n, are set to their optimal values. The cost of a small error in y^n depends on whether y^n is set too high or too low.[12] When y^n is set too low, the cost is approximately (to first order) equal to the size of the deviation times the left-hand derivative of profits with respect to y^n evaluated at the optimum. When y^n is set too high, the cost is the right-hand derivative times the size of the deviation. These derivatives are:

$$\text{Left-hand derivative} = \frac{\partial R}{\partial y} - \frac{\partial C_n}{\partial y^n} = \sum_{i \neq n} \frac{\partial C_i}{\partial y^i} \tag{4.7}$$

$$\text{Right-hand derivative} = \frac{\partial C_n}{\partial y^n}$$

As long as capacity is costly, these derivatives are not zero. Errors of fit incur first-order costs.

The intuitive reason for this conclusion is straightforward. When y^n is set too large, the extra capacity is useless. The loss incurred from this error is the cost of that extra capacity, as confirmed by the expression for the right-hand derivative. When y^n is too small, the limited capacity forces a reduction in the level of output with a marginal value of $\partial R/\partial y$ but also saves costs of $\partial C_n/\partial y^n$. This difference appears as the first expression for the left-hand derivative in Equation 4.7. Using the third adding-up condition in Equation 4.6, this condition can be transformed into the second

[12] This is because the total profits are not a differentiable function of y^n at the optimum point.

expression for the left-hand derivative. According to the second expression, the loss is just the same as if a small amount of useless extra capacity were built by all the other departments, because the shortfall in capacity at department n renders the extra capacities of the other departments unproductive.

The most important message, however, is not the particular forms that these costs take. Rather, it is that decisions concerning timing and scale are, as we have claimed, decisions with design attributes. Explicit coordination of these decisions by either a central coordinator or by committee meetings, rather than decentralized decisions loosely coordinated by prices, is predictably the norm for decisions of these kinds.

Ex Post Renegotiation The second commitment problem is related but more subtle. In some circumstances, it may be advantageous for both parties to renegotiate the contract ex post because what was efficient when the contract was first entered may not be so once actions have been taken or further information revealed. If the parties understand at the time they are crafting the original agreement that they will later face these incentives, however, they may not be able to draft the contract in a way that generates the desired behavior.

For example, many firms use stock options to motivate executives. These options give the executives the right to buy the firm's stock at a specified price in the future. The idea is that having the options will motivate the executives to work to get the market price of the stock above the price specified in the options so that they can make money buying the stock cheaply. However, the options will not have much effect on motivation if the price at which the option can be exercised is too high relative to the market price, because the goal of getting the stock price higher than the exercise price will seem unreachable. Thus, the exercise price is typically set moderately higher than the current market price. Now suppose that after options are issued, the price of the firm's stock falls drastically. Suddenly, the options are essentially worthless to the executives and, thus, are worthless as incentives. It might then make sense to issue new options reflecting the lower stock price, and many firms have done precisely this. If the executives forecast that this will happen again in the future, however, they will not be as worried about a fall in the stock's price as they otherwise would be. This dulls their motivation to improve the stock price.

A complication is that in some contexts it will turn out ex post that to maximize total returns the original terms should not be followed, and yet one party or the other may have an incentive to insist on the inefficient completion of the contract. For example, suppose a department store chain has contracted with a knitting firm to deliver 10,000 cotton sweaters in various colors. The sweaters are worth $20 a piece to the store, and they will cost the knitter $10 each to produce. The agreed price is $15, and meeting the order will absorb the full capacity of the knitter. Then the knitter gets an order from another chain for 20,000 polo shirts. Each shirt is worth $15 to this chain and can be produced for $5, but the knitter cannot produce both the sweaters and the shirts. Efficiency requires that the shirts be produced rather than the sweaters because this yields a total value of $200,000, whereas the surplus on the sweaters is only $100,000. Thus, the original contract should be abrogated. If this possibility had been foreseen in the original bargaining, then the parties would have efficiently agreed that the shirts should be produced instead of the sweaters. If the contract was simply to deliver the sweaters at the agreed price, however, then the first chain may insist on delivery. In this case, a possible solution is for the knitter to renege on its contract and pay damages of $50,000 to the first chain for its lost profits. However, this kind of solution is problematic if the values of the various options to the different parties are not commonly known to all, because then the appropriate level of damages is hard to determine.

Investments and Specific Assets

Although even relatively simple contracts can be subject to various problems and disputes, the most commercially important impacts of imperfect commitment arise when significant investments are required, because the amounts are significant and because the benefits may accrue over an extended period of time. In economic language, an **investment** is an expenditure of money or other resources that creates a potential continuing flow of future benefits and services. The potential flow itself is called an asset. The kind of assets that are most familiar are physical ones, like houses or machines, which provide a potential flow of housing or manufacturing services

Terrorism, Parenting, and Commitment

The governments of many countries have announced that they will not bargain with terrorists, even when hostages have been taken and their lives are threatened. This policy is designed to *prevent* terrorism: If potential terrorists expect that the world's governments will not deal with them, even to save the lives of their citizens, then there is less gain to kidnapping or other terrorist acts.

The problem with the policy is that it requires commitment to be credible. If hostages actually are taken, then the pressures to save their lives by negotiating with their captors are immense, no matter what policies were announced beforehand. This is especially so in countries with popularly elected governments and open mass communications, where the pain and grief of the victims' relatives is seen on television and where the governments are particularly responsive to citizens' concerns. Terrorists clearly understand the incentives to renege on the policy, and this led them to take many hostages from among the citizens of democratic countries but few from undemocratic ones.

Similar, if less dramatic, commitment problems arise repeatedly for parents raising children. For the children's own good, parents want them to do various things: avoid dangerous play, study hard, save money. To encourage the desired behavior, parents use threats of punishment, ranging from "If you play on the road, you'll get a spanking you'll never forget" to "Unless you save the money yourself, you can't have a new dress for high school graduation." Of course, the parents' pain in meting out the punishment may be at least as great as the child's in receiving it. The temptation is to forgive and omit the punishment because once the misbehavior has occurred punishment brings pain with no immediate gain.

Both these examples show how concern with the future effects of not following through with the promised response to misbehavior can stiffen resolve and lead the government or parent to absorb the costs of carrying out the promise. Reputation concerns sometimes lead them to carry out their threats, even when they have an otherwise attractive temptation to renege.

over time, or financial ones like savings bonds, which yield a flow of cash returns. Many assets are not physical, however. Patents and copyrights are assets that allow their owners to demand royalties or exclusive rights to reproduce books or software programs for a certain period of time. Investments in education can create an extremely important asset, **human capital**, which leads to an increased flow of future income as well as flows of equally important, if perhaps less tangible, benefits.

The kind of investments that are most problematic in terms of the incentives they create are investments in specific assets, that is, assets that are most valuable in one specific setting or relationship. When the value maximization principle applies, the *specificity* of an asset is measured as the percentage of investment value that is lost when the asset is used outside the specific setting or relationship.

An important special case of specific assets are cospecialized assets. Two assets are cospecialized if they are most productive when used together and lose much of their value if used separately to produce independent products or services.

For example, consider the electrical power generation industry. Most electrical

generating plants are located either near the customers they serve or near a source of energy. Coal-burning electrical plants, for example, tend to be located either near coal mines or near cities. The plants near cities most often receive their coal by railroad. Special rail lines are built to the mine in order to ship out their coal. These feeder rail lines are assets that are cospecialized with the coal mine itself. If the rail lines were not used to ship coal from the mine, there would be a great loss of economic value. Because coal is bulky and has a low value per unit of weight, it would probably be economically infeasible to ship the coal by means other than rail, so the mine without the rails is of greatly reduced value. Meanwhile, there is not much other use that could be made of a rail line that runs between a railroad junction and a coal mine other than to ship coal. Thus, absent the coal, the rails also are of little value. The total value of the mine and the rail line in their separate, alternative uses is much lower than is the value of the two assets when used together to provide coal to utilities.

In the case of a mine-mouth electric plant, which is located away from the city it serves and near the mouth of a particular coal mine, the mine and the plant are cospecialized assets. The mine-mouth plant relies on the output of the mine for its coal and would lose much of its value if the mine were closed. The mine would lose much or all of its value if its only customer—the generating plant—were closed. The plant and the mine in this case are cospecialized because they are far more valuable if used together to support the production of electricity than if the coal were diverted to other uses.

The problem that arises when investments are made to create cospecialized assets is that much of the value of the investment depends on the behavior of another asset owner with his or her own selfish interests. This opens the possibility of various sorts of (ex post) opportunistic behavior that endangers the investment. Mine owners who invest in expanding their mine, for example, may find themselves at the mercy of the electric utility, which might later threaten to use the plant only for backup of its nuclear or hydro-powered plants. It is not only the mine owners who need to worry about its specific investment, however. The utility owner might worry that after the plant is built, the mine owners will try to raise their price for coal. Once a mine-mouth plant is built, there is no alternative source of coal supply for the plant, so the utility owner may become vulnerable to demands for a price increase.

THE HOLD-UP PROBLEM The general business problem in which each party to a contract worries about being forced to accept disadvantageous terms later, after it has sunk an investment, or worries that its investment may be devalued by the actions of others, is called the **hold-up problem**. The party that is forced to accept a worsening of the effective terms of the relationship once it has sunk an investment has been *held up*.

Clearly, if contracts were complete, the hold-up problem would not arise: The parties could specify the full range of circumstances that might arise and could agree on the behavior to be followed in each of these. The usual mechanisms for contract enforcement would then prevent ex post opportunism. In the present situation, you would expect that the mine and the utility, faced with the potential hold-up problem, would at least enter into a long-term contract, so that the coal price level could be determined in advance. But at what level should the price be set? The coal mine owner might be worried about increasing costs of labor and mining equipment. Should the contract include an *escalator clause*, which automatically adjusts the price of the coal when some index of mining costs rises? Should the price of coal be tied to the spot market price of coal in the general area?

How reliable must the coal supply be? If the mine owner guarantees an uninterrupted supply of coal, it will be in a weakened bargaining position when the

coal miners' contracts come up for renewal. If not, the utility may worry that the mine owner will be too willing to weather a long strike. The mine and the utility have different interests.

What about future expansions? If the demand for energy increases, is the mine obliged to supply an increased amount of coal at the same price per ton? If its average and marginal cost curves slope upward, then the additional coal will be more expensive to provide. Should the price negotiations be left for later? Later, the mine will be a monopoly supplier; now, it is just one of several possible sources of coal for a new plant whose location has yet to be determined.

The hold-up problem that the coal-burning electric plant illustrates is that large, specific investments make asset owners vulnerable to opportunistic behavior by their contracting partners. This is not a problem in standard market theory where contracting is perfect and alternative suppliers are readily available. Perfect contracting implies that the parties can write adequate protections into the contract to prevent the kinds of problems that we have described. Besides, if there were many suppliers for all inputs, then there would be no need for extensive contracting. There would be no scope for price gouging because the prices would be set by a competitive market. There would be no concerns about supply interruptions because there would be alternative supplies of an identical or nearly identical input, and no worries about demand curtailment because there would be alternative uses for the asset. *It is the specificity of assets together with imperfect contracting that lies at the core of the hold-up problem.* Concern about these problems may lead to inefficiencies as firms, fearing that their investments will leave them vulnerable, refuse to make the efficient investment.

In a series of articles, Paul Joskow has investigated the nature of coal supply contracts to the electric utility industry.[4] His empirical research reveals that coal mines supplying mine-mouth plants are most frequently owned by the utility they serve and, in all other cases, have a long-term contract to supply the utility. According to Joskow, these contracts are typically quite complex, involving escalator pricing clauses of various kinds, and they do a good job of tracking the actual variations in the cost of supplying coal.

Another example of cospecialized assets can be found in the automobile industry. In the 1920s General Motors purchased their automobile bodies from an independent firm, the Fisher Body company. As the technology of automaking advanced and the car companies moved from wooden bodies to metal ones, General Motors began to design a new automobile plant to assemble their cars. In order to improve the reliability of supply and to reduce shipping costs, GM asked Fisher to build a new auto body plant adjacent to the new GM assembly plant. The plants would have no need for shipping docks; bodies would be transferred on the production line right from the Fisher plant to the General Motors plant. Fisher refused to make the requested investment, perhaps for fear that the new plant, so closely tailored to GM's needs, would be vulnerable to demands that GM might later make. The issue was eventually resolved by vertical integration: General Motors purchased Fisher Body.[5]

The hold-up problem is an example of **postcontractual opportunism**. It arises

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[4] Paul Joskow "Vertical Integration and Long-Term Contracts: The Case of Coal-Burning Electric Generating Plants," *Journal of Law, Economics, and Organization*, 1 (Spring 1985), 33–80; "Contract Duration and Durable Transaction-Specific Investments: The Case of Coal," *American Economic Review*, 77 (March 1987), 168–85; and "Asset Specificity and the Structure of Vertical Relationships: Empirical Evidence," *Journal of Law, Economics, and Organization*, 4 (1988), 95–117.

[5] See Benjamin Klein, Robert Crawford, and Armen Alchian, "Vertical Integration, Appropriable Rents, and the Competitive Contracting Process," *Journal of Law and Economics*, 21(1978), 297–326.

Table 5.1 **The possibility of grabbing creates a Prisoners' Dilemma game.**

		Firm A	
		Grab	Don't
Firm B	Grab	−1, −1	−2, 3
	Don't	3, −2	2, 2

because contracts are incomplete and imperfectly specified, so that the parties to the contract can exploit loopholes to gain an advantage over one another.

A Mathematical Example of the Hold-Up Problem A numerical example will help clarify the logic of the hold-up problem. Suppose that there are two firms, A and B, which have joined together to take advantage of a business opportunity. Each firm is called on to make an investment in an asset that is entirely worthless outside of the joint venture. These assets are therefore completely cospecialized.

To keep matters simple, suppose that the project requires that each party make an investment. The cost of the investments is 2 for each party, and the gross return from the investment is 8, yielding a net benefit of 4. To capture the possibility of ex post opportunism, suppose that the division of the gross benefits between the two firms can be affected by costly actions that each takes, but that it is not possible to make an enforceable contract governing these actions. Again, for simplicity, suppose the possible actions are "grab" or "don't grab," where grabbing costs 3. If neither firm grabs, then the gross returns are shared equally and each party enjoys a net benefit of 2. If both grab, the gross benefits are again shared equally, but each firm absorbs the additional cost of 3 associated with grabbing. This yields each firm a net payoff of $4 - 2 - 3 = -1$. Finally, if one firm, say A, grabs and the other does not, then it gets all of the benefits. The grabber's payoff is then $8 - 2 - 3 = 3$, and the other's payoff is -2. These payoffs are shown in Table 5.1; the payoff table has the form of a **Prisoners' Dilemma** game.[6]

Clearly, if the firms could contract about grabbing, they would contract never to do that: It is costly and creates no value. If contracting about grabbing is impossible, however, what should we expect the outcome to be?

The analysis begins by considering the decision facing firm A, assuming that the specific investments have already been made. Firm B is supposed to be making its decision at the same time, so there is no possibility that what B chooses will be influenced by A's actual choice. Suppose then that A hypothesizes that B will not grab. What are A's options? If A grabs, then A receives a payoff of 3, as computed earlier. If A refrains from grabbing, then it gets only 2. Thus, if A thinks B will refrain, its own selfish interests are best served by grabbing the returns for itself. In other words, A's highest payoff in the first column of Table 5.1 is achieved by grabbing. Now suppose that A hypothesizes that B will grab. Then by grabbing as well it gets a payoff of -1, whereas its payoff is -2 if it refrains. Once again, A does best by grabbing. Because A has the same selfishly best action no matter what it

[6] The Prisoners' Dilemma is the most famous and most studied strategic game. For a lively introduction to the subject, see Avinash Dixit and Barry Nalebuff, *Thinking Strategically* (New York: W.W. Norton, 1991).

expects B to do, a selfish A will always grab. A parallel analysis of B's decision indicates that a selfish B would do likewise, leading to payoffs of −1 for both.

So far, the analysis is the same as for any Prisoners' Dilemma game; the difference here, however, is that the firms do not have to play. If the firms recognize the incentives that they will face after the specific investments are made, they will want to allow for these in deciding whether to invest at all. In this example, it turns out that the threat of opportunism completely destroys the incentives to invest: If the firms cannot commit not to attempt to grab more than their share of the returns, no investment takes place.

This mathematical example is so stark that one might be tempted to believe that simple morality or business ethics could overcome the problem, and indeed it sometimes does. Much of the problem is that what is honest and fair is rarely as obvious in the rich settings of reality as it sometimes seems in classroom examples, and it is tempting for people to think that what serves their own interests is honest and fair. It is too risky to rely on others to act consistently contrary to their own selfish interests unless there is something that commits them to that kind of behavior.

Achieving Commitment

We have already mentioned relational and implicit contracts as a response to contractual incompleteness. These contracts serve to set expectations and establish decision processes to deal with the inevitable unforeseen circumstances while avoiding some of the difficulties of writing explicit detailed contracts. When cospecialized assets are a factor, a common response is for the same person or firm to own both. After all, there is little concern that a firm will hold itself up! As we see later, however, this ownership solution may also incur certain costs that limit its usefulness.

Another approach is to attempt to achieve commitment by noncontractual means. This may be worthwhile, but costly. William the Conqueror and Cortes used up valuable resources—perfectly good fleets. Apple presumably spent extra on construction costs for its overly specialized plant and in return got a facility that may become obsolete sooner than it otherwise would. Yet such expenditures may be the cost of commitment.

THE ROLE OF REPUTATIONS In other situations, concern with one's *reputation* may be an effective check on ex post opportunism, overcoming the temptations to renege or renegotiate. It may even achieve the same results as actual commitment. Bargaining with terrorists or not punishing naughty children establishes without a doubt that there is no commitment and invites further challenges. Renegotiating executive pay contracts makes less credible any claims that bad future performance will not be rewarded. Not paying legitimate bills or not fulfilling obligations also results in a reputation for untrustworthiness. In a world of costly and incomplete contracting, *trust* is crucial to realizing many transactions. Thus, the concern with getting a bad reputation that reduces future possibilities for profitable transactions can limit reneging. Effectively, it removes the incentives for opportunistic behavior by creating a cost offsetting the short-term gains of opportunistic behavior.

As we see in Chapter 8, the value of a reputation—and thus the costs incurred in building and maintaining a good one—depends on how often it will prove useful. This in turn is related to the *frequency* of similar transactions, the *horizon* over which similar transactions are expected to occur, and the transaction's profitability. The incentives to build and maintain a reputation are larger the more frequent the transaction, the longer the horizon, and the more profitable the transaction. One implication is that, in relational contracting, where there is an issue of which party should have the discretion to direct activities in unforeseen events, it should be the

one with the most to lose from a damaged reputation. This is likely to be the one with the longer horizon, the more visibility, the greater size, and the greater frequency of transactions. In the employment relationship with a stable, long-lived firm, this is more likely to be the firm than the employee.

Private Information and Precontractual Opportunism

Once whatever contingencies are going to be accounted for are established, the parties to a potential contract must reach agreement on a course of action and on the payments to be made. This involves some sort of bargaining.

Bargaining is a very complex process. It is also central to economic life. It was long traditional in economics to suggest that the outcomes of bargaining among small numbers of people are essentially indeterminate. Such hard-to-describe features as bargaining strength, credibility, guile, patience, and strategic insight would combine with initial conditions to determine the outcome. The efficiency principle would suggest that an efficient agreement should be reached, but predicting the actual result was too much for economic analysis.

Yet, with the hypothesis of no wealth effects, the value maximization principle means that the value-creating aspects of the agreement are determinate. Only the monetary transfers would be determined by bargaining strength and the like. Recall, however, that efficiency is defined relative to a set of feasible outcomes. What is feasible—and thus what is efficient—turns out to depend crucially on the informational conditions. When the costs and benefits of different plans to each party are known to that party alone, or when the likelihood of different possible outcomes are private information, then these *informational asymmetries* can prevent any agreement from being reached, even when an agreement would be efficient under complete information. Furthermore, even if an agreement is reached, it will not typically be efficient when judged by the standards of complete information, especially when the parties have an option not to participate.

Bargaining over a Sale

The possibility of failing to reach an agreement can arise in the simplest problems with private information about values. Consider two people, a buyer and a seller. The seller owns a unit of some good in which the buyer is interested. Each is privately informed about the value that he or she places on having the good. The buyer believes that the seller values the good either at \$2 or that she finds it worthless. The seller believes that the buyer values the good at either \$1 or \$3. The buyer assigns a probability of 0.2 to the seller's valuing the good at \$2, and correspondingly, a probability of 0.8 that it is worth \$0 to her. The seller assesses a probability of 0.2 that the buyer's valuation is \$1 and a probability of 0.8 that it is \$3 (see Table 5.2). Of course, the seller knows what the good is actually worth to her, and the buyer knows what it is worth to him.

If the actual valuations are \$1 for the buyer and \$2 for the seller, it is efficient for the good to remain with the seller—this maximizes the total value. This occurs with probability $.2 \times .2 = .04$. In all other circumstances, 96 percent of the time, efficiency demands that a sale occur and the good be transferred to the buyer.

If the valuations were known, there would not be much difficulty. Both parties would know what the good was worth to the other, and if there is a value gain from transferring the good, they ought to be able to settle on some payment from the buyer to the seller that makes both parties better off.

Table 5.2 Efficient Outcomes with Different Possible Valuations

	Seller's Value	
Buyer's Value	\$0 (Probability = .8)	\$2 (Probability = .2)
\$1 (Probability = .2)	Trade	No Trade
\$3 (Probability = .8)	Trade	Trade

INFORMATIONAL ASYMMETRIES AND STRATEGIC MISREPRESENTATION With the actual informational asymmetries, however, there are potential problems with the bargainers' trying to misrepresent their valuations to get a better price. If the buyer's actual value is \$3, he might like to insist that it is only \$1 in hopes of paying a lower price. If he can convince the seller that the good is worth only \$1 to him, then (provided that trade occurs) he cannot be expected to pay more than \$1. So, for example, by refusing any offer at a higher price, he may gain. Similarly, if the seller's value is actually \$0, she might try to insist that she actually values the good at \$2 so she would get a better price. Of course, misrepresentation of this sort carries the risk of the sale falling through because the parties convince each other that their valuations are \$1 and \$2, so that they believe there is no basis for mutually beneficial trade. This may be a risk worth taking, however.

To prevent this sort of misrepresentation, each party must do at least as well behaving straightforwardly and, in effect, "confessing" his or her actual valuation, as he or she does misrepresenting it. Further, because trade is voluntary, neither can be forced to trade if he or she does not find it personally advantageous. These conditions constrain the set of arrangements that are feasible by requiring that each party get at least some minimum amount of surplus from the transaction, where the particular amount depends on the party's actual valuation.[7] It may then be possible that there is simply not enough surplus to go around. This can prevent trade from occurring even though the buyer actually values the good more than the seller does. With private information, the full-information efficient solution may no longer be feasible.

EFFICIENT TRADE MAY BE UNATTAINABLE To illustrate this possibility, let us take the point of view of a mediator with the power to suggest the prices that will govern exchange when the parties represent their values in particular ways. If there is no pattern of prices that a neutral mediator could identify to promote efficient trade, then there is none that the parties, with their additional interests, could find for themselves.

The problem that must be solved is to discourage the buyer from falsely claiming a low value (in the hope of getting a low price) and the seller from falsely claiming a high value. A seller who claims a value of 2 must be offered a price of at least 2, or she will refuse to participate. (Such limitations on what is feasible that arise from individuals' options not to participate are called *participation* incentive constraints.) Offering a higher price in this instance only encourages the seller to make this claim. Therefore, as Table 5.3 shows, part of any solution to this problem must entail setting a price of 2 when the seller claims a value of 2 and trade takes place. Similarly, to

[7] This surplus is sometimes called an *informational rent*, that is, an excess return (rent) that is received because of the individual's private information.

Table 5.3 Prices That Best Encourage Honest Reporting and Efficient Trade

		Buyer's Value	
		1	3
Seller's Value	0	1	p
	2	No Trade	2

discourage the buyer from claiming a value of 1, the price should be set at 1 when that value is claimed and trade takes place.

Now, the problem is to find a price, p, to be paid when the buyer says he has a high value ($3) and the seller says her value is low ($0) that will encourage both to be truthful. Consider the seller first. If she reports honestly, she gets $1 when the buyer claims a value of $1. If the buyer is honest, this will happen 20 percent of the time. The seller gets p the rest of the time, where p is the price when the buyer's value is $3 and the seller's is $0. Thus, the seller's *expected* payoff when she plays honestly is $.2 \times 1 + .8 \times p$. If she were to claim her actual value for the good is $2, she would not trade when the buyer says his value is low, which would happen 20 percent of the time again. She would be left with the good in this circumstance, but this is worthless to her. The rest of the time she would get the price of $2 that we calculated earlier—the minimum consistent with her being willing to trade if $2 were her actual valuation, as she claims it is. Thus, her expected payoff from dissembling about her valuation is $.2 \times 0 + .8 \times \$2 = \$1.60$. Honesty pays if the first of these two expressions is larger, that is, if $.2 + .8p \geq 1.60$. This *incentive compatibility constraint* must hold if the seller is to find it worthwhile to report honestly. It implies that p must be at least $1.75.

A similar computation can be done for the buyer. If he reports honestly, he pays a price of $2 when the seller says her value is $2 and a price of p when she says her value is $0, garnering a benefit of $.2 \times (3-2) + .8 \times (3 - p) = \$2.60 - .8 \times p$. If the buyer dissembles, he trades only when the seller says her value for the good is $0, in which case he pays $1 and gets the good that is actually worth $3 to him. This would occur 80 percent of the time if the seller is straightforward about her announcement, so his expected payoff from dissembling is $.8(\$3 - \$1) = \$1.60$. To make it worthwhile for the buyer not to dissemble, we must have $\$2.60 - .8p \geq \$1.60$, or $p \leq \$1.25$.

Thus, to induce the seller to be straightforward, the price she gets when she says her value is $0 and the buyer says his is $3 must be at least $1.75, but in the same circumstance he cannot be asked to pay more than $1.25. The conclusion is that it is impossible to find any prices that always make it in the parties' individual interests to report truthfully and to trade whenever trade is, in fact, value increasing.

ACHIEVING EFFICIENCY DESPITE THE INFORMATIONAL ASYMMETRIES We conclude that, at least in this particular example with private information about values, it is not possible for trade to occur precisely when the buyer's value exceeds the seller's.[8] The

[8] You might object that we have not really shown this because we assumed that it was necessary to get the buyer and seller to reveal their true valuations rather than to misrepresent what the good was worth. Perhaps there is some scheme under which they do misrepresent, but trade nevertheless occurs just when the buyer's actual value exceeds the seller's. However, a remarkable result called the *revelation principle*

special characteristic of this example is that the risks to the seller of exaggerating her value to an honest buyer and to the buyer of misstating his value to an honest seller are both low: The sale will be consummated with 80 percent likelihood anyway.

If the probabilities were reversed in the example, so that the low value for the buyer and the high value for the seller both occurred with high probability (0.8 each), then it would be possible to get correct revelation and still have trade occur just when the buyer's value exceeds the seller's. For example, consider the incentives that exist if p is set to \$1.50 in Table 5.3. If the buyer reports truthfully and the seller dissembles when her actual value is \$0, then the seller's expected payoff is $.8 \times 0 + .2 \times \$2.00 = \$.40$, whereas telling the truth pays $.8 \times \$1.00 + .2 \times \$1.50 = \$1.10$. Honesty, in this case, is the best policy for the seller, and the same can be shown to be true for the buyer; the incentive compatibility constraints are met.

The crucial difference when the probabilities are reversed like this is that now there is a high probability (.8) that the buyer will in fact be dealing with a seller with whom there are no gains to trade unless the buyer's value is high, and similarly, the seller is likely to be dealing with a low-value buyer and so should not trade unless her value is low. This makes it more costly to dissemble, because dissembling now runs a higher risk of preventing trade. If the high-value buyer (or the low-value seller) is straightforward, he always gets to trade. If he tries to manipulate the price to his advantage, then 80 percent of the time he does not get to trade and misses out on these benefits. This makes it easier to meet the incentive compatibility constraints—they do not eliminate the full-information efficient outcome.

Incentive Efficiency

We have seen that private information sometimes, but not always, makes it impossible to achieve the full-information efficient outcome. Recall that we have defined efficiency relative to a set of people and a set of feasible outcomes. If incentive constraints are unavoidable, then we do not label the outcome as inefficient merely because a better outcome could be achieved in the absence of incentive constraints. Instead, a mechanism for determining allocations is efficient if there is no alternative method that is feasible in the presence of incentive constraints and that all the relevant parties prefer to the original mechanism. In much of the economics literature, mechanisms that are efficient when incentive constraints are recognized in this way are called **incentive efficient**. According to the efficiency principle, we ought to expect the bargainers to find incentive-efficient mechanisms. The key question then becomes how much value the parties can create for themselves when faced with incentive constraints.

INCENTIVE EFFICIENT BARGAINING WITH PRIVATE INFORMATION: AN EXAMPLE Another example of bargaining indicates what can be accomplished. Suppose again we have a buyer and a seller, but now each party's value for the good might take on any value between \$0 and \$1 with equal probability. Of course, the buyer knows the true value B that he places on the good, and the seller knows her value S. Suppose the two adopt the following simple rule to govern their bargaining: They will each simultaneously name a price; if the price named by the seller is less than that named by the buyer, the sale is made at a price midway between the prices; and if the seller's demand exceeds the buyer's offer, no trade is made and the two bargainers are committed to walk away.

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shows that this is not the case: Any pattern of outcomes that can be achieved using any mechanism that respects incentive constraints can in fact be achieved by a mechanism in which the buyer and seller report their values and are given incentives to do so honestly.

If each side named the actual value of the good to him or her, trade would immediately occur whenever there are gains from trade, that is, whenever the buyer's value B exceeded the seller's value S. It cannot be in the two parties' individual interests, however, to reveal their actual values: If the buyer is going to bargain "honestly," it is advantageous for the seller to play strategically and announce a higher asking price than her true value. Although this increases the chances that trade will not occur, the trades that are lost are ones that were hardly profitable anyway. For example, suppose the buyer is forthright but that a seller whose value is \$.50 reports a value of \$.52 instead. The only trades that will be lost on account of this misrepresentation occur in situations in which the buyer's value lies between \$.50 and \$.52 and therefore in which the price lies below \$.51, so at most \$.01 of profit is lost. On average, when a loss occurs, it amounts to just \$.005. Moreover, the chance of loss is just the 2 percent probability that the seller's value lies in the specified range, so the total expected loss from trades missed on account of the misrepresentation is .02 × \$.005 = \$.0001. Against this is a price increase of \$.01 that is enjoyed whenever the buyer's value is at least \$.52, which occurs 48 percent of the time. So, the expected extra profits are then \$.0048, for a net expected gain of \$.0047 (=\$.0048 − .0001).

Similarly, if the seller is forthright, the buyer gains by understating his valuation. From the buyer's point of view, claiming that the value is lower than it actually is involves a chance of blocking a trade that would be profitable, but as before the cost of a small misrepresentation is tiny. More often, the misrepresentation will lead to a reduced price for the buyer, and this effect is by far the larger one: Small misrepresentations always pay.

In actuality, the calculations we have just made are not the most relevant ones. If it is going to be in the seller's interest, for example, to misrepresent her value, then the buyer ought to account for that in determining his own misrepresentation. Actually calculating the bid and ask functions that the two parties would use if they correctly forecast one another's choices is somewhat complicated, but the result that emerges is that the two will systematically bid different amounts than their true values. As a result, trade will occur if and only if the buyer's valuation exceeds the seller's by 1/4.[9] This is suggestive of a typical pattern: *When incentive constraints are important in bilateral bargaining, trade takes place only if the gains from trade are sufficiently large.*

One might hope that some other bargaining arrangement would yield greater efficiency, but this is actually not possible. In the example considered, the specified bargaining rules actually maximize efficiency subject to the constraints imposed by the private information. For example, the parties could have initially decided to continue bargaining as long as one or the other wished to do so or until an agreement was reached. The inefficiency will still be manifested, however, either in the form of costly disagreements or as a costly delay until an agreement is reached.

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[9] To calculate the strategies that the parties will adopt if they correctly forecast one another's incentives to adopt strategic behavior, suppose that the seller expects the buyer's bid to be a linear function $b = ß_1 + ß_2 B$ of his true valuation and, in a corresponding fashion, that the buyer expects the seller's asking price to be a linear function $s = \sigma_1 + \sigma_2 S$ of S. Suppose too that the parties actually do decide to use rules of this form to decide how much to bid and ask. (Of course, correct revelation of their values occurs when they pick $ß_1 = \sigma_1 = 0$ and $ß_2 = \sigma_2 = 1$.) Then it is possible to calculate, for given assumed values of the σ's, how the buyer's expected payoff depends on his choice of ß, and similarly, how the seller's payoff depends on her choice of σ for any assumed values of ß. Maximizing each party's payoff with respect to its choices (by setting derivatives equal to zero) and then requiring that each correctly forecasts the other's choices yields the conclusion that $b = (1/12) + (2/3)B$ and $s = (1/4) + (2/3)S$. Then $b > s$ only if $B > S + 1/4$.

The existence of private information frequently means that some value-maximizing plans cannot be realized. Sometimes the potential gains from transacting are missed. Incentive-efficient arrangements minimize these losses, as in this example, where trade does not occur if the gains are small but does if they are sufficiently large.

Efficient Agreements with Large Numbers of Participants

Although the problem of incentives and private information can be serious even in a bilateral relationship, it becomes much more serious when the numbers of people who must agree begin to grow. When there is a diversity of interests, even moderate-sized groups often find it impossible in practice to reach a unanimously acceptable decision. In theory, with a large enough set of participants, it can be a virtual certainty that there are gains to be realized from an agreement, and yet there may be no agreement that meets the incentive and participation constraints, making it impossible to realize the potential gains.

INVESTMENT PROJECTS TO CREATE PUBLIC GOODS: AN EXAMPLE To illustrate the underlying theoretical problem, we consider a simple decision about whether to undertake an investment project to create a public good. The cost of the project is $1 in total, no matter how many people are benefitted. Each individual either wants to have the project done, placing a value of $2 on it, or is indifferent about it, getting a value of $0 from its being undertaken. The probability that the project is valued positively by any single individual is p, a number strictly between 0 and 1. Again, each individual knows his or her actual valuation, but this value is private information. Individuals are free to drop out of the group, but if they do so they do not get the benefits of the project.

If anyone wants the project done, value maximization demands that it should be carried out because the cost is $1 and the benefit is at least $2. If the group consists of a single individual, that person will surely execute the project. Suppose, however, that there are several people. Anyone forced to pay more towards the costs of the project than it is worth to him or her can drop out of the group and avoid paying. Thus, no one can be forced to pay more than $2, and someone who claims his or her value is $0 cannot be made to pay anything. In essence, everyone must agree to the plan for it to be undertaken. The danger is that people will try to free ride off the others, falsely claiming that they value the project at $0 and hoping that someone else will pay for it.

Suppose that if $m \geq 1$ out of the N people in the group say their values are high, then the project is built, with the costs being assigned randomly to one of the m people, or that the costs are split equally among the m people voting for the project. If there are two people in the group ($N = 2$), then we can express the incentive constraint that a person with a value of $2 be willing to acknowledge that as follows:

$$p(\$2 - \$.5) + (1 - p)(\$2 - \$1) \geq p(\$2) + (1 - p)(\$0)$$

The first term on the left is the probability p that the other person will say his or her value is high (assuming honest reporting by that person), times the gain the first individual receives in this case. This gain is the $2 he or she gets from having the project done, less the expected cost to him or her, which is $.50. The second term is the probability that the other person will report a value of zero, times the gain in this case, which is the benefit of the project less its cost. So the left-hand side is the expected return from reporting honestly. If the person dissembles, then the project is built and financed only when the other person says his or her value is $2 (which occurs with probability p), in which case that other person bears all the cost. The right-hand side is thus the expected return to dissembling. The inequality says that

being forthright is worthwhile. This inequality must hold if the project is to be built whenever it is value maximizing.

Solving for p, we see that p cannot be too large: $p \leq 2/3$. If p is too high, it appears too likely that the other person will volunteer to pay, and the individual under consideration cannot be induced to report his or her true valuation. It is precisely when it is likely that others would be willing to finance the project even without one person's participation that the person is most tempted to claim that he or she does not value the project and, as a result, that an efficient project will be delayed or left unbuilt.

The way to induce correct revelation is to reduce the probability that the project is carried out when the reported gains seem small. Let q be the probability that the project is carried out when only one person states that his or her value is \$2. Then the incentive constraint becomes

$$p(\$2 - \$\tfrac{1}{2}) + (1 - p)q(\$2 - \$1) \geq pq(\$2) + (1 - p)(\$0)$$

Now, if $p > 2/3$, q must be less than 1 to induce truthful reporting. For example, as p approaches 1, q must approach 3/4. Thus, 25 percent of the time when someone volunteers to pay, the project is not carried out. In other words, people will sometimes free ride and not volunteer to pay for the project, even when it is actually individually worthwhile to pay for it themselves.

If there are three people, the incentive constraint (calculated assuming that the project will be built whenever it is value maximizing) gets even tougher to meet. In this case, it is

$$p^2(\$2 - \$\tfrac{1}{3}) + 2p(1 - p)(\$2 - \$\tfrac{1}{2}) + (1 - p)^2(\$2 - \$1) \geq p^2(\$2) + 2p(1 - p)(\$2) + (1 - p)^2(\$0)$$

which requires p to be less than about 0.44. If p exceeds this value, the incentive constraint cannot be met if the project is to be built whenever its value exceeds its direct cost.

As N gets large, the incentive constraint becomes more and more difficult to meet—free riding becomes more and more tempting if the others are expected to report honestly. The essential intuition is that dissembling is costly only when the others' reports make the given individual *pivotal*—the project is built if he or she admits to having a high value for it, and not otherwise. As N gets large, it becomes a virtual certainty that someone else will value the project highly and, assuming honest reporting, will volunteer to pay for it. The cost of free riding, which is the chance that the misrepresentation prevents the project from being constructed, then becomes vanishingly small, and so misrepresentation cannot be prevented.

For example, when $N = 3$ and $p = 1/2$, the incentive constraints mean that the best that can be done (in terms of maximizing total expected value) is that the project is undertaken when all three or any two of the people say their values are high, but it is undertaken less than 90 percent of the time when only one person admits to having a high valuation (that is, the person is pivotal). Yet, this one person is individually willing to pay enough to finance the project.

As we discussed earlier, this analysis implies that it is not reasonable for people to assume that others will all report honestly. That recognition increases their own individual returns to honest reporting. The precise calculations are rather complex, but the conclusion for this example is that as the number of people involved in the decision becomes large, assuming that everyone behaves similarly, the probability that someone will volunteer to contribute to the project tends to a limit of approximately

two chances in seven, even though it is essentially certain that funding the project is value maximizing.

Bargaining Costs

Contracting requires reaching a mutually beneficial agreement. We have just seen that private information about the values of different options can prevent reaching an agreement on value-maximizing plans. One way to think of this is in terms of **bargaining costs**, the transaction costs of reaching mutually acceptable agreements. Bounded rationality imposes bargaining costs, even in the absence of any strategic behavior. It takes time and effort to imagine and list contingencies, to determine efficient courses of action, and to settle on divisions of costs and benefits. In fact, bounded rationality means that it may be impossible to carry out these tasks completely and perfectly. The resulting incomplete contracts then give rise to further transaction costs: The costs that are incurred in achieving commitment by noncontractual methods and the inefficiencies that result from attempts to protect against imperfect commitment.

When private information is a factor, there are additional costs. These can be substantial. They can prevent agreement or delay its realization considerably. In either case, simple value maximization computed without regard to the constraints imposed by incentives is a misleading basis for prediction.

In some bargaining situations, as we have already seen, private information may completely prevent agreement even though there are significant gains to be had from trade. Before studying this problem further, we first examine why people may choose to become privately informed and the special problems this poses.

Measurement Costs and Investments in Bargaining Position

In many exchanges, some important characteristic or quality of the good being traded is unknown to both the buyer or seller. For example, when an oil company buys rights to extract the petroleum on a piece of land, neither the company nor the seller typically knows how much oil will be found. Similarly, neither the purchaser or the seller of a piece of equipment, or both, may be unsure about how durable the equipment may be. Although it is conceivable that such a situation could sometimes be unproblematic—the parties might make a deal based on expected values—that is not always the most likely outcome. More often, it is possible and profitable for one or both parties to gain a bargaining advantage by investing in information about the quality of the good. For example, by becoming informed the buyer can better avoid paying a higher price for the good than what it is actually worth. Moreover, if one side has become informed, it is very likely to be in the other side's interest to absorb the costs of becoming informed as well because otherwise it will be subject to adverse selection or opportunistic misrepresentation. Thus, both sides may be led to incur the costs of measuring the good's worth.

These information costs are wasted from the standpoint of society as a whole. It is true that the information gathering will lead to a negotiated price that reflects the value more precisely, but that does not increase total wealth. A different price benefits one party only at the expense of the other. The good still ends up being transferred, just as efficiency required and just as it would have been had the information not been gathered. The costs of gathering that information are subtracted from the parties' total wealth; there is no offsetting benefit unless the information affects a productive decision or the allocation of the asset. In line with the value maximization principle,

you might therefore expect businesspeople to try to devise institutions that reduce this source of waste.

The De Beers Diamond Monopoly

The De Beers group, which runs the remarkably successful world diamond cartel, uses an unusual method to sell diamonds that has been interpreted in just this way—as an attempt to reduce excessive expenditures on information.[10] Buyers indicating an interest in purchasing particular amounts of particular sizes and qualities of stones are offered a packet of stones, called a "sight," that roughly corresponds to their indicated interests. The stones are graded and sorted on the basis of their gross characteristics only, without any attempt by De Beers to estimate their value closely. The sights are offered on a strict take-it-or-leave-it basis, according to a pricing formula based on these gross characteristics. Absolutely no negotiation over the price is permitted. Also, the content of the packet is not negotiable (except for correcting errors in grading). Furthermore, failure to take the sight as offered results in the buyer's being excluded from buying from De Beers in the future.

Although elements of this arrangement may represent De Beers' exploitation of its monopoly position (it controls over 80 percent of the world diamond supply), there is little reason to expect that even a monopolist would choose an inefficient distribution scheme. From a value-maximization perspective, it makes sense to look for a competing explanation of these details in terms of efficient exchange.

Just such an explanation has been offered. If bargaining about the price and contents of a packet were allowed, then there would be incentives for the buyers to spend resources in finely examining each stone to determine the most valuable gem that can be cut from each and to estimate the values of the uncut stones that way. De Beers would have to do the same to protect itself; that is, it would have to abandon its system of using only rough grading of the uncut stones. To minimize examination costs and thereby maximize total value, however, a careful examination should be made only by the final buyer, who must actually cut the stone, and *not* by the seller or the other potential buyers. Moreover, it does not much matter which diamond retailer purchases which stone. What is most important for efficiency is that the transaction be made and that wasteful information gathering be minimized. Forbidding negotiations eliminates the incentive for buyers to incur examination costs, and the practice of withdrawing the opportunity to buy in the future prevents the "leave it" option from being used as a bargaining device to try to improve future offers. Moreover, buyers benefit from knowing that the stones they are purchasing have not been "picked over" by other potential buyers, so that examining the stones is not necessary for self-protection. Both sides enjoy savings from this system of trade.

Similar arguments have been advanced to explain the practices of prepackaging fruits and vegetables in grocery stores rather than selling them individually, of supplying warranties with products, of "block-booking" a set of movies from a given studio rather than allowing theater owners to pick and choose the films they want, and of paying percentage royalties rather than a fixed fee to book authors. In each case, the specified practice reduces or eliminates the incentives to incur costs inspecting individual items even when the efficient outcome is for trade to occur. The practices are then interpreted as examples of the value maximization principle.

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[10] Roy Kenney and Benjamin Klein, "The Economics of Block Booking," *Journal of Law and Economics*, 26 (1983), 497–540.

Investing in Bargaining Advantages

These measurement costs are an example of the larger class of expenditures made to gain an advantage in bargaining or trade. Many such expenditures are typically unproductive, yet they are incurred because they have private value in shifting the distribution of the gains from trade. Another example comes from the international arms control arena. In the United States there have been repeated arguments for funding of enormously expensive weapons systems and the Strategic Defense Initiative ("Star Wars") on the basis that they provide "bargaining chips" in negotiations with the Soviet Union on arms control. The idea seems to be that these systems are being developed and the corresponding expenditures made not because deployment of the systems is actually worthwhile but so that they can be bargained away and leave the United States with an adequately mighty arsenal.

These investments in bargaining position carry direct costs that the bargainers may wish they could all avoid. In this regard, the time and energy so often spent haggling, posturing, and delaying agreement in attempts to influence the terms of the deal are related wastes. Moreover, to the extent that the potential contracting parties forecast that these costs will be incurred, they may be forestalled from ever attempting to reach an agreement.

Adverse Selection

Besides the incentive to misrepresent valuations, a second sort of incentive problem arising from precontractual informational asymmetries is known as *adverse selection*. The term was coined in the insurance industry. The selection of people who purchase insurance is not a random sample of the population, but rather a group of people with private information about their personal situations that makes it likely they will receive a higher-than-average level of benefit payments under the insurance policy. For example, if an insurance company were to issue an individual health insurance policy that covers the medical costs associated with pregnancy and delivery, it is a safe bet that the policy would be purchased disproportionately by women planning to bear children in the near future. Childbearing plans are a privately known, *unobserved characteristic* of the insurance buyer that has a huge effect on insurance costs. The offering of comprehensive individual pregnancy and delivery coverage is subject to such severe adverse selection that such coverages are no longer available in the United States, where most health insurance is privately provided.

In many countries, the inadequacy of private health insurance has led governments to nationalize the provision of health care, although with mixed results. In the United States, the private sector has responded by developing a new practice—group health insurance—that partially substitutes for the missing market in health-insurance contracts. In modern practice, most coverage for pregnancy and delivery benefits is supplied through employer-provided medical insurance plans, where coverage is automatic or compulsory for all members of the employee group. Because participation is not voluntary and because pregnancy and delivery benefits are packaged with other health-care benefits, the insurance company is able to insure a cross-section of the community and thus avoid adverse selection.

A second example of adverse selection arises in connection with warranties on automobiles. New cars typically come with a warranty under which the manufacturer pays for any problems (other than routine maintenance) that arise with the car over a certain period of time, such as one year or 12,000 miles. Some auto makers have experimented with selling an optional extended warranty covering nonroutine expenses over a longer period, such as five years or 50,000 miles. Insurance theory predicts

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Exercises

Food for Thought

1. We are all familiar with the experience of feeling anger when we have been cheated or otherwise mistreated, and many of us have gone out of our ways to exact retribution for such injustices, even when we expect never to see the malefactor again and we do not expect anyone else to be aware of our punishing the transgression. Apparently, we have a taste for vengeance. Most of us also seem to have a conscience, a preference for behaving honestly and fairly. Indeed, Robert Frank [*Passions Within Reason: The Role of the Emotions* (New York: W.W. Norton, 1988)] has suggested that early humans who had such preferences would have had an evolutionary advantage, so that such preferences should have become widespread as people who

were biologically "hard-wired" with these traits were more successful in competing to reproduce. What effect would the possibility that people might have such preferences have on the analyses in this chapter of commitment and bargaining?

2. In common parlance, we often speak of "signaling intentions." For example, a firm might cut prices to signal its intent to compete fiercely to defend a market against a competitor's expansion. The idea is to deter the competitor, who would not want to challenge the first firm if it will react aggressively. The difficulty is that if the competitor does sink costs in expanding, the first firm may not find it worthwhile to fight. There is thus an issue of credibility of the signal. This problem arises whenever the signaler still has an option after the fact about what actions it will take. What means are available to give credibility to signals of intent?

3. One of the most rapidly growing health concerns of much of the developed world is the spread of Acquired Immune Deficiency Syndrome (AIDS), with the number of reported victims growing annually at alarming rates. One of the few effective drugs to alleviate the symptoms of AIDS is AZT. Although this drug is inexpensive to produce, it is sold for a high price, earning large profits for its manufacturer at the expense of the already suffering AIDS victims. As a result, there has been a public outcry to void the patent on AZT and to allow it to be produced competitively, at lower cost. What damage might such an action cause? What protections exist for an inventor that might keep a government from seizing intellectual property? Are there any practical alternatives available to deal with the very real medical-social problem that avoid the damage you have identified?

Analytical Problems

1. In the model of an insurance market with adverse selection, suppose that the parameters entail $c\bar{x} < v < \bar{x}(1 + c)/2$. Show graphically that there is a price at which the insurer breaks even and some people are willing to purchase insurance. What is the formula for that price?

2. Suppose that, in the context of the example in the text of the Spence model of education as a signal, the high-ability workers are relatively more numerous, making up 80 percent of the work force, but that the costs of acquiring education and the productivities are as in the text (that is, is C_H = \$10 and C_L = \$20, while wages are \$50 for the high-ability workers and \$20 for the low). Determine how large E_H, the level of education chosen by the high-ability workers, must be to be a credible signal if the level chosen by the low-ability workers is 0. Show that the high-ability workers would be better off if there were no signaling (for example, if C_H were to rise to \$20).

3. The units of two neighbors in a condominium housing development, Able and Baker, overlook a small courtyard. The board of the condominium association is considering whether to plant a tree in the courtyard. The cost is \$40. Everyone agrees that Able values having the tree either at \$100 or at \$0. The probability that the tree is worth \$100 to Able is p, a number between zero and one. Similarly, Baker values the tree at either \$70 (which occurs with probability r) or \$0. Able and Baker are the only people who will be able to see and enjoy the tree. The board decides to ask each of them how much having the tree planted is worth to him or her personally. If both say that they value the tree then the board will have it planted and the cost shared equally between the two neighbors. If only one of them expresses a desire to have the tree planted, then it is still planted but that person is charged the whole cost. If neither claims the tree is desirable, it is not planted.

Derive the incentive compatibility constraints for each individual, and solve for the values of p and r that make it incentive compatible to offer to help pay for the

tree exactly when it is actually valuable to the individual concerned. Now let q_A be the probability that the tree is planted when only Able offers to pay, and q_B be the probability that it is planted when only Baker offers to pay. If $p = 0.7$ and $r = 0.8$, what are the largest values of q_A and q_B consistent with correct revelation of willingness to pay?

4. A used-car salesperson and a potential buyer negotiate over the price of an old car. The seller believes that the buyer values the car at \$500 with probability x and at \$1,000 with probability $(1-x)$. The buyer believes that the car is worth \$250 to the salesperson with probability $(1-x)$ and \$750 with probability x. Under what circumstances should trade occur? How does the probability of efficient trade depend on the value of x? If a mediator who wants to maximize the expected gains from trade specifies that trade should occur at a price of p when the seller claims a value of \$250 and the buyer claims a value of \$1,000, derive two conditions on p that ensure that both parties will report honestly. Solve for the value of x that makes both incentive constraints exactly hold as equalities. Note that for lower values of x, trade is more likely to be efficient but it cannot always be attained. Explain why.

5. A seller values a good at either \$0 with probability x or \$10 with probability $(1-x)$. The buyer does not know what the good is worth, but knows that he values the good at \$10 more than the seller does. Thus, it is certainly efficient for trade to occur, no matter what the seller's actual valuation of the good. Show that it is compatible with individual incentives for trade always to occur at a price of \$10, no matter what value the seller announces. Suppose now instead that the good is worth only \$5 more to the buyer than the seller. Is it possible to find prices (as functions of the seller's claimed value) so that the seller will honestly report her true value and both sides will be willing to trade at these prices? Why or why not?

6. Suppose X is an intermediate good in the production of a final good Y. Firm A has a patent on good X, which it produces at zero marginal cost. Firm B is considering buying a machine which it can use to turn up to 10,000 units of X into Y on a one-for-one basis. The machine costs \$30,000, once purchased it cannot be resold, and it wears out completely once 10,000 units of Y are produced. Firm B can sell good Y for \$5 per unit. Firm A offers to supply good X to firm B at \$1 per unit once B has bought the machine.

Suppose A and B can write a binding contract setting the delivery price of good X. Will B buy the machine? What profits are earned by each firm? Suppose instead no such contract is possible. Is A's promise credible? Will B invest in the machine? What are the resulting profits?

Now suppose that A can enter into a licensing agreement with another firm, C, under which C can produce good X, incurring a marginal cost of \$c per unit. Suppose that A gives C the license, and that the cost, c, is \$1.01. Is A's price promise to supply B at a price of \$1 now credible? Does A gain from giving away its technology? Explain why or why not. Suppose the cost c were higher than \$1.01, but less than \$2. What would you expect to happen now? What if c were less than \$1?

Cabral

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- Productivity measurement (for 3.20): §3.1 (p. 36–42)
- Firm boundaries; why firms are different: §3.4–3.5 (p. 54–59)
- Exercises 3.9, 3.11, 3.14, 3.20: Ch. 3, Exercises (p. 62–64)
- Externalities; the Coase theorem: §5.1 (p. 93–99)
- Exercise 5.1: Ch. 5, Exercises (p. 117–117)
- Sequential games: backward induction, commitment: §7.2 (p. 167–174)
- Exercises 7.6, 7.14: Ch. 7, Exercises (p. 180–182)
- Vertical integration; investment incentives: §13.1 (p. 334–340)
- Exercise 13.10: Ch. 13, Exercises (p. 346–346)

CHAPTER 1

What is industrial organization?

What is industrial organization? It might help to start by clarifying the meaning of "industrial." According to Webster's *New World Dictionary*, "industry" refers to "manufacturing productive enterprises collectively, especially as distinguished from agriculture" (definition 5 a). "Industry" also means "any large-scale business activity," as in the tourism industry, for example (definition 4 b).

This double meaning is a frequent source of confusion regarding the object of industrial organization. For our purpose, "industrial" should be interpreted in the sense of Webster's definition 4 b. That is, industrial organization applies equally well to the steel industry and to the tourism industry; as far as industrial organization is concerned, there is nothing special about manufacturing.

Industrial organization is concerned with the *workings of markets and industries, in particular the way firms compete with each other.* The study of how markets operate, however, is the object of microeconomics; as a Nobel prize winner put it, "there is no such subject as industrial organization," meaning that industrial organization is nothing but a chapter of microeconomics.[1] The main reason for considering industrial organization as a separate subject is its emphasis on the study of a firm's strategies that are characteristic of market interaction: price competition, product positioning, advertising, research and development, and so forth. Moreover, whereas microeconomics typically focuses on the extreme cases of monopoly and perfect competition, industrial organization is primarily concerned with the intermediate case of oligopoly, that is, competition between a few firms (more than one, as in monopoly, but not as many as in competitive markets).

For the above reasons, a more appropriate definition of the field would be something like "economics of imperfect competition." But the term "industrial organization" was adopted, and I am not the one to change it.

Industrial organization is concerned with the workings of markets and industries, in particular the way firms compete with each other.

1.1 AN EXAMPLE

Examples are often better than definitions. In this section, I look into the market for anti-ulcer, anti-heartburn drugs, a case that touches on a number of issues of interest to industrial organization. The case thus provides a useful introduction to the next section, where I look more systematically at the central questions addressed by industrial organization.

Until the late 1970s, there was no effective drug to treat ulcers; severe cases required surgery. Then a research project at Smith, Kline & French (which is now part of GlaxoSmithKline) culminated with the discovery of cimetidine, a wonder drug sold under the brand name Tagamet. The production cost of Tagamet was rather small compared to the price for which it was sold. In cases like this, the huge gap between price and cost is bound to raise a variety of issues. For example, it may seem unfair that many suffering patients be deprived of a cheap-to-produce drug simply because price is so high. Then again, without revenues from drugs such as Tagamet it seems impossible for firms like Smith, Kline & French to continue churning out blockbuster drugs.

The fact is that the market for anti-ulcer drugs was enormous. Although Smith, Kline & French had a patent for cimetidine (valid until 1997), this did not stop other pharmaceutical giants from coming up with alternative products: Zantac, introduced by Glaxo in 1983; Pepcid, introduced by Merck in 1986; and Axid, introduced by Eli Lilly in 1988.

For many industry experts, Zantac, Pepcid, and Axid were little more than a copycats (also known as "me too" drugs). In fact, reviews of clinical trials indicated that there was little difference in the success rates of the various drugs; in other words, one drug could easily substitute for any of the other ones. Why wasn't then price competition more intense? The answer is: advertising.

> BELLYACHE BATTLES. We knew that the battle for your bellyaches would be big, but we had no idea it would be so bloody. Hundreds of millions of dollars are being poured into advertising designed to establish brand loyalty for either Tagamet HB or Pepcid AC. Zantac 75 will join the fray shortly.
>
> These drugs were blockbusters as prescription ulcer treatments; now that they are available over the counter for heartburn their manufacturers have really taken off the gloves.[2]

Nor is this specific to anti-ulcer and heartburn drugs. Overall, the advertising budgets of large pharmaceutical companies are of the same order of magnitude as their research budgets. What matters is not just the product's worth, but also what consumers — and

doctors, who frequently act as agents for the final consumer — think the product is worth.

Glaxo emerged as the winner of this advertising battle. It "set out to make heartburn into an acute and chronic 'disorder' that came with serious consequences if not treated twice daily with the company's two-dollar pill."[3] By 1988, Zantac had overtaken Tagamet as the world's best-selling drug.

Zantac's patent expired in the late 1990s, paving the way for competition by generics. A generic is a chemically equivalent drug that is mainly sold under the chemical name (Ranitidine, in the case of Zantac) rather than under the brand name. Notwithstanding numerous claims that generic Zantac has the same effect as branded Zantac, the latter still manages to command a large market share while selling at a much higher price. In July 1999, shortly after patent expiry, discount drug seller RxUSA was quoting a 30-tablet box of 300 mg Zantac at $85.95. For a little more than that, $95, one could buy a 250-tablet box of 300 mg generic Zantac Ranitidine — that is, for 7.5 times less per tablet. More than a decade after patent expiry, price differences remain significant: in January 2014, 150 mg Zantac cost almost 40 cents per tablet, whereas the corresponding generic sold for about 8 cents. Another testament to the enduring value of a strong brand is that, in 2006, Boehringer Ingelheim paid more than $500 million dollars for the US rights to Zantac.

At the time of Zantac's launch, Glaxo was an independent company. Since then, it first merged with Wellcome to form GlaxoWellcome, then with SmithKline (which in turn resulted from a then recent merger) to form GlaxoSmithKline.[a] Frequently, these mergers are heralded as the sources of important synergies. For example, when GlaxoWellcome was formed, the merging parties argued that the combination of Wellcome's AZT and Glaxo's 3TC worked better against AIDS than either drug alone.[4] Critics, however, see it primarily as a source of greater market power: if you cannot beat the competitor, then buy it.

1.2 CENTRAL QUESTIONS

The example in the previous section illustrates several issues that industrial organization is concerned with (see below, in italics): for decades, GlaxoWellcome was a firm that commanded a significant degree of *market power* in the anti-ulcer and heartburn therapeutical segment (the relevant *market definition*). GlaxoWellcome, which resulted from the *merger* of Glaxo and Wellcome, established its position by means of a clever *R&D strategy* that allowed it to enter an industry already dominated by SmithKline; and by means of an aggressive *marketing strategy* that increased its market share. For a period of time, Zantac's position was protected by *patent rights*. This is no longer the case, meaning that *differentiating the product* with respect to the incoming rivals (generics producers) is now a priority.

In this section, I attempt to formulate the object of industrial organization in a more systematic way. One can say that the goal of industrial organization is to address

a. It's a good thing the merged firms did not keep all of their names, else we would have to spell GlaxoWellcomeSmithKlineFrench. An even more impressive example is given by Sanofi: if every one of its predecessor firms kept its name, the company would be called — take a deep breath — Sanofi Synthelabo Hoechst Marion Roussel Rhone Poulenc Rohrer Marion Merril Dow! (exclamation mark added).

the following four questions: (a) Do firms have market power? (b) How do firms acquire and maintain market power? (c) What are the implications of market power? (d) What role is there for public policy with regard to market power?

Since all of these questions revolve around the notion of market power, it may be useful to make this notion more precise. **Market power** may be defined as the ability to set prices above cost, specifically above incremental or marginal cost, that is, the cost of producing one extra unit.[b] So, for example, if GlaxoWellcome spends $10 to produce a box of Zantac and sells it for $50, then we say that it commands a substantial degree of market power.

Now for the questions.

■ **IS THERE MARKET POWER?** Understandably, this is an important question, in fact, a crucial one. If there is no market power, then there is little point in the study of industrial organization.

Over the years, many empirical studies have attempted to measure the extent of market power. Assuming that costs are proportional to output, a good approximation to the extent of market power can be obtained from data on prices, output and profit rates.[c] One study along these lines found that the extent of market power in the American economy is very low, a conclusion that follows from observing relatively low profit rates.[5] This finding is consistent with one of the central tenets of the so-called Chicago school of economics: as long as there is free entry into each industry, the extent of market power is never significant. If a firm were to persistently set prices above cost, a new firm would find it profitable to enter the market and undercut the incumbent. Therefore, market power cannot persist, the argument goes.[d]

Not every economist agrees with this view, either at a theoretical or at an empirical level. From an empirical point of view, an alternative approximation to the value of marginal cost is obtained by dividing the increase in cost from year t to year $t + 1$ by the increase in output in the same period. Based on this approach, a study estimates that prices may be as much as three times larger than marginal cost.[7]

Evidence from particular industries also suggests that the extent of market power may be significant. Take, for example, the US airline industry. A 1996 US government report analyzed average fares in 43 large airports. In 10 of these airports, one or a few airlines held a tight control over takeoff and landing slots. The report found that, on average, fliers were paying 31% more at these airports than at the remaining 33 airports.[8] In other words, the report suggests that airlines which manage to control the critical asset of airport access hold a significant degree of market power.

Further examples could be supplied. These would not necessarily be representative of what takes place in every market. To be sure, there are a large number of industries where firms hold little or no market power (cf Section 4.1). The point is that there are *some* industries where market power exists to a significant extent.

b. A rigorous definition of marginal cost and other cost concepts is given in Section 3.1. If costs are proportional to output, then marginal cost is equal to unit cost.

c. The profit rate, r, is given by revenues minus cost divided by costs: $r = (R - C)/C$, where R is revenues and C is costs. If costs are proportional to output, then costs are given by unit cost times output: $UC \times Q$, where UC is unit cost and Q is output. Revenues, in turn, are given by price times quantity: $R = P \times Q$, where P is price. It follows that $r = (P - UC)/UC$, so r is a good measure of the gap between price and unit cost (which in this case is also equal to marginal cost).

d. The theory of **contestable markets** formalizes this argument.[6]

■ **How do firms acquire and maintain market power?** Market power translates into higher profits. Creating and maintaining market power is therefore an important part of a firm's value-maximization strategy.

How do firms acquire market power? One possibility is to be legally protected from competition, so that high prices can be set without new competitors entering the market. For example, in the 1940s and 1950s Xerox developed the technology of plain-paper photocopying, and patented it. Given the legal protection provided by Xerox's patents, it could raise prices to a significant level without attracting competition (cf Box 15.1).

Firm strategy may also play an important role in establishing market power. Take, for example, the case of the British Sky Broadcasting Group (BSkyB), a leading firm in the British digital TV market (formed in 1990 by the merger of Sky Television and British Satellite Broadcasting). Attempting to preempt the competition, in 1999 BSkyB introduced an aggressive package that included a free set-top decoder box, free Internet access, and a 40% discount on telephone charges.[9] The plan, which was largely successful, was to create an early lead in installed base of subscribers, an early lead that would hopefully become permanent. A more recent example is given by Samsung Electronics. Attempting to break into the lucrative smartphone market, Samsung sold their iPhone-like handsets at significantly lower prices than Apple's; and took a bet by being among the first to use Google's new Android operating system. By 2012, Samsung had already captured a third of the smartphone market.[10]

Creating market power is only one part of the story. A successful firm must also be able to maintain market power. Patents expire. Imitation takes place. Protected industries are deregulated. What can incumbents do in order to maintain their position? The airline industry provides an example. American Airlines managed to drive out various competitors who attempted to enter into its Dallas/Forth Worth hub: Vanguard, Sun Jet, Western Pacific. Fares on the route between Dallas and Kansas City, for example, fell from $108 to $80 when Vanguard entered the market. After Vanguard exited, American gradually raised fares up to $147. Joel Klein, then head of the Antitrust Division at the Justice Department, said that American's strategy achieved more than just driving current rivals out of the market — it also sent a clear signal to potential future entrants: "A sophisticated economist compared it to choosing between two fields with 'no trespassing' signs. One has two dead bodies in it, the other has no dead bodies in it. Which field would you feel ready to trespass?"[11] Reputation for toughness is a reliable means of maintaining a position of market power.

A more recent example from the high-tech world is given by Apple Computer. By a combination of constant innovation and clever marketing, the Cupertino giant has managed to maintain a strong market position for a long time. Particularly important is Apple's "ecosystem" of devices and software: "its phones, tablets, computers, and the mobile and desktop operating systems that run them are blending into a single, inseparable whole."[12]

In different chapters of this text, especially in Chapters 11–16, I will examine a large set of strategies that firms may deploy in order to create and maintain their market power.

■ **WHAT ARE THE IMPLICATIONS OF MARKET POWER?** From the firm's point of view, market power implies greater profits and greater firm value. From a social welfare point of view — or from a policymaker point of view, if we believe policymakers pursue the collective good — the implications are more complicated.

The first-order effect of a high price is a transfer from consumers to firms:[e] for each extra dollar in price, each buyer is transferring one extra dollar to the seller. If regulators put a greater weight on consumer welfare than on profits, then this transfer should be seen as a negative outcome. In fact, antitrust and competition policies are to a great extent motivated by the goal of protecting consumers from these transfers (see the next question).[f]

But, in addition to a transfer effect, a high price also implies an inefficient allocation of resources. High airfares, for example, mean that there are potential fliers who refrain from buying tickets even though the cost of carrying them as passengers would be very low. From a social point of view, it would be efficient to fly many of these potential travelers: although the value they derive from flying is lower than the price (hence they don't fly), that value is greater than the cost of flying (which is much lower than price). The loss that results from the absence of these sales is the **allocative inefficiency** implied by market power.[g]

"The best of all monopoly profits is the quiet life:"[13] A monopolist does not need to be bothered with competition. More generally, firms with greater market power have less incentives to be cost efficient, one may argue. For example, for many years European airlines were known to be less efficient than North-American airlines. To a great extent, this efficiency gap resulted from the more intense competition in the North-American market.[h] In other words, market power implies a second type of inefficiency: **productive inefficiency**, which we define as the increase in cost that results from market power.[i]

When market power is artificially maintained by government intervention, a third type of inefficiency may result: rent seeking. By **rent seeking** I mean unproductive resources spent by firms in an attempt to influence policymakers. Consider, for example, the following news article regarding AT&T's effort to maintain its position in the cable television market:

> This summer, AT&T Corp. faced the specter of cities around the country requiring it to open its cable television lines to rival Internet companies . . . The threat never really materialized. Why not? It depends on whom you ask.
>
> AT&T attributes its success to its ability to explain the issues to local officials . . . [Others have a different opinion:] "It comes down to bribery or threats," says Greg Simon, co-director of Opennet Coalition, a group of companies that has launched its own lobbying effort to promote open access.[14]

Another example of large amounts of resources spent in an attempt to influence decisionmakers is the 1998 US v. Microsoft case. Netscape, Sun Microsystems, and Microsoft

e. By "first-order" I mean the effect that is quantitatively most significant.

f. An alternative perspective on antitrust and competition policy is that it serves to protect the interests of firms. See Section 5.4.

g. A rigorous definition of this concept is given in Section 4.4.

h. Since European airline deregulation in the 1990s the situation has changed considerably.

i. Again, we defer the more precise definition to a later chapter. The discussion of the above hypothesis (market power leads to productive inefficiency) can be found in Chapter 5.

itself would not have spent the vast amounts they did spend if the operating system industry were not as profitable as it is; thus the idea that rent seeking is a consequence of market power.

Along similar lines, a more recent example is provided by Amazon's effort to maintain its dominance in the e-book retail market. When its business strategy was challenged by Apple's "agency model," Amazon approached the US Department of Justice (DOJ) with extensive legal arguments that the DOJ later used to sue Apple for anticompetitive practices.[15,j]

To summarize, the above paragraphs support the view that market power, good as it might be for firms, is bad for society. First, it makes the owners of firms richer at the expense of consumers. Second, it decreases economic efficiency (allocative and productive efficiency). Third, it induces firms to waste resources in order to achieve and maintain market power. However, from a *dynamic* point of view, an argument can be made in favor of market power:

> As soon as we go into the details and inquire into the individual items in which progress was most conspicuous, the trail leads not to the doors of those firms that work under conditions of comparatively free competition but precisely to the doors of the large concerns.[16]

This argument is one of the central points of the Austrian school, led by its greatest exponent, J. Schumpeter, author of the above quotation. I will examine it in greater detail in Section 15.1. Like the Chicago school, the Austrian school has a very clear position when it comes to market power. However, whereas a Chicago economist would argue that market power does not exist, a Schumpeterian would rather say that market power exists — and it's a good thing that it does, for market power is a precondition for innovation and progress.

■ **IS THERE A ROLE FOR PUBLIC POLICY AS REGARDS MARKET POWER?** In the context of industrial organization, the primary role of public policy is to avoid the negative consequences of market power. Public policy in this area can be broadly divided into two categories: regulation and antitrust (or competition policy).[k] Regulation (economic regulation) refers to the case when a firm retains monopoly or near-monopoly power and its actions (e.g., the price it sets) are directly under a regulator's oversight. For example, ConEdison, a US electricity and gas supplier, needs regulatory approval in order to change its rates.

Antitrust policy (or competition policy) is a much broader field. The idea is to prevent firms from taking actions that increase market power in a detrimental way. A couple of examples may help.

- For the past two decades, Mars and Unilever have engaged in a series of legal cases in European courts. The issue is the legality of Unilever's exclusivity policies regarding retail ice cream sales. In many European countries, Unilever imposes fridge exclusivity: if a store accepts a fridge paid by Unilever, then

j. I will return to this case in Section 9.5.

k. The terminology "antitrust" is more common in the US, whereas "competition policy" is the corresponding European term; see Section 5.5.

the store can only use the fridge to stock Unilever products. Mars claims that exclusivity effectively makes it impossible for Mars to sell Snickers ice cream and related products as most stores have no space for more than one fridge. Unilever responds that it's their fridges and that they require return on an expensive investment. Different but similar cases in various European countries have come to different conclusions, sometimes favorable to Unilever, sometimes favorable to Mars (see also Section 12.1).

- In March 2011, AT&T announced that it planned to purchase T-Mobile USA, a smaller wireless operator. Five months later, the US Department of Justice (DOJ) formally announced that it would seek to block the takeover, arguing that it would increase market power substantially. At first AT&T gave signs that it would defy DOJ's decision, but eventually the bid was abandoned. Although the merger might have brought some benefits to consumers, competition between the would-be merging parties has also been a positive force. For example, in 2013 T-Mobile USA announced that it would pay contract cancellation fees of AT&T subscribers who wanted to switch to T-Mobile.

The above two examples provide an idea of the variety of situations that may fall under the scope of public policy. The overall rationale is to prevent and remedy situations where market power may reach unreasonable levels, to the detriment of society — consumers in particular. Over the course of the next chapters, we will examine several other areas for policy intervention motivated by the goal of curbing market power.

As was stated earlier, the Chicago school takes a very different approach. The claim is that, in a world of free competition, market power is never very significant. In fact, the few situations where market power does exist result precisely from government intervention. In other words, the Chicago school reverses the order of causation: it's not that market power prompts government intervention but the exact opposite — government intervention creates market power, protecting the interests of firms and not those of consumers. As Milton Friedman, a leader of the Chicago school, put it in the late 1990s:

> Because we all believed in competition 50 years ago, we were generally in favor of antitrust. We've gradually come to the conclusion that, on the whole, it does more harm than good. [Antitrust laws] tend to become prey to the special interests. Right now, who is promoting the Microsoft case? It is their competitors, Sun Microsystems and Netscape.[17]

To summarize,

> The central questions addressed by industrial organization are: (i) Is there market power? (ii) How do firms acquire and maintain market power? (iii) What are the implications of market power? (iv) Is there a role for public policy with regard to market power?

■ **Industrial policy.** In addition to regulation and antitrust (or competition policy), some countries have followed policies that are intended to promote particular firms or groups of firms. Of particular importance is **industrial policy**. The goal of industrial policy is very different from regulation and antitrust. Whereas the latter attempt to promote competition, the former is geared towards strengthening the market position of a firm or industry, namely with respect to foreign firms. For example, much of the success of Airbus Industrie, a consortium backed by four European countries, is the result of the support it has received from the respective governments over the past three or four decades. Starting from a market share of less than 10% in the 1970s, Airbus is now fighting head to head with Boeing, the industry's main competitor.

Industrial policy is generally not favored by economists. In practice, it amounts to governments picking winners among a number of potential firms and industries. But why should governments know better than the market who the promising firms and industries are? A frequent argument in support of industrial policy is the example of MITI, the Japanese Ministry of Industry and Foreign Trade. True, the prowess of the Japanese export sector is a success story and owes a great deal to the role played by MITI. For example, MITI's support was an important factor in the emergence of Japan as a leader in semiconductors. But together with the success stories there are also a fair number of flops: for example, the 1980s project to develop a "fifth generation computer," which would leapfrog the American counterparts, lead to very poor results.[18]

Even in the US — the most pro-market western economy — we find examples of failed industrial policy. Recently, the state of Rhode Island approved $75 million in loans for a fledgling video game company led by a former Major League Baseball star pitcher. The ill-fated company went bankrupt two years later and managed to rack up $150 million in debt in the process.[19]

In sum, although there are success stories (Airbus?), the overall record of governments meddling in business strategy is at best poor. For these reasons, and as a matter of consistency, when talking about public policy I will restrict attention to regulation and antitrust (or competition policy).

1.3 COMING NEXT...

There are 15 chapters to come, divided into four different parts. Part I is introductory in nature. It provides basic tools required for the study of IO (consumer behavior in Chapter 2, firm behavior in Chapter 3); and covers the extreme situations when markets are efficient (Chapter 4) and when they are not (Chapter 5). I conclude Part I by discussing advanced pricing strategies, though still in a context where strategic interaction is absent (Chapter 6). For readers with a background in the field of microeconomics, some of the material treated in Chapters 2–5 may be familiar and can be skipped without much loss.

Insofar as industrial organization is the study of imperfect competition, Parts II through IV make up the core of the text. Within these, Part II plays a central role, as it introduces the basic theory of oligopoly competition. I begin with an introduction to game theory (Chapter 7), an essential tool for studying strategic behavior; and then cover static models (Chapter 8) and dynamic models (Chapter 9) of oligopoly interaction.

Throughout most of the text, I assume a *given* industry structure. Part III takes one step back and examines the endogenous determinants of industry structure. I begin by looking at how technology and demand conditions influence market structure (Chapter 10), and then move on to examine the role played by mergers and acquisitions (Chapter 11) and firm strategy (Chapter 12).

Part IV extends the analysis by considering firm strategies beyond the simple pricing and output decisions examined in Parts II and III. These include vertical relations (Chapter 13); product differentiation and advertising (Chapter 14); and innovation (Chapter 15). I conclude with a chapter on networks, a phenomenon of increasing importance in the "new economy" (Chapter 16).

■ **A NOTE ON METHODOLOGY.** Many economists analyze industries with reference to a framework known as the **structure-conduct-performance (SCP) paradigm**.[20] First, one looks at the aspects that characterize market structure: the number of buyers and sellers, the degree of product differentiation, and so forth. Second, one pays attention to the typical conduct of firms in the industry: pricing, product positioning and advertising, and so forth. Finally, one attempts to estimate how competitive and efficient the industry is.

Underlying this system is the belief that there is a causal chain linking the above components: market structure determines firm conduct, which in turn determines industry and firm performance. For example, in an industry with very few competitors, each firm is more likely to increase prices or collude with its rivals. And higher prices have the performance implications we saw in the previous section.

Causality also works in the reverse direction. For example, a firm that does not perform well exits the market, so performance influences market structure. Likewise, a firm may price very low in order to drive a rival out of the market, an instance where conduct influences structure. Finally, government intervention and basic demand and supply conditions also influence the different components of the SCP paradigm.

In Chapter 10 I examine the relation between the different components in the structure-conduct-performance paradigm. However, most of the text centers on the analysis of firm conduct and how it influences firm and industry performance as well as market structure.[1]

SUMMARY

• Industrial organization is concerned with the workings of markets and industries, in particular the way firms compete with each other. • The central questions addressed by industrial organization are: (i) Is there market power? (ii) How do firms acquire and

1. It should be clear that the SCP paradigm is not a model that directly provides answers to the questions listed in the previous section. It is best thought of as a guide that allows one to analyze and understand the workings of different industries. Alternative frameworks have been proposed for the same or similar purposes. Examples include Michael Porter's **five-forces framework** for the analysis of industry competition. The five forces are: suppliers, buyers, substitute products, potential entrants, and competition between incumbent firms.[21]

maintain market power? (iii) What are the implications of market power? (iv) Is there a role for public policy with regard to market power?

Key concepts

• market power • contestable markets • allocative inefficiency • productive inefficiency • rent seeking • industrial policy • structure-conduct-performance (SCP) paradigm • five-forces framework

Review and practice exercises

■ **1.1. Competition and performance.** Empirical evidence from a sample of more than 600 UK firms indicates that, controlling for the quantity of inputs (that is, taking into account the quantity of inputs), firm output is increasing in the number of competitors and decreasing in market share and industry concentration.[22] How do these results relate to the ideas presented in this chapter?

Notes

1. Stigler, George J. (1969), *The Organization of Industry*, Homewood, Illinois: R D Irwin, p. 1.

2. *The People's Pharmacy* (http://homearts.com/depts/health/kfpeop18.htm).

3. Petersen, Melody (2008), *Our Daily Meds: How the Pharmaceutical Companies Transformed Themselves into Slick Marketing Machines and Hooked the Nation on Prescription Drugs*, New York: Sarah Crichton Books.

4. *The Scientist,* Vol. 9, No. 14, p. 3, July 10, 1995.

5. Harberger, Arnold C. (1954), "Monopoly and Resource Allocation," *American Economic Review* **44**, 77–87.

6. Baumol, William, John Panzar, and Robert Willig (1982), *Contestable Markets and the Theory of Industry Structure*, New York: Harcourt Brace Jovanovich.

7. Hall, Robert E. (1988), "The Relationship Between Price and Marginal Cost in US Industry," *Journal of Political Economy* **96**, 921–947.

8. *The Wall Street Journal Europe*, November 14, 1996.

9. *The Wall Street Journal Europe*, May 6, 1999.

10. *The Economist*, September 17, 2014.

11. *Financial Times*, 24 May 1999.

12. "Apple Strengthens the Pull of Its Orbit With Each Device," *The New York Times,* October 23, 2014.

13. Hicks, John (1935), "Annual Survey of Economic Theory: The Theory of Monopoly," *Econometrica* **3**, 1–20.

14. *The Wall Street Journal*, November 24, 1999.

15. *The Wall Street Journal*, September 11, 2014.

it obvious that firms are different?"). Hopefully, by the end of the chapter you will agree that the time reading it was well spent; and that the economics' assumption of a production-function-based profit-maximizing firm is not so bad after all.

3.1 THE FIRM'S PRODUCTION, COST, AND SUPPLY FUNCTIONS

At the risk of oversimplifying, we can think of a firm as a process of transforming inputs into outputs. This is easier to see for a firm that makes actual things. For example, a bagel bakery uses water, flour, and other ingredients, together with machinery (an oven) and labor (someone has to put it all together), to produce tasty bagels. Firms that offer services also go through a similar process. For example, a consulting firm uses hours of labor — many, many hours, I'm told — together with some capital (mainly laptop computers) and materials (paper and paper clips), to produce solid advice to corporations that need it.

The firm's **production function** is the mapping that tells us, for a given set of inputs, how much output a firm is able to produce. Normally, this depends on the particular firm, as some firms are more efficient than others at transforming inputs into outputs. It also depends on the quality of inputs, for example skilled versus unskilled labor.

Formally, we denote the production function by $f(x_1, ...x_n)$, where x_i the quantity of input i. For the purpose of this chapter, we consider two inputs: capital and labor. This is not to say that other inputs are not relevant: no matter how many ovens and oven operators you have, you cannot make bagels without flour. It's just that, for the purposes of illustrating the main principles, it suffices to consider two inputs. Moreover, in many examples (e.g., consulting services) these are indeed the main inputs into production (in other words, paper and paper clips are a small fraction of the consulting firm's operations).

What does the production function f look like? One way to answer this question is to answer a related question: what combinations of inputs lead to a given level of output? The answer is given by the **isoquant** curves, which look a lot like the indifference curves introduced in the previous chapter.

Figure 3.1 shows two possible extremes regarding isoquant curves. On the left, we have the case of a production process that operates on fixed proportions. For example, Air France uses planes and pilots to produce transportation services. Each plane requires a pilot (for simplicity, I ignore co-pilots in the present analysis). Having more pilots than planes does not allow Air France to carry more passengers (assuming the planes are already used all the time). Likewise, having more planes than pilots doesn't help either (until Google or someone else comes up with a pilotless plane).

Fixed-proportion production processes lead to right-angle isoquants, as shown on the left panel of Figure 3.1: with one plane and one pilot the firm produces one unit

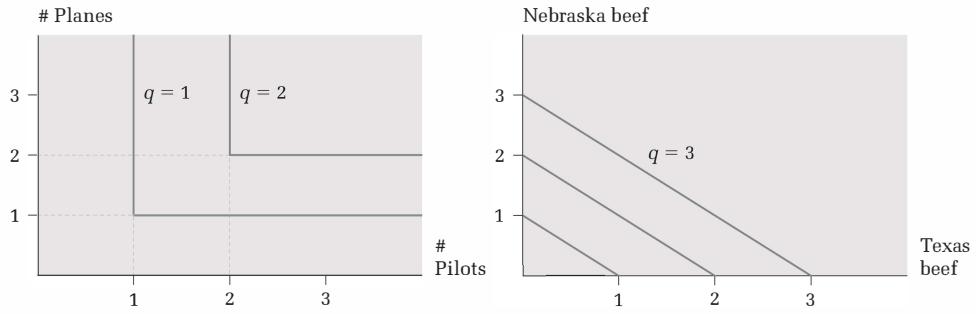

FIGURE 3.1
Production functions: two extreme cases

of output (if the plane carries 100 passengers, then let the output unit be hundreds of passengers). Given one plane, having more pilots does not help: total output is still one unit, and so all of these points belong to the same isoquant. With two planes and two pilots Air France can produce two units of output. More than two pilots given two planes or more than two planes given two pilots still leaves us at two units of output. More generally, the production function corresponding to fixed proportions is given by $f(x_1, x_2) = \min\{x_1, x_2\}$. We say that in this case inputs are **perfect complements**.

The right panel of Figure 3.1 shows the opposite extreme case: **perfect substitutes**. Suppose that McDonalds uses both Texas and Nebraska beef as input into its burgers. Moreover, for the purpose of this example, suppose that beef is the only ingredient into burgers (I am aware this is a strong assumption, but please bear with me). At the risk of offending the great states of Texas and Nebraska, it seems reasonable to assume that the quality of beef in these two states is similar. Therefore, the quantity and quality of McDonalds burgers depends on the total quantity of beef, not on the particular proportions of beef from Texas or Nebraska.

Perfect substitute inputs lead to straight isoquants, as in the right panel of Figure 3.1. One unit of output (burger) can be obtained with one unit of Texas beef and zero units of Nebraska beef; or one unit of Nebraska beef and zero units of Texas beef; or any combination therein.

Finally, the left panel in Figure 3.2 shows an intermediate case, that is, a case when inputs are neither perfect complements nor perfect substitutes. The particular isoquants in this figure are derived from a particular production function, which is known as the Cobb-Douglas production function and provides a good fit to the data in many settings. If capital (K) and labor (L) are the two main inputs in the production process, then the **Cobb-Douglas production function** is given by:

$$q = K^{\alpha} L^{\beta}$$

In the particular example depicted in Figure 3.2, I chose parameter values $\alpha = \frac{1}{2}$ and $\beta = \frac{1}{2}$ and plotted three different isoquants, corresponding to output levels 1, 2 and 3. Here's an interesting feature about these isoquants: for a given output level (e.g., $q = 4$), if I use less L then I need to use more K to compensate for the decrease in L;

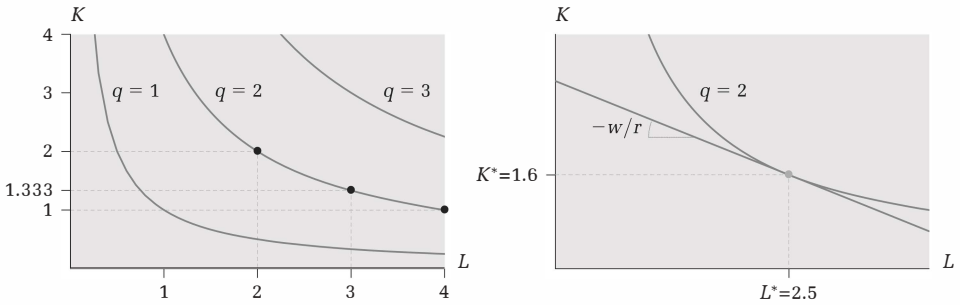

FIGURE 3.2
Isoquants and cost minimization

and the further I decrease L the greater the increase in K required to compensate for the decline in L. This latter property reflects the law of **diminishing marginal returns**, a feature that we saw in the context of consumer utility. It also works in terms of increases of L. Consider for example the isoquant corresponding to $q = 2$. One possible way of achieving this output level is to employ two units of K and two units of L. By increasing L to 3, I am able to drop K from 2 to 1.333, a drop of .666. If I further increase L from 3 to 4 (the same increment as before), I can now decrease K from 1.333 to 1, a drop of .333. The fact that .333 is lower than .666 is just another case of diminishing marginal returns at work.

Based on the comparison of the two panels in Figure 3.1 and based on the preceding analysis, we conclude that:

> Reflecting decreasing marginal returns, isoquants are convex curves; the closer complements two inputs are, the more convex the corresponding isoquants are.

Understanding the degree of complementarity or substitutability between two different inputs is important for a variety of reasons. For example, in the past few decades desktop computers have become a common feature in our daily lives. What impact has this had on the firms' demand for labor? One way to rephrase this question is: if we increase the quantity of one particular input (IT, specifically desktop computers) what impact does this have on the demand for labor? As often is the case in economics, the answer is: it depends. In this case, it depends on the type of labor. For example, at university departments the number of secretaries has declined considerably. I typed this book and I type all of my papers directly on my computer. Had I been born 20 or 30 years earlier, I would probably have written a manuscript and asked someone in the "typing pool" to turn it into a typescript. In sum, we may say that secretarial services and desktop computers are largely substitute inputs. But there are other types of labor that are complements with respect to IT investments, including desktop computers. The most obvious one is IT specialists: more desktops in the organization means more desktop problems; more

desktop problems require more people to solve those problems. In sum, we may say that IT specialists and desktop computers are largely complement inputs.

■ **PRODUCTIVITY.** The term **productivity** is used frequently to describe a firm's performance. Unfortunately, it means different things to different people and as a result may be a source of confusion. Some, for example, measure average labor productivity, which is given by $q\,p/L$, or revenue per worker (q is number of units produced, p is price per unit, and L is number of workers). However, this may not be the best measure of firm performance. Consider two firms, both with the same production function. If the first firm has a higher level of capital than the second firm, then it will also have a higher level of labor productivity — though we might argue that, having the same production function, the two firms are equally productive.

In this sense, a better measure of performance would be that of **total factor productivity** (TFP). Suppose that: (a) market price is given (and is the same for all firms in the industry); (b) each firm has a Cobb-Douglas production function with inputs K and L; (c) all firms have the same α and β coefficients but differ with respect to a multiplying coefficient:

$$q_i = \xi_i K_i^{\alpha} L_i^{\beta} \tag{3.1}$$

The coefficient ω_i would then be a better measure of a firm's performance, namely its total factor productivity. In fact, given two firms using the same quantity of inputs, the firm with a higher ω_i is able to produce a higher output level.

ESTIMATING TOTAL FACTOR PRODUCTIVITY. Taking logarithms on both sides of the production function (3.1), we get:

$$\ln q_i = \ln \omega_i + \alpha \ln K_i + \beta \ln L_i$$

or simply

$$\ln \omega_i = \ln q_i - \alpha \ln K_i - \beta \ln L_i \tag{3.2}$$

It can be shown (cf Exercise 3.15) that a profit maximizing firm chooses input levels such that the ratio between input costs (e.g., $w\,L$ in labor) and total revenue (i.e., $p\,q$) is equal to the respective coefficient in a Cobb-Douglas function (e.g., β for labor):

$$\alpha = \frac{r\,K}{p\,q}$$

$$\beta = \frac{w\,L}{p\,q}$$

The values of r, w, K, L, p, q can all be obtained from observable data. Together with the estimates of α and β, (3.2) gives us an estimate of the each firm's total factor productivity.

■ **COST MINIMIZATION.** The parallel between producer theory and consumer theory is more general than the point I made earlier. In consumer theory, we have a consumer who transforms consumption into utility — a kind of production function. The consumer's

problem we solved was to maximize utility given a certain income level and given prices for the various goods. In production theory, we ask a similar question: given a desired output level and given input prices, we determine the input mix that minimizes cost.

This problem is shown in the right panel of Figure 3.2. Suppose we want to achieve $q = 4$. Suppose moreover that the cost of capital is $r = 12.5$, whereas the cost of labor is $w = 8$. Then it can be shown (cf Exercise 3.16) that the optimal solution is $K = 1.6$ and $L = 2.5$, as shown in the figure. Graphically, the problem of cost minimization (right panel of Figure 3.2) is similar to the problem of utility maximization considered in Chapter 2. Consumers seek the highest utility level consistent with a certain budget set; firms seek the lowest cost consistent with a certain output level. In both cases, the solution is given by a point of tangency: in the case of consumers, the optimal solution corresponds to a point where the indifference curve is tangent to the budget line (which, as you will recall, has a slope equal to -1 times the ratio between the prices of the two goods in question); in the case of firms, the optimal solution corresponds to a point where the isoquant is tangent to a line with slope equal to -1 times the ratio of the two inputs' prices.

If the cost of capital is $r = 12.5$ and the cost of labor is $w = 8$; if the best input mix to produce two units of output is $K = 1.6$ and $L = 2.5$; then the lowest cost the firm needs to incur in order to achieve $q = 2$ is $12.5 \times 1.6 + 8 \times 2.5 = 40$. More generally, for a given set of input prices r, w and for a given output level q, we can follow the same steps and obtain the minimum cost required to achieve q. In this particular example we would get $C(q) = 20\,q$ (cf Exercise 3.16), an expression we denote the cost function. The reason we get such a simple expression is that we are considering a simple production function and, more important, we are assuming that both K and L can be adjusted at will, an assumption that does not always hold true. In general, things can get a little messier, as we will see later.

Before doing so, I should mention that, just as consumers have demand functions for products, firms have demand curves for inputs; and just as consumer demand can be characterized by its direct and cross price elasticities, so the firm's demand for inputs can be more or less elastic with respect to its price and the price of other inputs. In particular, the cross-price elasticity of the demand for inputs depends on the shape of the production function. If the production function is close to the extreme of perfect complements (left panel in Figure 3.1), then an increase in the cost of capital, for example, leads to a lower demand for capital *and* a lower demand for labor (since the quantities of capital and labor move hand in hand); that is, an increase in the cost of capital leads to a lower demand for labor: a negative cross-price elasticity, which as we saw in Chapter 2 denotes complements.

By contrast, if the production function is close to the perfect substitutes case (right panel in Figure 3.1), then depending on the price ratio of inputs the firm either only uses capital or only uses labor.[b] An increase in the cost of capital, for example, either keeps the demand for labor at zero or, if the ratio $-w/r$ "crosses over" the slope of the isoquant, switches all of the capital demand into labor demand; that is, an increase in the price of capital leads to a higher demand for labor: a positive cross-price elasticity, which as we saw in Chapter 2 denotes substitutes.

b. If the price ratio is exactly equal to the negative of the slope of the isoquants, then the firm is indifferent regarding the relative choice of capital and labor.

To see how this works in practice, consider two different industries (building hydroelectric dams and building aircraft) and two different countries (US and India). One important difference between the US and India is that w/r is considerably lower in India, mainly because labor costs are lower. In order to dig a dam, one can use many possible combinations of labor and machines (that is, we are closer to the perfect substitutes case in Figure 3.1). To the extent that these are close substitutes, in India the industry is very labor intensive (that is, the ratio K/L is very low), whereas, in the US, the industry is very capital intensive (that is, the ratio K/L is very high). By contrast, in order to build an aircraft, certain amounts of skilled labor *and* machinery are required (that is, we are closer to the perfect complements case in Figure 3.1). For this reason, the K/L ratio is relatively similar in the US and India even though the w/r ratio is much lower in India.

■ **COST FUNCTION.** As I mentioned earlier, the firm's **cost function**, typically denoted by $C(q)$, shows the least total cost of inputs the firm needs to pay in order to produce output q; that is, the cost of producing q assuming the firm does so efficiently. The cost function $C(q)$ leads to a series of related cost concepts:

- **Fixed Cost** (FC). The cost that does not depend on the output level.
- **Variable Cost** (VC). That cost which would be zero if the output level were zero.
- **Total Cost** (TC). The sum of fixed cost and variable cost.
- **Average Cost** (AC) (also known as "unit cost"). Total cost divided by output level.
- **Marginal Cost** (MC). The cost of one additional unit. In other words, the total cost of producing $q + 1$ units minus the total cost of producing q units of output. (Strictly speaking, this is the definition of incremental cost. The rigorous definition of marginal cost is the derivative of total cost with respect to the output level.)

To illustrate all of these cost concepts, let us consider a very simple example, that of a small, price-taking T-Shirt Factory. In order to produce T-shirts, a manager leases one machine at the rate of $20 per week. The machine must be operated by one worker. The hourly wage paid to that worker is as follows: $1 during weekdays (up to 40 hours), $2 on Saturdays (up to 8 hours), and $3 on Sundays (up to 8 hours). Finally, the machine — which is operated by the worker — produces one T-shirt per hour.

Assuming that current output (q) is 40 T-shirts per week, we have that:

- The fixed cost is given by the machine weekly lease. We thus have FC = $20.
- The variable cost is given by 40 T-shirts times one hour per T-shirt times $1 per hour, which equals $40.
- The average cost is (20+40)/40 = $1.50.

- The marginal cost is \$2. In fact, producing the 41st T-shirt in a given week would imply asking the worker to work on Saturday, which would be paid at the hourly rate \$2; and producing a T-shirt requires one hour of work.

These cost values were computed for a particular output level. However, both average cost and marginal cost depend on the output level. By computing the values of marginal cost and average cost for each output level, we get the marginal cost and average cost functions. Figure 3.3 depicts these functions for the particular case of the T-shirt factory. The more general case is given by the right panel in Figure 3.3.

What is the use of all of these cost concepts? Consider the following application. Suppose that Benetton, the sole buyer of T-shirts from our small factory, is offering a price of $p = \$1.80$ per T-shirt. Moreover, Benetton is willing to buy as many T-shirts as the factory wants to sell at that price (that is, the T-shirt factory is a price-taking firm). Given this offer, should the factory operate on Saturday?

At the current output of $q = 40$ T-shirts a week, average cost is given by \$1.50 (see above). This means that, at $p = \$1.80$, the factory is making money. It might seem that, for this reason, it is worth it to operate on Saturdays as well: "If you are making money at the current output level, produce more and you will make more money." As it turns out, this is wrong. What is relevant for the decision of whether or not to operate on Saturdays is the comparison between price and *marginal* cost, not the comparison between price and average cost. And since marginal cost of operating on Saturdays is \$2, it is not worth pursuing that strategy given that the selling price is only \$1.80.

In other words, although the factory is making money at output level $q = 40$ (because price is greater than average cost), profits would be lower if output were increased (because price is lower than marginal cost); the factory would lose money at the margin. (By "lose money at the margin" I mean lose money by producing an additional — a marginal — unit of output.)

Suppose now that Benetton (still the sole buyer) offers a price $p = \$1.30$ per T-shirt. No matter what the output level is, price falls below average cost. (Check this.) That is, no matter how much the factory produces, it will lose money. In fact,

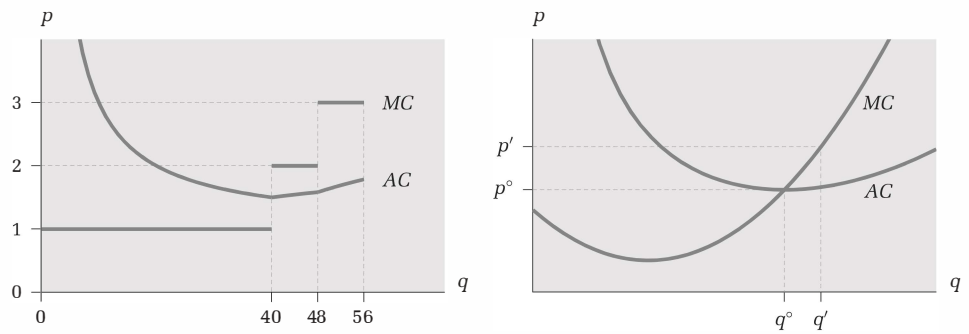

FIGURE 3.3
Marginal cost and average cost: T-shirt factory example (left) and general case (right)

take place, it is not necessary for takeovers to actually take place — it is sufficient for the *threat* of takeover to be in place.

In summary,

> Although management and ownership are normally separated, there are reasons to believe that deviations from profit maximization cannot be too large. These reasons include management incentive contracts, labor market discipline, product market discipline, and capital market discipline.

The precise meaning of "not too large," that is, the extent to which profit maximization is a good approximation, remains an unresolved empirical question.

3.4 WHAT DETERMINES A FIRM'S BOUNDARIES?

Why should firms be of the size they are; why not smaller or bigger? What does economic analysis have to say about firm size? It may be useful to divide this into two questions: (a) what determines the horizontal extension of the firm and (b) what determines the degree of vertical integration. By horizontal extension I mean how much of a given product does a firm produce and how many different products it offers. By vertical integration, I mean how many stages of the production process take place within the firm. So, for example, by acquiring Skoda, SEAT, and Bentley, Volkswagen increased its horizontal stretch. Volkswagen would increase the degree of vertical integration if it were to acquire a tire manufacturer, for example (tires being one of the inputs into car manufacturing).

The horizontal size of the firm is largely determined by costs. If average cost is U shaped (cf Figure 3.3) and there is free entry into the industry, then firms will tend to produce at the level where average cost is minimized (cf Section 4.1). For example, there is an optimal size for a cement plant that minimizes cost. Plants of much smaller size or much larger size would probably incur a higher average cost and be unable to survive for very long. However, empirical evidence suggests average cost functions are U shaped with a flat bottom (that is, "saucer" shaped). This implies that there is a range of output levels which attain the minimum average cost. In other words, costs may not entirely allow us to pin down the size of the firm.

The problem becomes more complicated if we consider the distinction between plant and firm. Production costs are related to plant operation. Suppose there is a unique output level which minimizes plant average costs and that a given firm owns two plants. It is possible that the firm's average cost is similar to the average cost of a firm owning one plant only. If so, then multiplant firms create a new dimension of indeterminacy of firm size. This indeterminacy may be resolved if we take managerial costs into consideration: very large conglomerates may simply be too large to be managed efficiently. This brings us back to the idea of U-shaped cost curves, with the difference that the independent variable is now number of plants rather than output level.

■ **THE VERTICAL BOUNDARIES OF THE FIRM.**[5] Perhaps a more interesting question regarding the boundaries of the firm is: why do we observe a great degree of vertical integration in some industries and very little in others? One of the most important decisions that a firm has to make is how to obtain its inputs: to make or to buy. In other words, to use the market (vertical separation) or to use the firm (vertical integration). The degree of vertical integration in a given industry results from the aggregation of these micro decisions at each stage of the production process. Understanding the nature of the make-or-buy decision contributes to understanding the nature of the firm as well as one the determinants of industry structure.

Consider the example of the US coal industry. About 80% of the coal is used for electricity generation. Since the transportation costs from mine to power generation plant are high, it pays to locate power plants near coal mines. Consider the decision of building a power plant near a coal mine. Since there are no other nearby sources of coal, we refer to such an investment as a **specific asset**: the power plant is worth a lot less if not coupled with this particular source of supply. For this reason, if the power plant owner is different from the coal mine owner, the relation between seller and buyer is subject to a **hold-up problem**: once the buyer pays for the relationship-specific asset, the seller can charge a higher price. One possible solution to this problem is vertical integration. In the US, mine-mouth power plants and the respective coal mine are typically owned by the same entity; and when they are not, the contractual relationship between seller and buyer is typically governed by long-term contracts.[6]

But vertical integration does not solve all of the incentive problems. If I buy from an external supplier, that supplier has strong incentives to perform: it knows that underperformance easily leads to supplier switching. If I buy the external supplier and make it part of my organization, that supplier (formerly external, currently internal) now has weaker incentives, for the threat of switching to another supplier is likely smaller.

Both the extremes of complete vertical integration and complete vertical separation imply incentive problems. Sometimes, the optimal solution may lie between the extremes. One possibility is that of **tapered integration**, whereby a given input is bought from an affiliated supplier *and* from an independent one. Examples of tapered integration include soft-drink bottling for the Coca-Cola and Pepsi-Cola companies; and crude oil supply to oil refineries. A second intermediate system is that of **franchising**, a system that has been used in a variety of industries, from fast-food (e.g., McDonald's) to designer clothing (e.g., Stefanel). Franchising combines the benefits of vertical integration (specific investments are paid by the mother company) with the benefits of vertical separation (franchisees retain most of the profit they generate, and thus have strong incentives to be efficient). Finally, the Japanese system of supplier contracting also corresponds to an intermediate solution: the firm and its suppliers establish a long-term, informal relationship that falls short of full-scale vertical integration but goes a long way towards providing suppliers with the right incentives to invest in assets specific to the relationship with their customers.

In other words, the definition of the boundaries of the firm is far from being a well-defined problem with a well-defined answer. Consider, for example, the Japanese *keiretsu*, that is, a family of firms, typically from related industries, with cross shareholdings. Should a keiretsu be considered a firm in and of itself? Probably not — but neither is it a simple collection of independent firms. Moreover, it is not uncommon to find two otherwise similar firms that differ significantly in the way they are organized. For example, the Italian clothes manufacturer Benetton depends largely on a franchise system for its retail sales, whereas Spanish rival Zara is wholy vertically integrated. Examples like these suggest that there may be other determinants of firm structure that lead some firms to choose differently from others.

While the problems of firm organization are extremely complex, both in theory and in practice, it seems safe to summarize that:

> The horizontal boundaries of the firm are largely determined by cost considerations (including managerial costs). The vertical boundaries result from the balance between investment incentives (specific assets) and performance incentives.

3.5 WHY ARE FIRMS DIFFERENT?

Casual observation of the real world reveals that firms are different from each other: different in size, different in scope, and so forth. In particular, we observe that firm performance (e.g., profit rate) varies enormously across firms. This is not entirely surprising if we consider that different firms may be of very different size or belong to different industries. Why should we expect the profit rate of Boeing to be the same as that of a coffee shop?

Empirical observation suggests, however, that firms in the same industry and of similar dimension also perform very differently. Only 20% of the variance in firm profit rates can be explained by variables that relate to firm size, the type of industry in which the firm operates, and so forth.[7] Moreover, differences in firm performance seem to be very persistent. If we consider a set of firms that are 20 or more years old, we will observe that the firms that are more profitable today were, on average, more profitable 20 years ago.[8]

What is the source of the 80% variability in firm performance that is left to explain? Why do some firms hold a **sustained competitive advantage**? It may be useful to consider the analogy with car racing; say, Formula One racing. It is quite clear that there are significant differences between different Formula One teams and that those differences seem to persist from race to race. One rather obvious explanation is that not all drivers have the same ability. In order for the Sauber team to match the performance of the Mercedes team, the former might need to recruit a driver like Nico Rosberg

(who currently races for Mercedes). But such driver seems difficult to find. Likewise, in the world of business, there may be **impediments to imitation** that allow some firms to perform persistently better than others.[9] An obvious limit to imitation is given by legal restrictions. In the late 1960s and early 1970s, Xerox was the dominant player in the photocopier industry thanks to its patent on plain-paper photocopying. Once Xerox started licensing its technology, differences in performance with respect to competitors decreased considerably (see Chapter 15).

If Sauber cannot find another Nico Rosberg, it should at least be able to imitate the best features of the Mercedes car (which seems to be faster than the Sauber car, even controlling for driver skills). Most features of Formula One cars are not patented. However, there are so many aspects in which a Mercedes car differs from an Sauber car that it would be difficult to understand which one is responsible for the superior performance of the former. Something similar occurs with firms as organizations. For many years, Toyota's procurement system was considered clearly superior to its rivals'. However, the system involved so many different aspects that it was difficult to pin down what the crucial success factors were. More generally, the problem is even more serious: most of the features of any given organization are *tacit knowledge*, capabilities that are developed by experience and rarely written down; capabilities that are difficult to express formally as an algorithm or a set of rules. Even if one of Toyota's rivals were to recruit some key employees and managers from Toyota, the latter would have difficulty in expressing their knowledge in the new organization, let alone implementing it. In strategy jargon, we say this is a case of **causal ambiguity**.[10]

Finally, even at times when the Mercedes car/driver duo is slower than rivals, the Mercedes team has frequently turned out to be the winner. In these cases, success is largely due to **strategy**. Strategic or tactical decisions include whether to stop once or twice during a race, when to stop, what kind of tires to use, etc. — and how to react to the choices made by rival teams. In a business context, there are many dimensions in which strategy can have a lasting effect on firm performance: entry timing, capacity expansion, mergers and acquisitions, technological improvement, special contracts with customers and suppliers, not to mention pricing and advertising.

Of the three sources of competitive advantage listed above, strategy is the one we will focus on in the remaining chapters of the book. This is not to say that the other ones are not important. In fact, the argument can be made that firm **culture** — a broad term that would include tacit knowledge and other aspects which cannot be easily imitated — is an equally important source of competitive advantage. But the focus of industrial organization is on competition between firms, not so much the way a firm is internally organized.

A related source of heterogeneity across firms is the quality of its management. Empirical evidence suggests that (a) management quality is positively associated with firm performance ("management matters"); and (b) management quality varies considerably across firms. A recent study based on surveys of thousands of firms across the globe measured management quality by averaging a series of scores related to production logistics, inventories, performance evaluation, etc.[11]

Figure 3.5 plots the the distribution of values for the US and India. One first noticeable feature is the difference between the US and India. The average score for the US, μ_{US}, is well above 3, whereas the average score for India, μ_I, is well below 3. However, equally impressive is the variability of management quality within the US and within India. Some of the best-managed Indian firms are quite well managed: about 16% are better managed than the average US firm. By contrast, very few US firms have a score lower than 2, whereas about 20% of Indian firms fall below this threshold.

The preceding paragraph addresses point (b), that is, the idea that management quality varies considerably across firms. It does not address point (a), that is, the idea that management matters. To investigate the latter, one might regress firm performance (e.g., total factor productivity, as defined in Section 3.1), on the management quality score. Such a regression yields a large and statistical significant correlation. This is consistent with the proposition that management quality affects firm performance; however, it does not imply that there is a causal relation. The best way to test causality is to run a **field experiment**. One such experiment randomly split Indian textile firms in two groups and offered one of the groups (treatment group) consulting services from Anderson Consulting. As a result of these consulting services, the treatment group implemented important changes in its management practices, corresponding to an increase in the measurement quality score; moreover, these firms showed significant improvements in productivity (for example, a substantial drop in the number of defective products).

In conjunction with all of the factors mentioned above, one must also stress the importance of **history** in determining firm performance — and the persistence of differences in firm performance. Consider the example of competition in the large commercial aircraft market during the 1970s. The three competitors were Boeing (B747), McDonnell Douglas (DC10), and Lockheed (L1011). One important aspect of competition in this

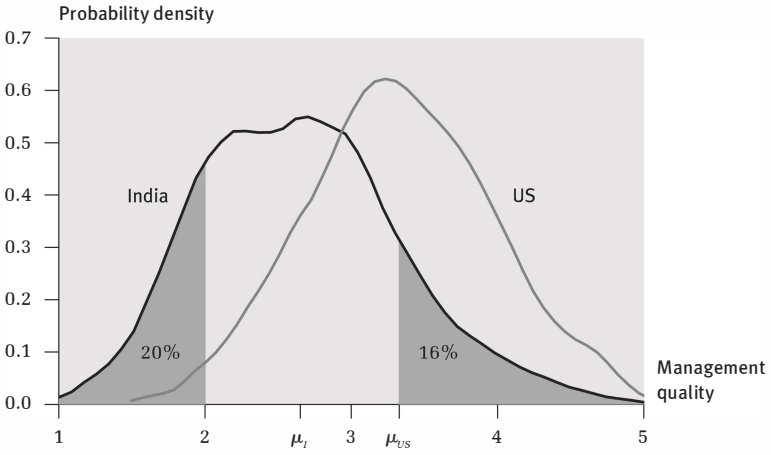

FIGURE 3.5
Management quality in India and in the US

industry is the existence of a steep **learning curve**: the more aircraft units a firm produces, the lower the cost of producing an aircraft unit. The name of the game is therefore to move down the learning curve as quickly as possible.

At the start, all three competitors maintained approximately equal market shares. But in the early 1970s, Lockheed's main engine supplier, Rolls-Royce, began to experience a series of technical and financial difficulties. This slowed down Lockheed's production rate and the speed at which it moved down the learning curve. When Lockheed tried to get back in the game, later in the 1970s, it was too late, for its competitors were already too competitive. McDonnell Douglas did not have much better luck. A series of crashes in 1980 severely hit consumer confidence in the DC10. In retrospect, this turned out to be a statistical coincidence, not the result of a fundamental problem with the DC10 design. But again, by the time McDonnell Douglas tried to get back in the game it was too late: Boeing had reached a superior degree of cost efficiency that allowed it to enjoy a sustainable competitive advantage over its rivals.[j]

This is an example where historical events (Rolls-Royce's problems in the early 1970s, the DC10 crashes in 1980) had an effect that lasted beyond the time when they occurred. In particular, these are events that, together with the learning curve phenomenon, have lead to a persistent asymmetry in firm performance between Boeing and its rivals. Other examples could be found, involving network externalities (see Box 16.3), consumer switching costs, and so forth. The point is that sometimes, instead of insisting on a general theory of firm heterogeneity, one simply has to accept the importance that history has in shaping an industry and the firms that it includes. I resume this discussion in Section 10.1. For now we conclude with a summary of the main points above:

> Firm performance varies a great deal. Firms are different because of impediments to imitation, causal ambiguity, firm strategy, management quality, or historical events.

Summary

• Reflecting decreasing marginal returns, isoquants are convex curves; the closer complements two inputs are, the more convex the corresponding isoquants are. • Marginal cost is the appropriate cost concept to decide how much to produce, whereas average cost is the appropriate cost concept to decide whether to produce at all. • The profit maximizing output level is such that marginal revenue equals marginal cost. • The lower the price elasticity of demand (in absolute value) the higher the price-cost margin should be set: • Although management and ownership are normally separated, there are reasons to believe that deviations from profit maximization cannot be too large. These reasons include management incentive contracts, labor market discipline, product market discipline, and capital market discipline. • The horizontal boundaries of the firm are largely

j. Eventually, Lockheed exited from the civil aviation market. McDonnell Douglas, in turn, was later acquired by Boeing. In the meantime, Airbus appeared on the scene, challenging Boeing's dominance.

(c) What is the profit-maximizing number of 3DTVs for Las-O-Vision to produce each day? What price does Las-O-Vision charge per 3DTV? What is its daily profit?

■ **3.6. MONSANTO.** With reference to the discussion in Box 3.1:

(a) How do you know that cutting the price of Roundup was a good idea for Monsanto?

(b) How might you estimate the elasticity of demand and the profit-maximizing price for 1995? Do you think Monsanto set the right price?

(c) If cutting price was a good idea, why didn't Monsanto do it earlier?

■ **3.7. WINDOWS.** Microsoft is the dominant player in the market for desktop operating systems. Suppose each copy of Windows is sold for $50. Suppose moreover that the marginal cost of production and shipping is $5. What value of the demand elasticity is consistent with this price? Does it make sense? What elements of the OS market may be missing from the elasticity rule?

■ **3.8. PROFIT MAXIMIZATION.** Explain why the assumption of profit maximization is or is not reasonable?

■ **3.9. CAR PARTS.** Two parts in an automobile taillight are the plastic exterior cover and the light bulb. Which of these parts is a car company more likely to manufacture in-house? Why?

■ **3.10. JET ENGINES.** There are three main suppliers of commercial jet engines, Pratt & Whitney, General Electric, and Rolls-Royce. All three maintain extensive support staff at major (and many minor) airports throughout the world. Why doesn't one firm service each airport? Why do all three feel they need to provide service and support operations worldwide themselves? Why don't they subcontract this work? Why don't they leave it entirely to the airlines?

■ **3.11. SMART CAR.** The Smart car was created as a joint venture between Daimler-Benz AG and Swatch Group AG. Although Micro Compact Car AG (the name of the joint venture) was originally jointly owned, in November of 1998 Daimler-Benz AG took complete control by buying Swatch's share.[12] The deal put an end to a very stressed relationship between Daimler and Swatch. What does Section 3.4 suggest as to what the sources of strain might have been?

■ **3.12. FRANCHISE RETAILING.** Empirical evidence from franchise retailing suggests that, even when stores have similar characteristics, the mother company resorts to a mix between company-owned stores and franchised ones.[13] How can this be justified?

■ **3.13. BODY SHOP.** The UK Body Shop franchise network consists of three types of stores: franchised, company owned, and partnership stores. All stores that are distant from headquarters by more than 300 miles are franchised. More than half of the company-owned stores are within 100 miles of headquarters.[14] How can you explain these facts?

■ **3.14. INTEL.** Explain why Intel has maintained, if not increased, its competitive advantage with respect to rivals. Indicate the explanatory power of the different causes considered in the text (impediments to imitation, causal ambiguity, strategy, history).

CHALLENGING EXERCISES

■ **3.15. INPUT SHARES.** Consider a firm with production function (Cobb-Douglas)

$q = \xi K^{\alpha} L^{\beta}$

Suppose that output price p and input prices r, w are given. Show that a profit maximizing firm chooses inputs levels such that the ratio between labor costs and total revenue wL/pq is equal to β.

■ **3.16. COST MINIMIZING INPUT MIX.** Consider a firm with production function (Cobb-Douglas)

$q = K^{\frac{1}{2}} L^{\frac{1}{2}}$

Suppose that one unit of capital costs $r = 12.5$, whereas one unit of labor costs $w = 8$.

(a) Determine the optimal input mix that leads to an output of $q = 2$.

(b) Determine the firm's cost function, that is, the minimum cost required to produce output q.

■ **3.17. PRICING WITH CAPACITY CONSTRAINTS.** Consider again the case of a monopoly facing linear demand and constant marginal cost. Demand is:

$q = a - bp,$

and marginal cost is $MC = c$. Suppose moreover that the monopolist has a limited capacity of K. In other words, it must be $q \leq K$. What is the optimal price?

■ **3.18. OPTIMAL BIDDING.** You are one of two companies bidding to try to win a large construction project. Call your bid b. You estimate that your costs of actually performing the work required will be \$800k. You are risk neutral. You will win if and only if your bid is lower than that of the other bidder. You are not sure what bid your rival will submit, but you estimate that the rival's bid is uniformly distributed between \$1 and \$2 million.[k] What bid should you submit?

k. By "uniformly distributed between a and b" we mean that all values between a and b are equally likely.

■ **3.19. COMPETITIVE PRESSURE.** Suppose that a firm's profits are given by $\pi = \alpha + \phi(e) + \epsilon$, where α denotes the intensity of product market competition, e effort by the manager, and ϵ a random shock. The function $\phi(e)$ is increasing and concave, that is, $d\phi/de > 0$ and $d^2\phi/de^2 < 0$.

In order for the firm to survive, profits must be greater than $\underline{\pi}$. The manager's payoff is $\beta > 0$ if the firm survives and zero if it is liquidated, that is, if profits fall short of the minimum target. The idea is that if the firm is liquidated, then the manager loses his or her job and the rents associated with it.

Suppose that ϵ is normally distributed with mean μ and variance σ^2, and that $\mu > \underline{\pi}$. Show that increased product market competition (lower α) induces greater effort by the manager, that is, $de/d\alpha < 0$.

APPLIED EXERCISES

■ **3.20. PRODUCTIVITY.** Find firm level data on output and inputs (both quantity and prices). Use this information to estimate each firm's total factor productivity, following the method outlined in Section 3.1 (cf in particular Equation 3.2).

NOTES

1. Lerner, A. P. (1934), "The Concept of Monopoly and the Measurement of Monopoly Power," *The Review of Economic Studies* **1** (3), 157–175.

2. Adapted from "A Weed Killer Is a Block to Build On," by David Barboza, *The New York Times*, August 2, 2001.

3. This section is partly based on Holmstrom, Bengt R, and Jean Tirole (1989), "The Theory of the Firm," in Schmalensee and Willig (Eds.), *Handbook of Industrial Organization*, Amsterdam: North-Holland

4. Hermalin, Benjamin E., and Michael S. Weisbach (1988), "The Determinants of Board Composition," *Rand Journal of Economics* **19**, 589–606; and Hermalin, Benjamin E., and Michael S. Weisbach (1998), "Endogenously Chosen Boards of Directors and Their Monitoring of the CEO," *American Economic Review* **88**, 96–118.

5. The theory presented in this section is based on Williamson, Oliver E. (1975), *Markets and Hierarchies*, New York: The Free Press; and on Grossman, Sanford J., and Oliver D. Hart (1986), "The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration," *Journal of Political Economy* **94**, 691–719.

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9. Rumelt, Richard P. (1984), "Towards a Strategic Theory of the Firm," in R. Lamb (Ed), *Competitive Strategic Management*, Englewood Cliffs, NJ: Prentice Hall. Three other important references for this and the topics to follow are: Penrose, E. T. (1959), *The Theory of the Growth of the Firm*, Oxford: Blackwell; Wernerfelt, B. (1984), "A Resource-Based View of the Firm," *Strategic*

CHAPTER 5

Market Failure and Public Policy

The perfect competition model is based on so many assumptions that it is not difficult to find examples where it fails. In this chapter, I consider three important sources of "market failure:" externalities, imperfect information, and market power. Externalities exist when the actions of an economic agent (consumer, firm, etc.) have an impact on other agents that go beyond the effect on market price. This may seem a bit vague, but hopefully the examples in Section 5.1 will make things clear; for now, let me just say that pollution and climate change provide a prominent example of externalities. The idea of imperfect information should be more straightforward: sometimes firms don't know everything about consumers and consumers don't know everything about firms. Health care is an example that comes to mind, but as we will see in Section 5.2 there are many, many more examples. Finally, market power refers to firms that have the ability to set prices, sometimes very high prices: your local supermarket, your airline, and your cable provider are just three examples; I am sure you can think of more examples (and I will suggest a few in Section 5.3).

One half of the chapter is devoted to these sources of market failure, whereas the other half focuses on public policy instruments to correct situations of market power: market regulation (Section 5.4), competition policy (Section 5.5), and firm regulation (Section 5.6).

5.1 EXTERNALITIES AND MARKET FAILURE

Where can you find beer for 5 cents a bottle? My best answer: try a class reunion with 100 diners. Let me explain. When you and your 99 colleagues go out for a meal, chances are you will decide to split the total bill: it would be far too complicated to keep

100 individual tabs. When it comes to decide whether to order that second or third beer, if you are an economist, then the way you'll reason is as follows: one more beer, five more dollars on the total tab; that's five cents for me — not a bad deal! Lest you think this is a purely theoretical consideration, economists have actually estimated this effect: even for a party as small as four, splitting the bill may lead to an increase in the total tab by as much as 40%.[1]

In economics, this problem is referred to as the **free riding problem**. The point is that an agent's decisions (for example ordering an extra beer at a split-the-bill dinner) does not take into account the costs imposed on other agents (99% of that extra beer is effectively paid by them). As a result, the total quantity consumed at the dinner is likely to be higher than socially efficient: even if the marginal cost of a beer is, say, 30 cents, there will be a lot of beer consumed for which willingness to pay is lower than social cost.

■ **TYPES OF EXTERNALITIES.** More generally, we say there is an **externality** when an agent's actions have an effect on other agents that goes beyond the market transaction per se. This effect can be negative (e.g., when I smoke people around me suffer) or positive (e.g., when I plant flowers in my yard, everyone who passes by benefits). I next consider some of the most common types of externalities.

- The **tragedy of the commons** is the situation where a common resource is overused with respect to the socially optimal level. The name dates back to the grazing fields in England that were shared by several flock owners; but there are some more recent and economically important cases, such as fisheries and clean air. For example, when a fishing boat decides how much cod to catch in New Foundland, the owner compares his own benefits and his own costs, ignoring the effect that this will have on the overall stock of cod. In this case, the market equilibrium involves an output level that is too high from a social point of view, possibly even leading to species extinction or at least overfishing.[a]

- Here's an externality that most of us complain about on a daily basis: **congestion**. If you are used to fly out of a busy airport like New York's LaGuardia or London's Heathrow, you know the drill: "Good morning folks, this is your captain speaking, we are now number 17 for takeoff." Which means an extra half hour taxiing on the tarmac. Excessive congestion results from an externality: when an airline decides to schedule a flight during rush hour, it does not take into account the extra delay it will impose on all flights departing right after.

- I've been giving examples of negative externalities, but there are also examples of positive externalities. An extreme case is given by **public goods**. If I build a park, I create a benefit for myself, but others can enjoy it at no extra

a. The tragedy of the commons may also be felt in a large multi-division organization where division managers and heads are rewarded only for the performance of their own divisions, thus failing to take into account the impact of their actions on the firm as a whole. See Exercise 5.6.

cost — a positive externality.[b] National defense, health, and education are also examples of investments and expenses that have some element of public good.[c]

As these examples show, externalities abound. This is bad news for the Fundamental Theorem, and it begs a number of questions: how poorly does the market solution work in the presence of externalities? Are there any means of restoring optimality — or something close to optimality — in the presence of externalities? I begin with the economists' favorite solution to the problem: externality-correcting taxes.

■ **SOCIAL COST AND PIGOU TAXES.** In markets with externalities, the Fundamental Theorem fails to hold: market price is no longer the right guide for consumers and producers; something else besides the "invisible hand" is required to establish social efficiency. What is this "something else"? Many economists will tell you that the solution is to apply a **Pigou tax**.

To motivate the idea, consider the problem of climate change. We may dispute the precise numbers, but we all agree that industrial production is an important factor in carbon dioxide emissions, which in turn contribute to climate change and related effects. What does this mean in economics terms? Figure 5.1 depicts the supply and demand curves in a given industry. Recall that, in a competitive industry, the supply curve reflects the sellers' marginal cost curves, whereas the demand curve — specifically, the inverse demand curve — reflects the buyers' willingness to pay for the product in question. The free market equilibrium implies price p° and output q°. By the Fundamental Theorem, this is the output level such that a marginal buyer's willingness to pay is just equal to the marginal seller's marginal cost: there is no trade that would increase total gains from trade; total surplus is maximized.

Suppose however that each unit of output implies a social marginal cost of $c(q)$, that is, a value that changes as q changes (in the figure, and in most real-world cases,

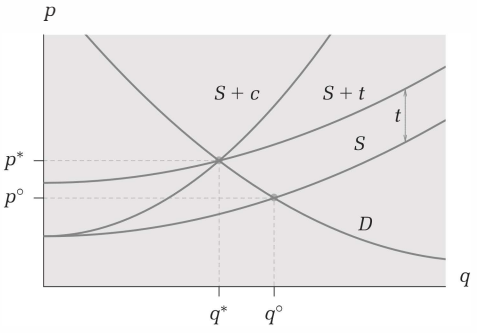

FIGURE 5.1
Market equilibrium and social optimum with externalities

b. In other words, public goods are non-rival (and non-excludable) goods: the consumption by one person does not impede the consumption of the same good by another person.

c. One important distinction is between public goods with exclusion and public goods with no exclusion. For example, I can restrict access to a park that I build, but it would be hard to stop residents in a given area from enjoying clean air, if clean air is provided.

the social marginal cost is increasing in q). Then the total marginal cost of producing a qth unit is given the sum of production marginal cost and social marginal cost, the curve $S + c$ in Figure 5.1. By the same reasoning as in the Fundamental Theorem, the socially optimal output level — the level that maximizes net benefits, including the extra social marginal cost — is given by the point where $S + c$ crosses the demand curve. This corresponds to output level q^*. Not surprisingly, q^* is lower than q°: whenever there is a negative externality, the equilibrium output level is greater than the socially optimal output level.

Is there any hope for the market? Enter economist Arthur C. Pigou, whose answer is yes, so long as we impose an output tax (if the externality is negative) or a output subsidy (if the externality is positive).[d] Consider Figure 5.1 again. Suppose that a tax t is levied on suppliers for each output unit. This implies an upward shift in the supply curve, from S to $S + t$. The market equilibrium is now given by q^*, the socially optimal output level.

A relative success story in applying Pigou's ideas is taxation of gasoline consumption in Europe, where a Pigou tax has shifted consumption levels closer to the social optimum. See Box 5.1 for details. A less successful story is the idea of a "Google tax," which is gaining traction in several European countries. See Box 5.2 for details. The idea is to tax Internet content aggregators (such as Google) that list snippets of content (e.g., news items) created by others. The "Google tax" is more about fairness than it is about efficiency. In fact, Pigou would recommend that both newspapers and Google be *subsidized*, since both content creation and links to content create a positive externality (see Box 5.2). However, such policy would face tremendous difficulties (e.g., deciding who will pay for the subsidy).

■ **ALTERNATIVE SOLUTIONS TO THE EXTERNALITY PROBLEM.** Notwithstanding economists' preferences, Pigou taxes are by no means the only way to deal with externalities. One important alternative is direct **regulation** of the externality-creating activity. For example, smoking creates an externality in the sense that it bothers — and harms — nearby non-smokers. In many countries around the world, this problem has been solved simply by banning smoking in public places.

In many settings, social norms play the role of government regulations. Even if you are splitting the restaurant bill, you will probably not order the Beluga caviar with Dom Perignon champagne on the side. Why not? Often going out to restaurants is a repeated game (cf Chapter 7): we tend to go out with the same groups of friends or colleagues, and so a norm of behavior (with a consequent threat of punishment) can arise. If you order caviar, it may be last time you do so with this group of friends, so think twice about it.

d. I am assuming that the amount of pollution is proportional to output. More generally, the tax should be on the amount of the polluting amount.

■ **THE COASE THEOREM.** One of the most creative ways of solving the inefficiency created by externalities may be to let the interested parties negotiate. The **Coase Theorem** — named after its originator, economist Ronald Coase — states that, if **property rights** are

Box 5.1 Approaches to reducing gasoline consumption: US and EU

In 2013, 1 liter of gasoline cost €1.53 at an average European Union (EU) pump, of which 58% corresponded to taxes and duties. In the United States, average price at the pump was $3.49 per gallon. Considering that one gallon equals 3.785 liters and that the yearly average exchange rate was 0.783 euros per dollar, this comes to about €.72 per liter, about one half of the price in Europe. For US readers, the European price was about $7.40 per gallon, about twice the US price.

Not surprisingly, gasoline consumption (both per person and per car) is much lower in the EU than in the US. Arguably, the value of q in Europe is much closer to q^* in the EU than in the US (cf Figure 5.1). This is partly because the price elasticity of the demand for gasoline is different from zero, but also because the demand for cars is very different in the EU than in the US. Reacting to high gasoline prices and the demand for low-consumption cars, European cars are smaller and consume less gasoline than their American counterparts. In 2013, average fuel efficiency in the US was 32 miles per gallon, whereas the EU showed a whopping 45. For European readers, this corresponds to 5.2 liters per 100 kilometers (EU) against 7.6 liters per 100 kilometers (US).

Various US governments have made efforts to increase automobile fuel efficiency, for example enacting Corporate Average Fuel Efficiency (CAFE) standards, regulations that force automakers a minimum average fuel efficiency. However, the regulated firms have managed to keep the minimum standards at "reasonable" levels. Moreover, various loopholes make CAFE standards less effective than one might think: for example, gas-guzzling SUVs are treated as light trucks (and thus excluded from the average), which effectively provides an "escape" for consumers and manufacturers who want to avoid regulation.

In sum, fuel efficiency policy suggests that price may be a better instrument than quantity regulation. It also provides an example of how the market-based US may at times be less market-based then the more government-centered EU. Finally, having established how the US and the EU differ in the way they regulate gasoline consumption, one is naturally led to the question of why they differ: Section 5.4 provides a partial answer.

properly assigned and negotiations are costless, then all externalities will be "internalized," so that the market solution (cum negotiations) leads to an efficient solution.[2]

Consider, for example, the case of a steel plant dumping waste in the river. Society might decide that (a) downstream parties have a property right to clean water, or (b) that the plant has the right to dispose of its waste as it sees fit. Property rights can be established in either way. But once those rights have been clearly established (and so long as they are properly enforced), then individuals can bargain over how to exercise those rights. If downstream parties have the right to clean water, then the steel plant can pay them in order to get permission to pollute. Alternatively, if the plant has the property rights, then downstream parties can compensate the steel plant to restrict its pollution inducing activities.

BOX 5.2 THE EUROPEAN "GOOGLE TAX"

If you "Googled" "ebola" in Spain sometime in October 2014, you would probably find a link to the newspaper *El País*, as well as a snippet of the news item in question. If you clicked through the link you'd be directed to the site containing the complete story. In the process, several firms made money off your clicks and eyeballs: *El País* included ads on the page where the story was displayed; Google, too, included ads on the search page you started from; and online ads are paid for (a lot, since they tend to be targeted).

In October 2014, Günther Oettinger, the EU's incoming Digital Commissioner, floated the idea of an EU-wide "Google-tax." This tax would require Internet search engine providers to pay a fee for displaying copyrighted materials on their sites. "If Google takes intellectual property from the EU and makes use of it, the EU can protect this property and demand that Google pay for it," said Oettinger.

This would not be the first attempt by European regulators to extract rents from the Internet giant. In 2006, a Belgium court ordered Google to pay Copiepresse, a consortium of Belgium newspapers, for the right to include news snippets on its search pages, "under penalty of a daily fine of €2,000,000 per day of delay." Google complied — not by paying but rather by removing the links from its pages. This led to a sharp decrease in advertising revenues for the Copiepresse affiliated newspapers. After several years of "link wars," the Belgians eventually caved in: in 2011, Copiepresse allowed Google to re-include their sites in Google's searches.

The newspapers claim their content is being stolen; Google defends itself by asserting that each month it originates billions of views to newspapers' websites. It may be that both are right, that is, both parties stand to gain from snippet display and links to the content source; as author Cory Doctorow aptly put it, "the net is the natural home of positive externalities." At the same time, it can be argued that Google makes relatively more money than the linked sites: many users coming from Google are known as "one hit wonders," users who are more likely to hit the "back" button than to continue browsing the site they were directed to from Google.

In October 2014, Spain created a "Google tax" similar to the Belgian one. The "link wars" are far from over.

Suppose that the value that the plant gets from the opportunity of dumping waste in the river is greater than the cost it imposes on downstream residents. Then, in equilibrium, negotiations will lead to an agreement whereby waste dumping takes place. Notice that this equilibrium is achieved regardless of the initial property rights allocation. If the factory owns the rights, then residents have no incentive to buy the rights from the factory. If, in turn, the residents own the rights, then the factory will make them a proposal they cannot refuse. Conversely, if the value that the plant gets from the opportunity of dumping waste in the river is greater than the cost it imposes on downstream residents, then in equilibrium negotiations will lead to an agreement whereby waste dumping does take place. Once again, this outcome arises regardless of the initial property rights allocation.

The reasoning underlying the Coase Theorem serves as an additional illustration of the difference between efficiency and fairness. The Fundamental Theorem and the

Coase Theorem are about efficiency: resource allocation that maximizes gains from trade (total surplus). These theorems have little to say about fairness. From an efficiency point of view, it makes no difference who owns the rights to a clean river (factory or residents). This is not to say, however, that both property rights allocations are equally fair.

Here's another example of the difference between efficiency and fairness: many cities around the world have banned smoking from public places, including bars. But suppose that some fabulously wealthy person came to town and really, really wanted to light up in a small bar, to the point where he was willing to pay $1 million to each person in the bar. Would it be efficient to let him do so? Would it be fair?

This section can be summarized by stating that:

> Market externalities imply market failure. Pigou taxes and other mechanisms may reestablish equilibrium efficiency.

In the remainder of the book we will come across various instances of externalities. Particularly important are externalities in competition (cf Chapter 8) and externalities in the relation between firms along the value chain (cf Chapter 13).

5.2 IMPERFECT INFORMATION

Suppose you want to buy a 2010 Honda Accord. By consulting the Internet or some other source, you find out that the value of a very well kept car is about $20,000. In practice, you know that some owners were not that careful, so that the actual value could be anywhere between $20,000 and $0 (zero being a really bad lemon). For simplicity, let's assume that, from the perspective of an uninformed buyer (you), the value could be anywhere between $0 and $20,000. This means that, on average, you should be willing to pay $10,000 (or less, if you are risk averse, a possibility I presently ignore).

Suppose that sellers, unlike you, are informed about the real value of the car they consider selling. If you are unwilling to pay more than $10,000 for a car, then the sellers whose cars are in best condition (and thus are worth more) will be unwilling to sell. For example, suppose that all sellers whose cars are worth more than $12,000 decide to stay out. If that is the case, and if you find out about it, then you shouldn't be willing to pay more than $6,000. But wait: if this is all that you are willing to pay, then more sellers are unwilling to sell, again those whose cars are in better condition.

This unraveling process of **adverse selection** may go on and on, possibly to the extreme that the market collapses completely.[3] That's a rather unfortunate outcome. As we saw in Section 4.4, under perfect competition all beneficial trades take place. In the present context, there may be many socially profitable trades (where the value of a specific car is greater for buyer than for seller) that do not take place. The key departure from the Fundamental Theorem (Section 4.4) is that perfect competition assumes perfect information, whereas in the present case you (the buyer) are uninformed about the

KEY CONCEPTS

• free riding problem • externality • tragedy of the commons • congestion • public goods • Pigou tax • regulation • Coase Theorem • property rights • adverse selection • asymmetric information • advantageous selection • moral hazard • patent • copyright • network effects • Harberger triangle • allocative inefficiency • excess burden • productive efficiency • rent seeking • residual demand • monopoly power • regulation • normative theory • capture theory • competition policy • antitrust • price fixing • merger policy • abusive practices • natural monopoly • firm regulation • regulatory capture • average cost pricing • rate-of-return regulation • regulatory lag • low-power incentive mechanism • high-power incentive mechanism • price cap regulation • bottleneck • essential facility • access price • Efficient Component Pricing Rule

REVIEW AND PRACTICE EXERCISES

■ **5.1. FRONT YARDS.** Front yards, if well tended, generate positive externalities for a house's neighbors. Do you think this is an important externality? Is the market solution inefficient?

■ **5.2. AT&T.** The long-run demand elasticity of AT&T in the period 1988–1991 was estimated to be around -10.[17] Assuming the estimate is correct, what does this imply in terms of AT&T's market power at the time?

■ **5.3. MONOPOLY POWER.** "The degree of monopoly power is limited by the elasticity of demand." Comment.

■ **5.4. WINDOWS.** Is the Windows operating system an essential facility? What about the Intel Pentium microprocessor? To what extent does the discussion in Section 5.6 on essential facilities (vertical integration, access pricing) apply to the above examples?

CHALLENGING EXERCISES

■ **5.5. CARBON TAX.** Consider an industry with demand $q = 1 - p$ and supply $q = p$. Suppose that each unit of output implies one unit of CO2 added to the atmosphere and a marginal social cost of e, where e is the total level of emmissions.

(a) What is the level of CO2 emission at the market equilibrium?

(b) What is the socially optimal level of CO2 emissions?

(c) Determine the Pigou tax that achieves the social optimum.

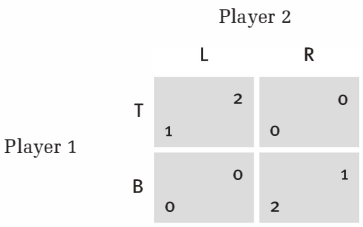

Figure 7.6
Multiple Nash equilibria

illustration for this game is the process of standardization. Strategies T and L, or B and R, correspond to combinations of strategies that lead to compatibility. Both players are better off under compatibility. However, Player 1 prefers compatibility around the standard (B-R), whereas Player 2 prefers compatibility over the other standard. More generally, this example is representative of a class of games in which (a) players want to coordinate, (b) there is more than one point of coordination, (c) players disagree over which of the two coordination points is better.[j]

Equilibrium multiplicity can be problematic for an analyst: if the game's equilibrium is a natural prediction of what will happen when players interact, then multiplicity makes such prediction difficult. This is particularly the case for games that have not only two but many different equilibria. Consider the following example: two people are asked to choose independently and without communication where and when in New York City they would try to meet one another. If they choose the same meeting location, then they both receive $100; otherwise they both receive nothing.[2] Despite the plethora of possible meeting and time locations, which correspond to a plethora of Nash equilibria, a majority of subjects typically choose Grand Central Station at noon: the most salient traffic hub in New York City and the most salient time of the day.

Although there is an uncountable number of Nash equilibria in the "meet me in New York" game, some are more "reasonable" than others. Here, "reasonable" is not meant in the game theoretic sense: all equilibria satisfy the conditions for a Nash equilibrium. Rather, it is meant in the sense that, based on information that goes beyond the game itself, there may be **focal points** on which players coordinate even if they do not communicate.[k] Game theory and the concept of Nash equilibrium are useful tools to understand interdependent decision making; but they are not the only source of useful information to predict equilibrium behavior.

j. Standardization problems are further discussed in Sections 16.3 and 16.4.

k. One game theorist defined a focal point as "each person's expectation of what the other expects him to expect to be expected to do" (cf Endnote 2).

7.2 SEQUENTIAL GAMES[l]

In the previous section, I justified the choice of simultaneous-choice games as a realistic way of modeling situations where observation lags are so long that it is as if players were choosing strategies simultaneously. When the time between strategy choices is

l. This and the next section cover relatively more advanced material which may be skipped in a first reading of the book.

sufficiently long, however, the assumption of sequential decision making is more realistic. Consider the example of an industry that is currently monopolized. A second firm must decide whether or not to enter the industry. Given the decision of whether or not to enter, the incumbent firm must decide whether to price aggressively or not. The incumbent's decision is taken *as a function* of the entrant's decision. That is, first the incumbent observes whether or not the entrant enters, and then decides whether or not to price aggressively. In such a situation, it makes more sense to consider a model with sequential rather than simultaneous choices. Specifically, the model should have the entrant — Player 1 — move first and the incumbent — Player 2 — move second.

The best way to model games with sequential choices is to use a game tree. A **game tree** is like a decision tree only that there is more than one decision maker involved. An example is given by Figure 7.7, where strategies and payoffs illustrate the case of entrant and incumbent described above. In Figure 7.7, a square denotes a **decision node**. The game starts with decision node 1. At this node, Player 1 (entrant) makes a choice between e and $\bar{e}$, which can be interpreted as "enter" and "not enter," respectively. If the latter is chosen, then the game ends with payoffs $\pi_1 = 0$ (entrant's payoff) and $\pi_2 = 50$ (incumbent's payoff). If Player 1 chooses e, however, then we move on to decision node 2. This node corresponds to Player 2 (incumbent) making a choice between r and $\bar{r}$, which can be interpreted as "retaliate entry" or "not retaliate entry," respectively. Games which, like Figure 7.7, are represented by trees are also referred to as games in **extensive form**.[m]

This game has two Nash equilibria: $(e, \bar{r})$ and $(\bar{e}, r)$. Let us first check that $(e, \bar{r})$ is indeed a Nash equilibrium, that is, no player has an incentive to change its strategy given what the other player does. First, if Player 1 chooses e, then Player 2's best choice is to choose $\bar{r}$ (it gets 20, it would get -10 otherwise). Likewise, given that Player 2 chooses $\bar{r}$, Player 1's optimal choice is e (it gets 10, it would get 0 otherwise).

Let us now check that $(\bar{e}, r)$ is an equilibrium. Given that Player 2 chooses r, Player 1 is better off by choosing $\bar{e}$: this yields Player 1 a payoff of 0, whereas e would yield -10. As for Player 2, given that Player 1 chooses $\bar{e}$, its payoff is 50, regardless of which strategy it chooses. It follows that r is an optimal choice (though not the only one).

Although the two solutions are indeed two Nash equilibria, the second equilibrium does not make much sense. Player 1 is not entering because of the "threat" that

m. From this and the previous sections, one might erroneously conclude that games with simultaneous choices must be represented in normal form and games with sequential moves in extensive form. In fact, both simultaneous and sequential choice games can be represented in both the normal and extensive forms. However, for simple games like those considered in this chapter, the choice of game representation considered in the text is more appropriate.

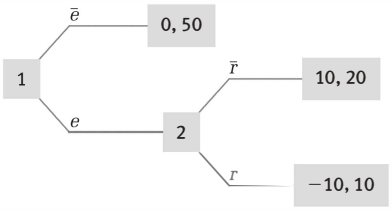

FIGURE 7.7
Extensive-form representation: the sequential-entry game

Player 2 will choose to retaliate. But, is this threat credible? If Player 1 were to enter, would Player 2 decide to retaliate? Clearly, the answer is "no": by retaliating, Player 2 gets -10, compared to 20 from no retaliation. We conclude that $(\bar{e}, r)$, while being a Nash equilibrium, is not a reasonable prediction of what one might expect to be played out.

One way of getting rid of this sort of "unreasonable" equilibria is to solve the game backward; that is, to apply the principle of **backward induction**. First, we consider node 2, and conclude that the optimal decision is $\bar{r}$. *Then*, we solve for the decision in node 1 *given the decision previously found for node 2*. Given that Player 2 will choose $\bar{r}$, it is now clear that the optimal decision at node 1 is e. We thus select the first Nash equilibrium as the only one that is intuitively "reasonable."

Solving a game backward is not always this easy. Suppose that, if Player 1 chooses e at decision node 1 we are led not to a Player 2 decision node but rather to an entire new game, say, a simultaneous-move game as in Figures 7.1 to 7.6. Since this game is a part of the larger game, we call it a **subgame** of the larger game. In this setting, solving the game backward would amount to first solving for the Nash equilibrium (or equilibria) of the subgame; and then, given the solution for the subgame, solving for the entire game. Equilibria which are derived in this way are called **subgame-perfect equilibria**.[3] For simple game trees like the one in Figure 7.7, subgame-perfect equilibria are obtained by solving the game backwards, that is, backwards induction and subgame perfection coincide.

■ **The value of credible commitment.** In the game of Figure 7.7, the equilibrium $(\bar{e}, r)$ was dismissed on the basis that it requires Player 2 to make the "incredible" commitment of playing r in case Player 1 chooses e. Such threat is not credible because, *given* that Player 1 has chosen e, Player 2's best choice is $\bar{r}$. But suppose that Player 2 writes an enforceable and not renegotiable contract whereby, if Player 1 chooses e, then Player 2 must choose r. The contract is such that, were Player 2 not to choose r and choose $\bar{r}$ instead, Player 2 would incur a penalty of 40, lowering its total payoff to -20.[n]

The situation is illustrated in Figure 7.8. The first decision now belongs to Player 2, who must choose between writing the bond described above (strategy b) and not doing anything (strategy $\bar{b}$). If Player 2 chooses $\bar{b}$, then the game in Figure 7.7 is played. If instead Player 2 chooses b, then a different game is played, one that takes into account the implications of issuing the bond.

Let us now compare the two subgames starting at Player 1's decision nodes. The one on top is the same as in Figure 7.7. As we then saw, restricting to the subgame perfect equilibrium, Player 2's payoff is 20. The subgame on the bottom is identical to the one on top except for Player 2's payoff following $(e, \bar{r})$. The value is now -20 instead of 20. At first, it might seem that this makes Player 2 worse off: payoffs are the same in all cases except one; and in that one case payoff is actually lower than initially. However, as I will show next, Player 2 is better off playing the bottom subgame than the top one.

Let us solve the bottom subgame backward, as before. When it comes to Player 2 to choose between r and $\bar{r}$, the optimal choice is r. In fact, this gives Player 2 a payoff of

n. This is a very strong assumption, as most contracts are renegotiable. However, for the purposes of the present argument, what is important is that Player 2 has the option of imposing on itself a cost if it does not choose r. This cost may result from breach of contract or from a different source.

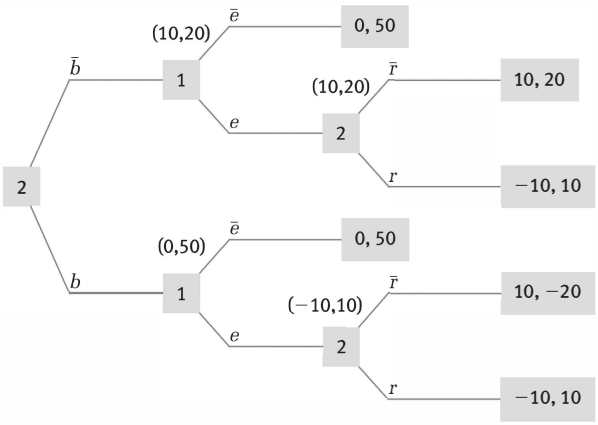

FIGURE 7.8
The value of commitment

10, whereas the alternative would yield −20 (Player 2 would have to pay for breaking the bond). Given that Player 2 chooses r, Player 1 finds it optimal to choose $\bar{e}$: it is better to receive a payoff of zero than to receive −10, the outcome of e followed by r. In sum, the subgame on the bottom gives Player 2 an equilibrium payoff of 50, the result of the combination $\bar{e}$ and r.

We can finally move backward one more stage and look at Player 2's choice between b and $\bar{b}$. From what we saw above, Player 2's optimal choice is to choose b and eventually receive a payoff of 50. The alternative, $\bar{b}$, eventually leads to a payoff of 20 only.

This example illustrates two important points. First it shows that:

> A credible commitment may have significant strategic value.

By signing a bond that imposes a large penalty when playing $\bar{r}$, Player 2 credibly commits to playing r when the time comes to choose between r and $\bar{r}$. In so doing, Player 2 induces Player 1 to choose $\bar{e}$, which in turn works in Player 2's benefit. Specifically, introducing this **credible commitment** raises Player 2's payoff from 20 to 50. Therefore, the value of commitment in this example is 30.

The second point illustrated by the example is a methodological one. If we believe that Player 2 is credibly committed to choosing r, then we should model this by changing Player 2's payoffs or by changing the order of moves. This can be done as in Figure 7.8, where we model all the moves that lead to Player 2 effectively precommiting to playing r. Alternatively, this can also be done as in Figure 7.9, where we model Player 2 as choosing r or $\bar{r}$ "before" Player 1 chooses its strategy. The actual choice of r or $\bar{r}$ may occur in time after Player 1 chooses e or $\bar{e}$. However, if Player 2 pre-commits

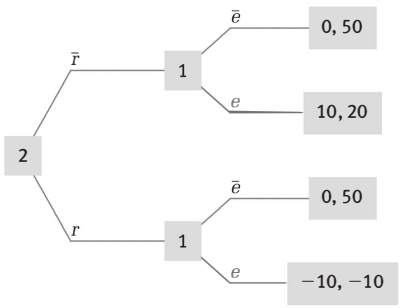

FIGURE 7.9
Modeling Player 2's capacity to precommit

FIGURE 7.10
A game with long-run and short-run strategy choices: timing of moves

to playing r, we can model that by assuming Player 2 moves first. In fact, by solving the game in Figure 7.9 backward, we get the same solution as in Figure 7.8, namely the second Nash equilibrium of the game initially considered.

■ **SHORT-RUN AND LONG-RUN.** To conclude this section, I should mention another instance where the sequence of moves plays an important role. This is when the game under consideration depicts a long-term situation where players choose both long-run and short-run variables. For example, capacity decisions are normally a firm's long-term choice, for production capacity (buildings, machines) typically lasts for a number of years. Pricing, on the other hand, is typically a short-run variable, for firms can change it relatively frequently at a relatively low cost.

When modeling this sort of strategic interaction, we should assume that players choose the long-run variable first and the short-run variable second. Short-run variables are those that players choose *given* the value of the long-run variables. And this is precisely what we get by placing the short-run choice in the second stage.

This timing of moves is depicted in Figure 7.10. This figure illustrates yet a third way of representing games: a time-line of moves. This is not as complete and rigorous as the normal-form and extensive-form representations we saw before; but it proves useful in analyzing a variety of games.

In a real-world situation, as time moves on, firms alternate between choosing capacity levels and choosing prices, the latter more frequently than the former. If we want to model this in a simple, two-stage game, then the right way of doing it is to place the capacity decision in the first stage and the pricing decision in the second stage. The same principle applies more generally when there are long-run and short-run strategic

variables. In the following chapters, we will encounter examples of this in relation to capacity/pricing decisions (Chapter 8), entry/output decisions (Chapter 10), and product positioning/pricing decisions (Chapter 14).

7.3 REPEATED GAMES

Many real-world situations of strategic behavior are repeated over an extended period of time. Sometimes, this can be modeled by an appropriate static model. For example, in the previous section we saw how a two-stage game can be used to model competition in long-term and short-term variables. Consider, however, the strategic phenomenon of **retaliation**, that is, the situation whereby a player changes its action in response to a rival's action. Clearly, this cannot be achieved in a static, simultaneous-move game, for in such a game there is no time for a player to react to another player's actions.

A useful way to model the situation whereby players react to each other's moves is to consider a repeated game. Consider a simultaneous-choice game like the one in Figure 7.1. Since in this game each player chooses one action only once, we refer to it as a **one-shot game**. A **repeated game** is defined by a one-shot game — also referred to as **stage game** — which is repeated a number of times. If repetition takes place a finite number of times, then we have a finitely repeated game, otherwise we have an infinitely repeated game.

In one-shot games, strategies are easy to define. In fact, strategies are identified with actions. In repeated games, however, it is useful to distinguish between actions and strategies. Consider again the game depicted in Figure 7.1, a version of the "prisoner's dilemma." As we saw earlier, it is a dominant strategy for Player 1 to choose B and it is a dominant strategy for Player 2 to choose R; the unique Nash equilibrium thus implies a payoff of 4 for each player.

Now suppose that this one-shot game is repeated indefinitely. Specifically, suppose that, after each period actions are chosen and payoffs distributed, the game ends with probability $1 - \delta$. For simplicity, suppose that players assign the same weight to payoffs earned in any period. Now the set of *strategies* is different from the set of actions available to players in each period. In each period, Player 1 must choose an action from the set $\{T, B\}$, whereas Player 2 must choose an action from the set $\{L, R\}$; these sets are the players' action sets. Strategies, however, can be rather complex combinations of "if-then" statements where the choice at time t is made dependent on what happened in previous periods; in other words, strategies can be history dependent. Given the conditional nature of strategies, the number of strategies available to a player is considerably greater than the number of actions it can choose from.

Specifically, consider the following strategies for Players 1 and 2: if in the past Player 1 chose T and Player 2 chose L, then continue doing so in the current period. If at any time in the past either of the players made a different choice, then let Player 1 choose B and Player 2 choose R in the current period.

Can such a pair of strategies form a Nash equilibrium? Let us first compute the equilibrium payoff that Player 1 gets playing the equilibrium strategies. Recall that (T, L) induces a payoff of 5 for Player 1. Total expected payoff is therefore given by:

$$V = 5 + \delta 5 + \delta^2 5 + ... = \frac{5}{1-\delta}$$

Notice that I successively multiply payoffs by δ for this is the probability that the game will continue on (that is, after each period is played, the game ends with probability $1 - \delta$). Now consider the possibility of deviating from the equilibrium strategies. If at a given period a given player chooses B instead of T, then expected payoff is given by:

$$V' = 6 + \delta 4 + \delta^2 4 + ... = 6 + \frac{4\delta}{1-\delta}$$

In order for the proposed set of strategies to be a Nash equilibrium, it must be that $V \geq V'$, that is,

$$\frac{5}{1-\delta} \geq 6 + \frac{4\delta}{1-\delta}$$

which is equivalent to $\delta \geq \frac{1}{2}$. Given the symmetry of the payoff function, the computations for Player 2 are identical to those of Player 1. We thus conclude that, if the probability that the game continues on into the next period, δ, is sufficiently high, then there exists a Nash equilibrium whereby players pick T and L in every period (while the game is still going on).

Intuitively, the difference between the one-shot game and the indefinitely repeated game is that, in the latter, we can create a system of intertemporal rewards and punishments that induces the right incentives for players to choose T. In one-shot play, the temptation for Player 1 to play B is too high: it is a dominant strategy, it yields a higher payoff regardless of what the other player does. In repeated play, however, the short-term gain achieved by choosing B must be balanced against the different continuation payoff each choice induces. The equilibrium strategies are such that, if Player 1 chooses B in the current period, then it gets 4 in every subsequent period. We thus have a gain of 1 in the short run $(6 - 5)$ against a loss of 1 in every future period $(5 - 4)$. Whether future periods matter or not depends on the value of δ. If δ is very high, then the loss of 1 in every future period counts very heavily, and Player 1 prefers to stick to T and forego the short-term gain from choosing B.

We conclude that:

> Because players can react to other players' past actions, repeated games allow for equilibrium outcomes that would not be an equilibrium in the corresponding one-shot game.

As I will show in Chapter 9, this idea of "agreements" between players that are enforced by mutual retaliation plays an important role in explaining the workings of cartels and collusive behavior. More generally, many agreements in a variety of social situations are

based not on formal contracts but rather on the trust that stems from repeated relationships. For example, in the diamond districts in Antwerp, New York City, and Tel-Aviv many high-value transactions are based on unwritten contracts, sometimes referred to as **relational contracts**.

Even more generally, the phenomenon of cooperation in society is frequently explained as the equilibrium of a prisoner's dilemma: individually and in the short run, each society member has an incentive to take advantage of the others. In practice, however, he or she refrains from doing so because the costs from being shunned by other members in the future outweigh the short-term gains.

To conclude, a note on concepts and terminology. There are two different interpretations for the role played by the parameter δ introduced earlier. One is that the game will end in finite time but the precise date when it will end is unknown to players; all they know is that, after each period, the game ends with probability $1 - \delta$. This is the interpretation I considered in this section and corresponds to the term *indefinitely*-repeated game. An alternative interpretation is that the game lasts for an infinite number of periods and that players discount future payoffs according to the discount factor δ: a dollar next period is worth δ dollars in the current period. This is the interpretation corresponding to the term "*infinitely*-repeated game."[o] Formally, if players are risk-neutral, then the two interpretations lead to the same equilibrium computation. I prefer the indefinitely-repeated game interpretation and terminology. However, it is rarely used.

7.4 INFORMATION

Consider the pricing dilemma faced by a health insurance company: a low price leads to small margins, but a high price runs the risk of attracting only high-risk patients. What should the insurance company do? Specifically, consider the following sequential game between an insurance company and a patient. First the insurance company decides whether to charge a high premium or a low premium; then, the patient decides whether or not to accept the company's offer.

So far, this looks like a sequential game like the ones considered in Section 7.2. But now we add a twist: the patient may either be a low-risk patient or a high-risk patient; moreover, the patient knows what type he is but the insurance company does not. Specifically, the insurance company believes that with probability 10% the patient is a high-risk patient. For a high-risk patient, health insurance is worth \$20,000 a year, whereas for a low-risk patient, the value is \$3,000. For the insurance company, the cost of providing insurance is \$30,000 for a high-risk patient and \$1,000 for a low-risk patient.

Notice that, based on the prior probabilities that the patient is high or low risk, the insurer estimates that the average valuation is given by $10\% \times 20 + 90\% \times 3 = 4.7$, whereas the average cost of insuring a patient is given by $10\% \times 30 + 90\% \times 1 = 3.9$. Based on these average, the insurance company is considering two possible prices:

o. The two interpretations are not mutually exclusive.

50%. Unlike Mars, Hershey knows the value of *b*. Draw the tree for this new game and determine its equilibrium.

■ **7.6. HERNAN CORTÉZ.** In a message to the king of Spain upon arriving in Mexico, Spanish navigator and explorer Hernan Cortéz reports that, "under the pretext that [our] ships were not navigable, I had them sunk; thus all hope of leaving was lost and I could act more securely." Discuss the strategic value of this action knowing the Spanish colonists were faced with potential resistance from the Mexican natives.

■ **7.7. HDTV STANDARDS.** Consider the following game depicting the process of standard setting in high-definition television (HDTV).[4] The US and Japan must simultaneously decide whether to invest a high or a low value into HDTV research. If both countries choose a low effort than payoffs are (4,3) for US and Japan, respectively; if the US chooses a low level and Japan a high level, then payoffs are (2,4); if, by contrast, the US chooses a high level and Japan a low one, then payoffs are (3,2). Finally, if both countries choose a high level, then payoff are (1,1).

(a) Are there any dominant strategies in this game? What is the Nash equilibrium of the game? What are the rationality assumptions implicit in this equilibrium?

(b) Suppose now the US has the option of committing to a strategy ahead of Japan's decision. How would you model this new situation? What are the Nash equilibria of this new game?

(c) Comparing the answers to (a) and (b), what can you say about the value of commitment for the US?

(d) "When pre-commitment has a strategic value, the player that makes that commitment ends up 'regretting' its actions, in the sense that, given the rivals' choices, it could achieve a higher payoff by choosing a different action." In light of your answer to (b), how would you comment this statement?

■ **7.8. FINITELY REPEATED GAME.** Consider a one-shot game with two equilibria and suppose this game is repeated twice. Explain in words why there may be equilibria in the two-period game which are different from the equilibria of the one-shot game.

■ **7.9. AMERICAN EXPRESS'S SPINOFF OF SHEARSON.** In 1993, American Express sold Shearson to Primerica (now part of Citigroup). At the time, the *Wall Street Journal* wrote that,

> Among the sticking points in acquiring Shearson's brokerage operations would be the firm's litigation costs. More than most brokerage firms, Shearson has been socked with big legal claims by investors who say they were mistreated, though the firm has made strides in cleaning up its backlog of investor cases. In 1992's fourth quarter alone, Shearson took reserves of $90 million before taxes for "additional legal provisions."[5]

When the deal was completed, Primerica bought most of Shearson's assets but left the legal liabilities with American Express. Why do you think the deal was structured this way? Was it fair to American Express?

■ **7.10. SALE OF BUSINESS.** Suppose that a firm owns a business unit that it wants to sell. Potential buyers know that the seller values the unit at either \$100 million, \$110 million, \$120 million, ... \$190 million, each value equally likely. The seller knows the precise value, but the buyer only knows the distribution. The buyer expects to gain from synergies with its existing businesses, so that its value is equal to seller's value plus \$10 million (In other words, there are gains from trade.) Finally, the buyer must make a take-it-or-leave-it offer at some price p. How much should the buyer offer?

CHALLENGING EXERCISES

■ **7.11. FIRST-PRICE AUCTION.** Consider the following auction game. There are two bidders who simultaneously submit bids b_i for a given object. Bidder i values the object at v_i; it knows its own value but not the other bidder's value. It is common knowledge that valuations v_i are uniformly drawn from the unit interval, that is, $v_i \sim U[0,1]$.

(a) Suppose that Bidder 1 expects Bidder 2's bid to be uniformly distributed between 0 and $\frac{1}{2}$. What is Bidder 1's optimal bid function (that is, bid as a function of valuation v_1)?

(b) If Bidder 2 expects Bidder 1 to follow the strategy derived in part (a), what is Bidder 2's belief about Bidder 1's bid levels?

(c) Determine the bidding game Nash equilibrium (assuming there is only one).

■ **7.12. AD GAMES.** Two firms must simultaneously choose their advertising budget; their options are H or L. Payoffs are as follows: if both choose H, then each gets 5; if both choose L, then each gets 4; if firm 1 chooses H and firm 2 chooses L, then firm 1 gets 8 and firm 2 gets 1; conversely, if firm 2 chooses H and firm 1 chooses L, then firm 2 gets 8 and firm 1 gets 1.

(a) Determine the Nash equilibria of the one-shot game.

(b) Suppose the game is indefinitely repeated and that the relevant discount factor is $\delta = .8$. Determine the optimal symmetric equilibrium.

(c) (challenge question) Now suppose that, for the first 10 periods, firm payoffs are twice the values represented in the above table. What is the optimal symmetric equilibrium?

■ **7.13. FINITELY REPEATED GAME.** Suppose that the game depicted in Figure 7.1 is repeated T times, where T is known. Show that the only subgame perfect equilibrium is for players to choose (B, R) in every period.

■ **7.14. CENTIPEDE.** Consider the game in Figure 7.13.[6] Show, by backward induction, that rational players choose d at every node of the game, yielding a payoff of 2 for Player 1 and 0 for Player 2. Is this equilibrium reasonable? What are the rationality assumptions implicit in it?

■ **7.15. ADVERTISING LEVELS.** Consider an industry where price competition is not very important: all of the action is on advertising budgets. Specifically, total value S (in dollars) gets splits between two competitors according to their advertising shares. If a_1 is firm 1's advertising investment (in dollars), then its profit is given by:

$$\frac{a_1}{a_1 + a_2} S - a_1$$

(The same applies for firm 2). Both a_1 and a_2 must be non-negative. If both firms invest zero in advertising, then they split the market.

(a) Determine the symmetric Nash equilibrium of the game whereby firms choose a_i independently and simultaneously.

(b) Determine the jointly optimal level of advertising, that is, the level a^* that maximizes joint profits.

(c) Given that firm 2 sets $a_2 = a^*$, determine firm 1's optimal advertising level.

(d) Suppose that firms compete indefinitely in each period $t = 1, 2, \ldots$, and that the discount factor is given by $\delta \in [0, 1]$. Determine the lowest value of δ such that, by playing grim strategies, firms can sustain an agreement to set a^* in each period.

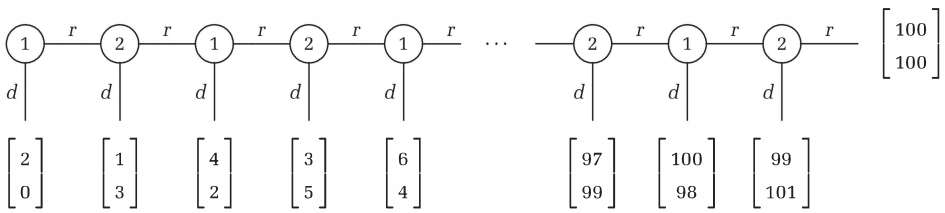

FIGURE 7.13
The centipede game. In the payoff vectors, the top number is Player 1's payoff, the bottom one Player 2's.

a firm that sells to other firms. In fact, there are cases when most of the market power is on the buyer's side: large supermarket chains, for example, have a great degree of market power with respect to suppliers.

These reasons justify the separate treatment of **vertical relations** between firms. By "vertical relations" I mean relations between two firms in sequence along the value chain, as in the examples above. Normally, I will refer to the case of a manufacturer selling to one or several retailers. However, the analysis in this chapter applies more generally to cases when an **upstream firm** (e.g., cement producer, flour producer) sells to a **downstream firm** (e.g., concrete producer, bakery).

The chapter consists of three sections. First, I consider the possibility of vertical integration between upstream and downstream firms. Next, I discuss vertical restraints as a means to achieve similar results without the need to resort to full integration. Finally, I examine public policy towards vertical integration and vertical restraints.

13.1 VERTICAL INTEGRATION

Consider a structure consisting of an upstream firm (M) and a downstream firm (R). Firm R is best thought of as a retailer, while firm M could be the manufacturer of the product, a wholesaler, or the supplier of an important input to firm R. For example, M could be an oil refiner and R a gas station.

Suppose there is a demand for the final product (supplied by R), given by $D(p)$. Regarding the production technology, let us assume the simplest possible structure: in order to produce one unit of output, R needs one unit of input. In fact, under the manufacturer/retailer interpretation this is the right assumption: in order to sell one TV set, a retailer must obtain exactly one TV set from the manufacturer. Suppose that R has no costs in addition to the wholesale price, w, that it pays its supplier.[a] Finally, firm M has a constant marginal cost c.

The situation is illustrated in Figure 13.1. Wanting to make a positive profit, the wholesaler sets w above marginal cost c, that is $w > c$. The retailer, in turn, sets optimal output such that its marginal cost, w, equals its marginal revenue. This results in retail price p^R. In this equilibrium, firm M earns a variable profit equal to area A, whereas firm R earns a variable profit equal to area B. Together, the "vertical chain" (firms M and R combined) earn a profit $A+B$.

Now suppose that firms M and R are integrated as firm $M\&R$ with the single goal of maximizing joint profit. This new monopolist inherits firm R's demand curve, $D(p)$, and firm M's marginal cost, c. Its optimal choice is given the intersection of marginal revenue and marginal cost. This corresponds to output level $q^{M\&R}$ and retail price $p^{M\&R}$.[b]

In the vertical integration equilibrium, firm $M\&R$ makes a profit equal to areas $A + C$ in Figure 13.1. Since C is greater than B, we conclude that firms M and R increase their joint profit by vertically integrating. The idea is that, under vertical integration, the wholesale price is simply a transfer price; therefore, it has no impact on retail price formation. For this reason, firm $M\&R$ is able to set the level of p that maximizes total

a. Later, I will consider more complex contractual relationships between manufacturer and retailer; for the time being, I consider the simplest case of linear pricing (cf Section 6.3).

b. Retail price happens to be equal to firm M's optimal wholesale price under vertical separation. This is true when the demand curve is linear, but is not true more generally.

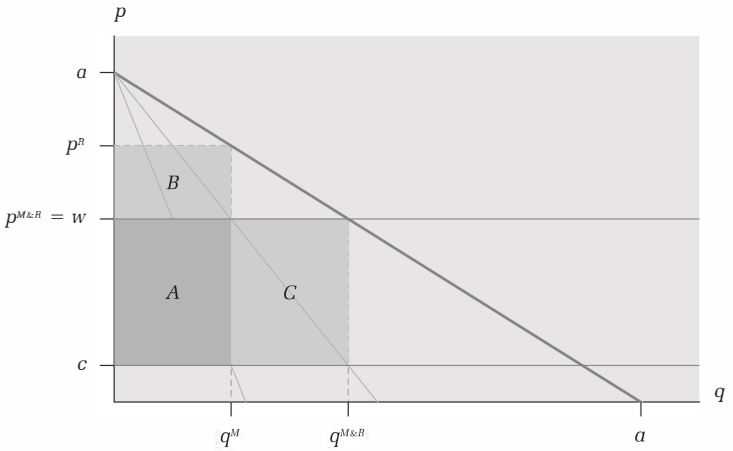

FIGURE 13.1
Double marginalization. If M and R are two different firms, then M sets a wholesale price w and R sets a retail price p^R. If M and R are part of the same firm, then retail price is given by $p^{M\&R}$ (which, if the demand curve is linear, is equal to w).

value along the value chain. By contrast, under vertical separation, firm M sets wholesale price w above marginal cost c; and firm R determines retail price p based on a cost (w) that is higher than the "real" marginal cost (c).

We conclude that:

> If a manufacturer sets a wholesale price to a vertically separated retailer, then their joint profits are lower, and retail price is higher, than under vertical integration.

This problem, known as the **double marginalization problem**, provides a first justification for vertical integration.[1] Under vertical separation, there are two monopoly pricing decisions in effect. If the only contractual instrument that M and R can use is the wholesale price, then two monopoly margins will be added to marginal cost, resulting in a price that is greater than monopoly price. As a consequence, the combined profits of M and R are lower than if they were integrated. Exercise 13.7 illustrates this problem and more.

DOUBLE MARGINALIZATION: FORMAL ANALYSIS. The retailer's problem is identical to a monopolist with marginal cost w. Suppose final demand is given by $D = a - p$. From Section 3.2, we know that optimal price is given by:

$$p = \frac{a + w}{2}$$

which results in:

$$q = \frac{a - w}{2}$$

For the manufacturer, this implies a profit of:

$$\pi_M = (w - c)\, q = (w - c)\, \frac{a - w}{2}$$

BOX 13.1 REGIONAL SPORTS NETWORKS[3]

Nearly 90% of the 116.4 million television households in the US subscribe to multichannel television. Distributors such as Comcast or TimeWarner negotiate affiliate fees with content providers such as Disney or ESPN. These are typically linear contracts; that is, they correspond to a per-subscriber fee.

Among the $30 billion per year in fees, about $4.1 billion correspond to regional sports networks (RSNs). RSNs are particularly interesting from an empirical research point of view because there is variation in ownership and carriage patterns. For example, in 2007, Comcast Sports Northwest (wholly owned by Comcast) was carried by Comcast but not by DirectTV or Dish; and independently-owned YES (Yankees Entertainment and Sports) was carried by TimeWarner and DirectTV but not by Dish.

This variation allows us to answer the question: does it matter if distributor and content provider are vertically integrated? The evidence suggests that, when there is vertical integration, (a) the price paid by consumers is lower; (b) the likelihood that the RSN is carried by the distributor is higher; and (c) the likelihood that the RSN is carried by rival distributors is lower.

The manufacturer's problem is identical to a monopolist with marginal cost c and demand $q = \frac{1}{2}(a - w)$. From Section 3.2, we know that optimal price is given by:

$$w = \frac{a + c}{2}$$

Figure 13.1 considers the case when $c = 1$ and $a = 9$. This results in $w = 5$, $p^R = 7$, $q^M = 2$, and $q^{M\&R} = 4$. In terms of profit levels, $A = 8$, $B = 4$, and $C = 8$. (Check this.) It follows that vertical integration increases total profits by $C - B = 4$ (a 33% increase).[c]

The movie industry provides an example of the effects of double marginalization: in the 1940s and 1950s, US movie ticket prices were lower at theaters owned by studios.[2] Eventually, a US Supreme Court decision forced the studios to sell their theater assets. Clearly, the court judged that double marginalization was not the only relevant effect. I will return to this in Section 13.3.

Box 13.1 provides additional evidence, this time regarding the US cable TV industry. When distributor (firm R) and supplier (firm M) are owned by the same entity (vertical integration), the price paid by consumers is lower. This is consistent with the idea that vertical integration alleviates the double marginalization problem. The evidence also suggests that the likelihood that a given regional sports network is carried by a distributor (that is, the firm M's product is sold by firm R) is greater under vertical integration. This too corroborates the double-marginalization-effect story, as shown in Exercise 13.6.

c. In fact, consumer welfare is also lower under vertical separation, since price is greater than monopoly price.

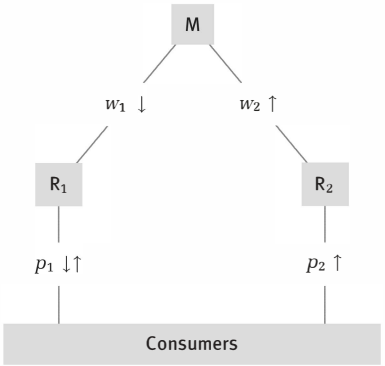

FIGURE 13.2
Vertical integration with downstream competition:
arrows indicate expected sign of change in variables following integration between M and R_1.

■ **DOWNSTREAM COMPETITION.**[4] Things get a little more complicated if there is more than one downstream firm. Figure 13.2 depicts the case when an upstream firm (manufacturer) M sells to two downstream firms (retailers) R_1 and R_2. Let w_i be the wholesale price paid by retailer R_i and let p_i be its retail price. Suppose that firm M merges with retailer R_i; what impact do we expect this to have on prices?

The first effect we would expect is a decrease in w_1. This corresponds to the double-marginalization effect considered earlier: once M and R_1 have a single profit maximization goal, one of the intermediation margins, w_1, ceases to exist. There may still be a transfer price w_1 used for accounting purposes; but as a decision problem, firm $M\&R_1$ no longer considers two price margins.

The second effect we would expect is an increase in w_2. Firm R_2, which was a customer of firm M, is now also a *rival* of firm $M\&R_1$. An increase in w_2 has an effect on firm M's revenues, an effect that previously firm M was not taking into account. Now that firm M is united to firm R_1, an increase in w_2 also has an effect on downstream competition: it puts pressure on R_2 to increase p_2, which in turn helps R_1, which in turn helps $M\&R_1$. In other words, an increase in w_2 is a way of **raising rivals' costs**, a strategy we discussed in Section 12.2.

Finally, when it comes to p_1 we find conflicting effects. Vertical integration eliminates double marginalization, which in turn provides a downward pressure on p_1 (as we saw earlier in this section). However, vertical integration also has a **competition softening** effect that tends to push price up. When R_1 considers an increase in price, it balances an increase in margin with a decrease in market share: the basic pricing trade-off we considered in Section 3.2. Now that R_1 is part of $M\&R_1$, it should consider an additional effect: by increasing price and losing market share, R_1 is giving R_2 some market share. This in turn implies that R_2 must order more input; and to the extent that $w_2 > c$, this extra input order increases the profits of the combined $M\&R_1$.

Overall, we expect the profits of $M\&R_1$ to increase and the profits of R_2 to decrease. Regarding consumers, we have conflicting effects: getting rid of double marginalization is good, but softening downstream competition is bad; in principle, the net effect can go either way.[d]

■ **INVESTMENT INCENTIVES.**[5] So far we discussed pricing incentives along the value chain. In some cases investment incentives are also important. Suppose that firm M comes up with a new electronic gadget worth v to consumers, but consumers will only learn about it if firm R makes an investment (for example, training its sales force to sell the new gadget). Alternatively, suppose that firm R comes up with a new car model worth v to consumers, but production is only possible if firm M makes an investment (for example, designing and building a mould required to produce a car part). Suppose moreover that these investments are **specific assets**, a concept introduced in Section 3.4: the sales force training is only useful for selling firm M's gadget; and the mould is only useful for making firm R's car part.

Let us continue with the example of the car manufacturer R outsourcing a part to manufacturer M. Frequently there are many contingencies which are difficult to predict and write a contract on (for example, how much it will it cost to produce the part or how many cars will be demanded). Accordingly, at some point in the future the exact terms of the transaction between M and R will need to be determined; whereas M's investment decision must be made from the get-go. Then we have a problem.

Specifically, suppose that the investment costs $c = 5$, whereas the total value created by the investment is $v = 8$ (a value which is appropriated by firm R upon sale). Clearly, this is a worthwhile investment. However, once the investment is made it is effectively a sunk cost, which in turn induces a **hold-up** problem (cf Section 3.4): suppose that R makes a take-it-or-leave-it offer to M, either $p^H = 6$ or $p^L = 3$. The situation is represented by the game in Figure 13.3. The only subgame-perfect equilibrium of this game (cf Section 7.2) is for R to offer p^L and for M to choose $\bar{a}$. This is an inefficient equilibrium: total payoff is zero, whereas the choice of a would lead to a total payoff of 3.

In this context, vertical integration may help solve the hold-up problem created by investments in specific assets: suppose that R acquires M. Now decisions such as the investment level, the number of parts, etc., are made by R, whereas M's manager simply executes orders. Taking the $M\&R$ group together, R's manager correctly determines that the investment a is worthwhile (it costs 5 and generates benefits of 8); and accordingly, the efficient level of investment is chosen.

If vertical integration solves some problems, it may also create new ones. Continuing with the same example: upon acquisition by the downstream firm, M's manager is at the orders of R's manager. As we saw in Section 3.3, this creates an **agency** problem: Firm R's manager wants Firm M (now a division of $M\&R$) to act in R's interest; but Firm R may not know exactly what to order, since Firm M's manager is better informed than Firm R's manager.

d. All of the above analysis assumes that firms compete in prices. The case when firms are quantity setters in analyzed in Exercise 13.9.

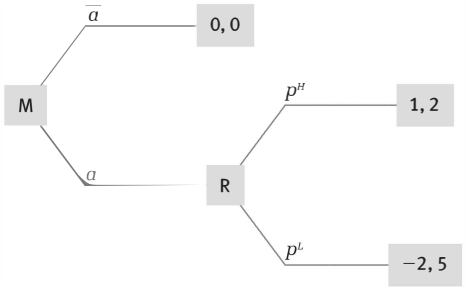

FIGURE 13.3
Specific assets and hold-up:
the upstream firm, M, decides whether to make an investment in a specific asset;
next, the downstream firm, R, decides what price to pay for the asset.

In sum,

> When investments in specific assets are at stake, vertical integration alleviates the hold-up problem but increases the agency problem.

The above analysis of the hold-up problem is based on a very stylized example. However, the basic ideas are far more general. For example, we could assume that, instead of a take-it-or-leave-it offer by R, firms M and R negotiate over transfer payments and split the cake (that is, the value at stake). If M chooses a, then a value $v = 8$ is at stake. If M and R split the cake, then R gets 4. This is still not sufficient to compensate for its initial investment of 5, so the unique subgame-perfect equilibrium is, again, for M not to invest.[e]

■ **VERTICAL INTEGRATION: THE EVIDENCE AND MORE THEORY.** In many industries, the last decades of the twentieth century witnessed a significant increase in the practice of **outsourcing**, purchasing inputs from an outside supplier (even though those inputs are available internally). In fact, currently in the US almost one-half of upstream establishments make no internal shipments to their downstream integrated establishments.[6] This evidence runs counter to the idea of vertical integration as a means to increase efficiency in vertical transactions (namely by reducing double marginalization).

Much of the outsourcing shift was directed at foreign suppliers (that is, outsourcing took the form of offshoring). In this sense, one explanation for the trend is simply that firms were seeking cheaper suppliers. However, recently much of that outsourcing has shifted back to the US, and the lack on internal flows is still very noticeable.

If vertical ownership is not associated to vertical flows, why then do firms vertically integrate? One explanation is that integration allows for the internal transfer of intangible inputs such as management ability, marketing know-how, intellectual property, and R&D capital. This would then suggest that the effects of vertical integration

e. Exercise 13.10 considers still another, namely the case of a continuum of choices.

have less to do with strategic interaction in setting prices and output levels than with organizational efficiency issues.

However, before discarding strategic behavior as a driver of vertical integration, we should note that, even if vertical integration is not associated with internal flows, it does imply the *option* of internal flows. In other words, the advantage of being vertically integrated with a potential supplier is that it forces outsource suppliers to be more efficient and bid more aggressively — or else the buyer switches to the internal source.[7]

13.2 VERTICAL RESTRAINTS

In the previous section, we saw how vertical integration may increase the efficiency of pricing along a value chain. But: is vertical integration *required* for efficient pricing? In many cases it is not; in this section, I show how a variety of **vertical restraints** may also do the job.

■ **NONLINEAR PRICING.** Suppose that, in addition to determining a wholesale price, the upstream firm, M, can also set a fixed fee f to be paid by the downstream firm, R, in case it wants to do any business with M. As in Section 6.3, we call the pair (f, w) a **two-part tariff**. Often, the fixed fee is also called **franchise fee** (although, in fact, franchise contracts involve more than just setting a fixed fee). More generally, as in Section 6.3, contracts of this type are an example of a **nonlinear pricing**.[f]

Suppose that the manufacturer sets $w = c$ and $f = \pi^M$, where $\pi^M \equiv (p^M - c)\,D(p^M)$ is monopoly profit for a vertically integrated firm. In terms of Figure 13.1, this corresponds to setting a fixed fee equal to the area $A + C$. There are three points to notice about this nonlinear contract: first, from the point of view of the firms, this contract is efficient, that is, it maximizes their joint profits. This is so because the marginal cost taken into consideration by the retailer (the wholesale price w) is equal to the true marginal cost (c); accordingly, the retailer sets price at the optimal level, that is, at the level that a vertically integrated monopolist would set (p^M). Second, by pricing this way, the retailer receives total gross profits $\pi = \pi^M$ (gross of the franchise fee); therefore, it is willing to pay up to $f = \pi^M$ as a fixed fee. Finally, the manufacturer receives zero in terms of variable profit (because it sells at marginal cost) but is able to recover all monopoly profits through the fixed fee.

We thus conclude that,

> If nonlinear contracts are possible, then the optimal solution under vertical separation is identical to that under vertical integration.

In fact, a more general point is that, *allowing for fixed fees, the upstream firm's profit maximization problem is essentially equivalent to maximizing joint profits of the*

f. Strictly speaking, the amount paid, $f + wq$, is a linear function of the quantity bought. The key point of a nonlinear contract is that price per unit ($w + f/q$ in the case of a two-part tariff) is not constant.

Now suppose that Microsoft and Intel sit down to negotiate an agreement to sell Windows and Pentium chips as a package to computer OEMs for a package price of p_{MI}.

(b) What package price would maximize Microsoft's and Intel's combined profits? By how much would an agreement between Microsoft and Intel boost their combined profits?

(c) Would final consumers benefit from such an agreement between Microsoft and Intel, or would they be harmed? What about computer OEMs? Relate your answer to your calculations in parts (a) and (b), and explain the economic principles underlying your answer.

■ **13.8. TWO-PART TARIFFS.** Suppose that a manufacturer sells to n retailers by means of a two-part tariff (f, w): a fix fee f and a wholesale price w. Explain the intuition of the result that the greater the degree of retailer competition, the greater the optimal wholesale price.

■ **13.9. DOWNSTREAM COURNOT COMPETITION.** Consider an industry with one upstream firm and n downstream firms that compete à la Cournot. Show that the optimal wholesale price is strictly between marginal cost and monopoly price.

■ **13.10. ASSET SPECIFICITY AND VERTICAL INTEGRATION.** The value created by the collaboration between firms M and R is given by $v = \sqrt{e_M} + \sqrt{e_R}$, where e_M and e_R are investments (in \$) by firms M and R. After investment levels have been chosen, M and R negotiate the division of v, each firm keeping 50%.

(a) Determine the equilibrium values of e_i $(i = M, R)$.

(b) Suppose that Firm M and R merge. Determine Firm $M\&R$'s optimal choices of e_i.

(c) Compare total payoff under vertical separation and under vertical integration.

(d) Comment on your answer to the previous question in light of the discussion in Section 13.1, Box 13.1, and Section 5.1.

NOTES

1. Spengler, Joseph (1950), "Vertical Integration and Antitrust Policy," *Journal of Political Economy* **58**, 347–352.

2. Gil, Ricard (2015), "Does Vertical Integration Decrease Prices? Evidence from the Paramount Antitrust Case of 1948," *American Economic Journal: Economic Policy* **7** (2), 1–1.

Varian Calc

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- Pareto efficient allocations; the contract curve: §32.3 (p. 605–609)
- 32 RQ 3: Ch. 32, Review Questions (p. 627–628)
- Commitment; hold up; bargaining (Rubinstein); the ultimatum game: Ch. 30, "Games of Commitment" to "T
- Quasilinear preferences and the Coase theorem: §35.2 (p. 670–672)
- 35 RQ 1–2: Ch. 35, Review Questions (p. 687–687)

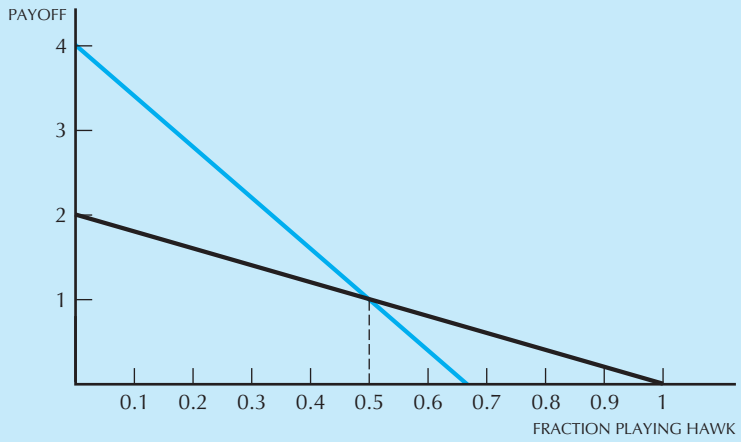


Figure 30.5 **Payoffs in the hawk-dove game.** The payoff to hawk is depicted in color; the payoff to dove is in black. When $p > 1/2$, the payoff to hawk is less than dove and vice versa, showing that the equilibrium is stable.

30.6 Games of Commitment

The previous examples involving games of cooperation and competition have been concerned with games with **simultaneous moves**. Each player had to make his or her choice without knowing what the other player was choosing (or had chosen). Indeed, games of coordination or competition can be quite trivial if one player knows the other's choices.

In this section we turn our attention to games with **sequential moves**. An important strategic issue that arises in such games is **commitment**. To see how this works, look back at the game of chicken described earlier in this chapter. We saw there that if one player could force himself to choose straight, the other player would optimally choose to swerve. In the assurance game, the outcome would be better for both players if one of them moved first.

Note that this committed choice must be both irreversible and observable by the other player. Irreversibility is part of what it means to be committed, while observability is crucial if the other player is going to be persuaded to change his or her behavior.

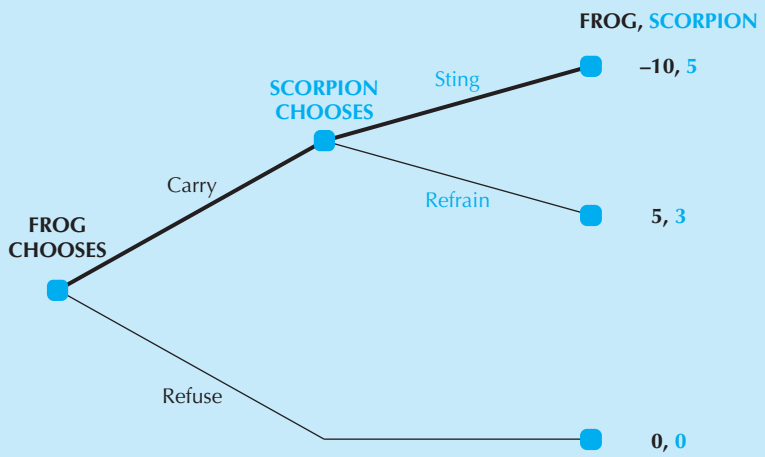
The Frog and the Scorpion

We begin with the fable of the frog and the scorpion. They were standing on the bank of the river, trying to figure out a way across. "I know," said

the scorpion "I will climb on your back and you can swim across the river." The frog said, "But what if you sting me with your stinger?" The scorpion said, "Why would I do that? Then we would both die."

The frog found this convincing, so the scorpion climbed on his back and they started across the river. Halfway across, at the deepest point, the scorpion stung the frog. Writhing in pain, the frog cried out, "Why did you do that? Now we are both doomed!" "Alas," said the scorpion, as he sank into the river, "it is my nature."

Let's look at the frog and the scorpion from the viewpoint of game theory. Figure 30.6 depicts a sequential game with payoffs consistent with the story. Start at the bottom of the game tree. If the frog refuses the scorpion, both get nothing. Looking up one line, we see that if the frog carries the scorpion, he receives utility 5, for doing a good deed, and the scorpion receives a payoff of 3, for getting across the river. In line where the frog is stung, he receives a payoff of −10, and the scorpion gets a payoff of 5, representing the satisfaction from fulfilling his natural instincts.



The frog and the scorpion. If the frog chooses to carry the scorpion, the scorpion will choose to sting him and both will die.

Figure 30.6

It is best to start with the final move of the game: the scorpion's choice of sting or refrain. Stinging has a higher payoff to the scorpion because "it is his nature" to sting. Hence the frog should rationally choose to refuse to carry the scorpion. Unfortunately, the frog didn't understand the scorpion's payoffs; apparently, he thought that the scorpion's payoffs

looked something like those in Figure 30.7. Alas, this mistake was fatal for the frog.

A smart frog would figure out some way to make the scorpion commit to not stinging. He could, for example, tie his tail. Or he could hire a hit frog, who would retaliate against the scorpion's family. Whatever the strategy, the critical thing for the frog to do is to change the payoffs to the scorpion by making stinging more costly or refraining more rewarding.

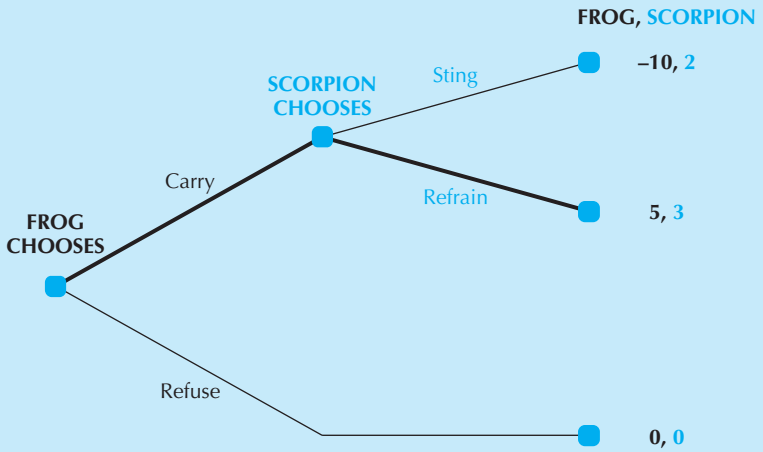


Figure 30.7 **The frog and the scorpion**. With these payoffs, if the frog chooses to carry the scorpion, the scorpion will not choose to sting him, and both will make it across the river safely.

The Kindly Kidnapper

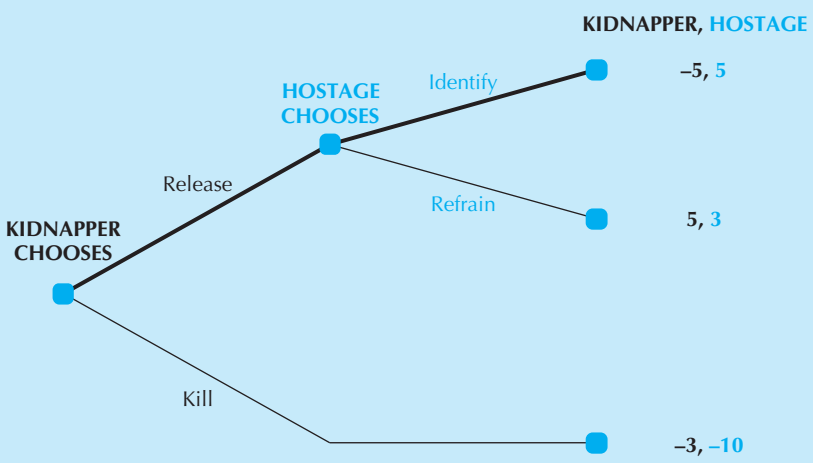
Kidnapping for ransom is a big business in some parts of the world. In Columbia, it is estimated that there are over 2,000 kidnappings for ransom per year. In the former Soviet Union, kidnappings rose from 5 in 1992 to 105 in 1999. Many of the victims are Western businesspeople.

Some countries, such as Italy, have laws against paying ransom. The reasoning is that if the victim's family or employers can commit themselves not to pay ransom, then the kidnappers will have no motive to abduct the victim in the first place.

The problem is, of course, once a kidnapping has taken place, a victim's family will prefer to pay the kidnappers, even if it is illegal to do so. Hence penalties for paying ransom may not be effective as a commitment device.

Suppose some kidnappers abduct a hostage and then discover that they can't get paid. Should they release the hostage? The hostage, of course, promises not to reveal the identity of the kidnappers. But will he keep this promise? Once he is released, he has no incentive to do so—and every incentive to try to punish the kidnappers. Even if the kidnappers want to let the hostage go, they can't do so for fear of being identified.

Figure 30.8 depicts some possible payoffs. The kidnapper would feel bad about killing the hostage, receiving a payoff of −3. Of course, the hostage would feel even worse, receiving a payoff of −10. If the hostage is released, and refrains from identifying the kidnapper, the hostage gets a payoff of 3 and the kidnapper gets a payoff of 5. But if the hostage does identify the kidnapper, he gets a payoff of 5, leaving the kidnapper with a payoff of −5.



Kidnap game. The kidnapper would like to release the hostage, but if he does, the hostage will identify him.

Figure 30.8

Now it is the hostage who has the commitment problem: how can he convince the kidnappers that he won't renege on his promise and reveal their identity?

The hostage needs to figure out a way to change the payoffs of the game. In particular, he needs to find a way to impose a cost on himself if he identifies the kidnappers.

Thomas Schelling, an economist at the University of Maryland who has worked extensively on strategic analysis in dynamic games, suggests that the hostage might have the kidnappers photograph him in some embarrassing act and leave them with the photos. This effectively changes the payoffs

from his subsequently revealing the identity of the kidnappers, since they then have the option of revealing the embarrassing photograph.

This sort of strategy is known as an "exchange of hostages." In the Middle Ages, when two kings wanted to ensure a contract wouldn't be broken, they would exchange hostages such as family members. If either king broke the agreement, the hostages would be sacrificed. Neither wanted to sacrifice their family members, so each king would have an incentive to respect the terms of their contract.

In the case of the kidnapping, the embarrassing photo would impose costs on the hostage if it were released, thereby ensuring that he will stick to his agreement not to reveal the identity of the kidnappers.

When Strength Is Weakness

Our next example comes from the world of animal psychology. It turns out that pigs quickly establish dominance-subordinateness relations, in which the dominant pig tends to boss the subordinate pig around.

Some psychologists put two pigs, one dominant, one subordinate, in a long pen.[2] At one end of the pen was a lever that would release a portion of food to a trough located at the other end of the pen. The question of interest was this: which pig would push the lever and which would eat the food?

Somewhat surprisingly the outcome of the experiment was that the dominant pig pressed the lever, while the subordinate pig waited for the food. The subordinate pig then ate most of the food, while the dominant pig rushed as fast as it could to the trough end of the pen, ending up with only a few scraps. Table 30.9 depicts a game that illustrates the problem.

Table 30.9

Pigs pressing levers.

		Dominant Pig: Don't press lever	Dominant Pig: Press lever
Subordinate Pig	Don't press lever	0, 0	4, 1
	Press lever	0, 5	2, 3

[2] The original reference is Baldwin and Meese, "Social Behavior in Pigs Studied by Means of Operant Conditioning," (*Animal Behavior,* (1979)). I draw on the description of John Maynard Smith, *Evolution and the Theory of Games* (Cambridge University Press, 1982).

The subordinate pig compares a payoff of (0, 4) to (0, 2) and concludes, sensibly enough, that pressing the lever is dominated by not pressing it. Given that the subordinate pig doesn't press the lever, the dominant pig has no choice but to do so.

If the dominant pig could refrain from eating all the food and reward the subordinate pig for pressing the lever, it could achieve a better outcome. The problem is that pigs have no contracts, and the dominant pig can't help being a hog!

As in the case of the kindly kidnapper, the dominant pig has a commitment problem. If he could only commit to not eating all the food, he would end up much better off.

Savings and Social Security

Commitment problems aren't limited to the animal world. They also show up in economic policy.

Saving for retirement is an interesting and timely example. Everyone gives lip service to the fact that saving is a good idea. Unfortunately, few people actually do it. Part of the reason for the reluctance to save is that individuals recognize that society won't let them starve, so there is a good chance they will be bailed out later on.

To formulate this in a game between the generations, let's consider two strategies for the older generation: save or squander. The younger generation likewise has two strategies: support their elders or save for their own retirement. A possible game matrix is shown in Table 30.10.

Intergenerational conflict over savings. **Table 30.10**

		Younger Generation: Support	Younger Generation: Refrain
Older Generation	Save	2, –1	1, 0
	Squander	3, –1	–2, –2

If the older generation saves and the younger generation also supports them, the old folks end up with a utility level of 2 and the young folks end up with −1. If the older generation squanders and the younger generation supports them, the elders end up with a utility of 3 and the young folks end up with −1.

If the younger generation refrains from providing support to their elders and the older generation saves, the old folks get 1 and the young folks get 0. Finally, if the old folks squander and the young folks neglect them, each ends up with utility of −2, the old folks from starving and the young folks from having to watch.

It is not hard to see that there are two Nash equilibria in this game. If the old folks choose to save, then the young folks will choose optimally to neglect them. But if the old folks choose to squander, then it is optimal for the younger generation to support them. And of course, given that the younger generation will support their elders, it is optimal for their elders to squander!

However, this analysis ignores the time structure of the game: one of the (few) advantages of being old is that you get to move first. If we draw out the game tree, the payoffs become those in Figure 30.9.

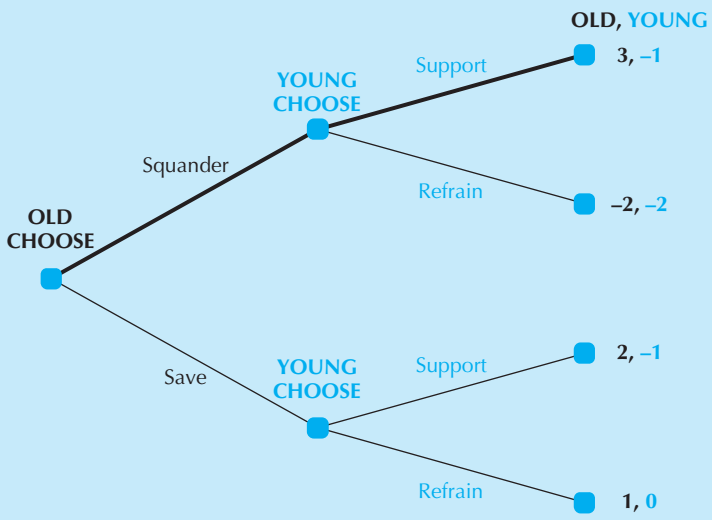


Figure 30.9 **The savings game in extended form.** Knowing that the younger generation will support them, the older generation chooses to squander. The subgame perfect equilibrium is (support, squander).

If the oldsters save, the youngsters will choose to neglect them, so the oldsters end up with a payoff of 1. If the oldsters squander, they know that the youngsters won't be able to bear watching them starve, so the oldsters end up with a payoff of 3. Hence the sensible thing for the oldsters to do is to squander, knowing they will be bailed out later on.

Of course, most developed countries now have a program like the U.S. Social Security program that forces each generation to save for retirement.

EXAMPLE: Dynamic inefficiency of price discrimination

Recall the definition of first-degree price discrimination: the seller prices the good so that it extracts the entire consumer surplus from the buyer. We argued in Chapter 14 that this was efficient since all valuable trades were carried out.

Despite this, there are cases where first-degree price discrimination can lead to inefficiency. Suppose an salesman is such a good bargainer that he can extract the entire consumer surplus from each customer—he is an expert at first-degree price discrimination. But here's the problem: if everyone knows that he can extract all the consumer surplus, why would they ever want to buy something from him? At best, they would just break even.

This is a bit like the example of the dominant pig in the previous section. The dominant pig ended up in a worse position because it could not commit to sharing. Well, people are (usually) more intelligent than pigs, so they realize that if they want some repeat business they should make sure their customers get some surplus out of the transaction.

To do this it is necessary to find a way to commit to offering a good deal to the customers, perhaps by offering a coupon that can be used after the negotiation has taken place or by having a posted price that is available to everyone.

Hold Up

Consider the following strategic interaction. You hire a contractor to build a warehouse. After the plans are approved and the construction is almost done, you realize that the color is bad, so you ask the contractor to change the paint, which involves a trivial expense. The contractor comes back and says: "That change order will be $1500, please."

You recognize that it will cost you at least that much to delay completion until you can find a painter, and you really do want the new color, so, muttering under your breath, you pay the cost. Congratulations, you have been held up!

Of course, contractors are not the only party at fault in this sort of game. The clients can "hold up" their payment as well, causing lots of grief for the contractor.

The game tree for the hold-up problem is depicted in Figure 30.10. We suppose that the value the owner places on having the new paint is $1500 and that the actual cost of painting is $200. Starting at the top of leaves

of the tree, if the contractor charges $1500, it will realize a profit of $1300, and the client gets a net utility of zero.

If the client looks for another painter, it will cost him $200 to pay the painter and, say, $1400 in lost time. He gets the color he wants which is worth $1500, buy has to pay $1600 in direct costs and delay costs, leaving him with a net loss of $100.

If the contractor charges the client the actual cost of $200, he breaks even and the client gets a $1500 value for $200, leaving him with a net payoff of $1300.

As can be seen, the optimal choice for the contractor is to extort the payment, and the optimal choice for the customer is to give in. But a sensible client will recognize that change orders will occur in any project. Because of this, the client will be reluctant to hire contractors with a reputation for extortion which is, of course, bad for the contractor.

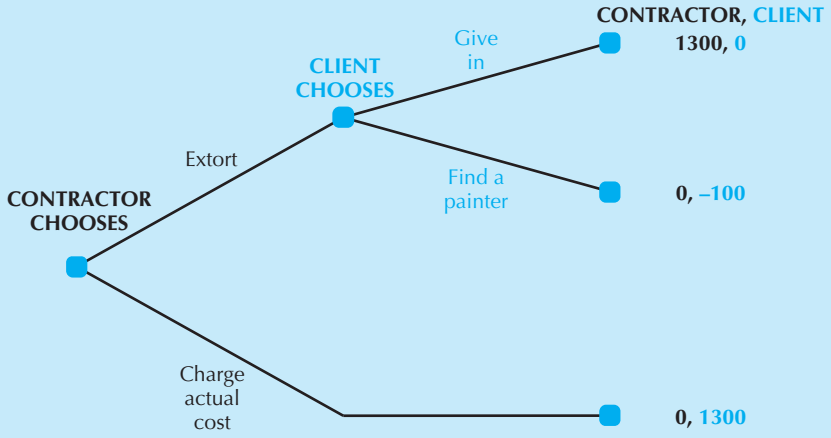


Figure 30.10 **The hold-up problem**. The contractor charges a high price for the change since the client has no alternative.

How do firms solve the hold-up problem? The basic answer is contracts. Normally, contractors negotiate a contract specifying what kinds of change orders are appropriate and how their costs will be determined. Sometimes there are even arbitration or other dispute resolution procedures built into the contracts. A lot of time, energy, and money goes into writing contracts just to make certain that hold up won't occur.

But contracts aren't the only solution. Another way to solve the problem is through commitment. For example, the contractor might post a bond

guaranteeing timely completion of the project. Again, there will generally be some objectively specified terms about what constitutes completion.

Another important factor is reputation. Obviously, a contractor who persistently tries to extort his customers will get a bad reputation. He won't be hired again by this customer, and he certainly won't get good recommendations. This reputation effect can be examined in a repeated game context in which hold up today will cost the contractor in the future.

30.7 Bargaining

The classical bargaining problem is divide the dollar. Two players have a dollar that they want to divide between them. How do they do it?

The problem, as stated, has no answer since there is too little information to construct a reasonable model. The challenge in modeling bargaining is to find some other dimensions on which the players can negotiate.

One solution, the **Nash bargaining model**, takes an axiomatic approach by specifying certain properties that a reasonable bargaining solution should have and then proving that there is only one outcome that satisfies these axioms.

The outcome ends up depending on how risk averse the players are and what will happen if no bargain is made. Unfortunately, a full treatment of this model is beyond the scope of this book.

An alternative approach, the **Rubinstein bargaining model**, looks at a sequence of choices and then solves for the subgame perfect equilibrium. Luckily the basic insight of this model is easy to illustrate in simple cases.

Two players, Alice and Bob, have \$1 to divide between them. They agree to spend at most three days negotiating over the division. The first day, Alice will make an offer, Bob either accepts or comes back with a counteroffer the next day, and on the third day Alice gets to make one final offer. If they cannot reach an agreement in three days, both players get zero.

We assume Alice and Bob differ in their degree of impatience: Alice discounts payoffs in the future at a rate of α per day, and Bob discounts payoffs at a rate of β per day. Finally, we assume that if a player is indifferent between two offers, he will accept the one that is most preferred by his opponent. This idea is that the opponent could offer some arbitrarily small amount that would make the player strictly prefer one choice and that this assumption allows us to approximate such an "arbitrarily small amount" by zero. It turns out that there is a unique subgame perfect equilibrium of this bargaining game.

We start our analysis at the end of the game, right before the last day. At this point Alice can make a take-it-or-leave-it offer to Bob. Clearly, the optimal thing for Alice to do at this point is to offer Bob the smallest possible amount that he would accept, which, by assumption, is zero. So if

the game actually lasts three days, Alice would get \$1 and Bob would get zero (i.e., an arbitrarily small amount).

Now go back to the previous move, when Bob gets to propose a division. At this point Bob should realize that Alice can guarantee herself \$1 on the next move by simply rejecting his offer. A dollar next period is worth α to Alice this period, so any offer less than α would be sure to be rejected. Bob certainly prefers $1-\alpha$ now to zero next period, so he should rationally offer α to Alice, which Alice will then accept. So if the game ends on the second move, Alice gets α and Bob gets $1-\alpha$.

Now move to the first day. At this point Alice gets to make the offer and she realizes that Bob can get $1-\alpha$ if he simply waits until the second day. Hence Alice must offer a payoff that has at least this present value to Bob in order to avoid delay. Thus she offers $\beta(1-\alpha)$ to Bob. Bob finds this (just) acceptable and the game ends. The final outcome is that the game ends on the first move with Alice receiving $1-\beta(1-\alpha)$ and Bob receiving $\beta(1-\alpha)$.

The first panel in Figure 30.11 illustrates this process for the case where $\alpha=\beta<1$. The outermost diagonal line shows the possible payoff patterns on the first day, namely, all payoffs of the form $x_A+x_B=1$. The next diagonal line moving toward the origin shows the present value of the payoffs if the game ends in the second period: $x_A+x_B=\alpha$. The diagonal line closest to the origin shows the present value of the payoffs if the game ends in the third period; the equation for this line is $x_A+x_B=\alpha^2$. The right-angled path depicts the minimum acceptable divisions each period, leading up to the final subgame perfect equilibrium. The second panel in Figure 30.11 shows how the same process might look with more stages in the negotiation.

It is natural to let the horizon go to infinity and ask what happens in the infinite game. It turns out that the subgame perfect equilibrium division is

$$\text{Payoff to Alice} = \frac{1-\beta}{1-\alpha\beta}$$

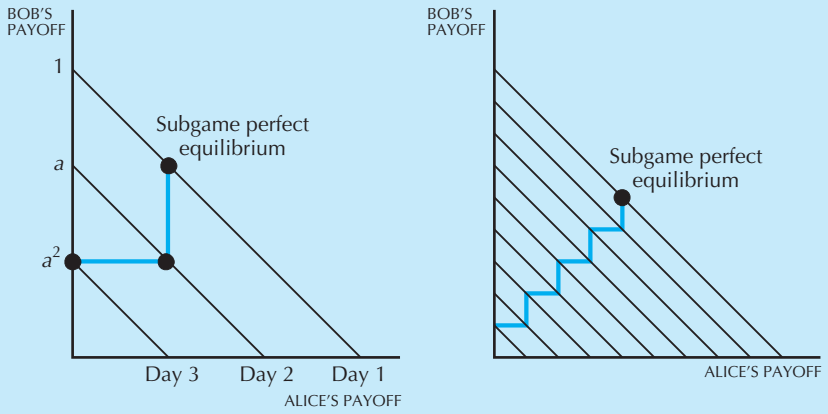
$$\text{Payoff to Bob} = \frac{\beta(1-\alpha)}{1-\alpha\beta}.$$

Note that if $\alpha=1$ and $\beta<1$, then Alice receives the entire payoff.

The Ultimatum Game

The Rubinstein bargaining model is so elegant that economists rushed to test it in the laboratory. They found, alas, that elegance does not imply accuracy. Naive subjects (i.e., noneconomics majors) aren't very good at looking ahead more than one or two steps, if that.

In addition, there are other factors that cause problems. To see this, let us examine a one-step version of the bargaining model described above.



A bargaining game. The heavy line connects together the equilibrium outcomes in the subgames. The point on the line that is furthest out is the subgame perfect equilibrium.

Figure 30.11

Alice and Bob still have $1 to divide between them. Alice proposes a division, and, if Bob agrees, the game ends. The question is, what should Alice say?

According to the theory, she should propose something like 99 cents for Alice, 1 cent for Bob. Bob, figuring that 1 cent is better than nothing, accepts, and Alice goes home happy that she studied economics.

Unfortunately, it doesn't work out like that. A more likely outcome is that Bob, disgusted by the paltry 1 cent, says "No way," and Alice ends up with nothing. Alice, recognizing this possibility, will tend to sweeten the offer. In actual experiments, the average offer for U.S. undergraduates is about 45 cents, and this offer tends to be accepted most of the time.

The offering players are behaving rationally, in the sense that the 45 cent offer is pretty close to maximizing the expected payoff, given the observed frequency of rejection. It is the receiving players who behave differently than the theory predicts, since they reject small offers, even though this makes them worse off.

There are many proposed explanations for this. One view is that too small an offer violates **social norms** of behavior. Indeed, economists have found quite significant cross-cultural differences in behavior in ultimatum games. Another, not inconsistent view, is that receivers get some utility payoff from hurting the offerers, in retaliation for the small offer. After all, if all you are losing is a penny, the satisfaction of striking back at the other player is pretty attractive by comparison. We will the ultimatum game in more detail in the next chapter.

Summary

1. A player's best response function gives the optimal choice for him as a function of the choices the other player(s) might make.

2. A Nash equilibrium in a two-person game is a pair of strategies, one for each player, each of which is a best response to the other.

3. A mixed strategy Nash equilibrium involves randomizing among several strategies.

4. Common games of coordination are the battle of the sexes, where both players want to do the same thing rather than different things; the prisoner's dilemma, where the dominant strategy ends up hurting both players; the assurance game, where both players want to cooperate as long as they think the other will cooperate; and chicken, where players want to avoid doing the same thing.

5. A two-person zero-sum game is one where the payoffs to one player are the negative of the payoffs to the other.

6. Evolutionary games are concerned with outcomes that are stable under population reproduction.

7. In sequential games, players move in turn. Each player therefore has to reason about what the other will do in response to his or her choices.

8. In many sequential games, commitment is an important issue. Finding ways to force commitment to play particular strategies can be important.

REVIEW QUESTIONS

1. In a two-person Nash equilibrium, each player is making a best response to what? In a dominant strategy equilibrium, each player is making a best response to what?

2. Look at the best responses for row and column in the section on mixed strategies. Do these give rise to best response functions?

3. If both players make the same choice in a coordination game, all will be well.

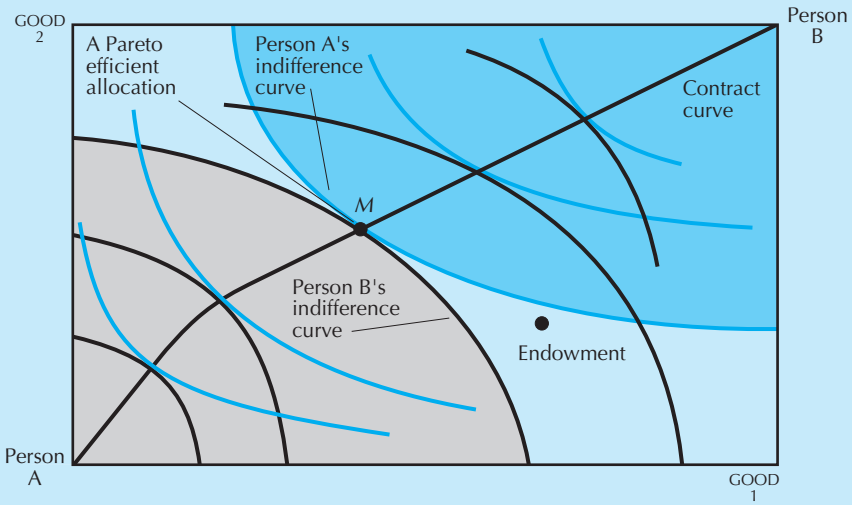
4. The text claims that row scores 62 percent of the time in equilibrium. Where does this number come from?

Now we can repeat the same analysis at the point M. We can draw the two indifference curves through M, construct a new lens-shaped "region of mutual advantage," and imagine the traders moving to some new point N in this region. And so it goes ... the trade will continue until there are no more trades that are preferred by both parties. What does such a position look like?

32.3 Pareto Efficient Allocations

The answer is given in Figure 32.2. At the point M in this diagram the set of points above A's indifference curve doesn't intersect the set of points above B's indifference curve. The region where A is made better off is disjoint from the region where B is made better off. This means that any movement that makes one of the parties better off necessarily makes the other party worse off. Thus there are no exchanges that are advantageous for both parties. There are no mutually improving trades at such an allocation.

An allocation such as this is known as a **Pareto efficient** allocation. The idea of Pareto efficiency is a very important concept in economics that arises in various guises.



A Pareto efficient allocation. At a Pareto efficient allocation such as M, each person is on his highest possible indifference curve, given the indifference curve of the other person. The line connecting such points is known as the contract curve.

Figure 32.2

A Pareto efficient allocation can be described as an allocation where:

1. There is no way to make all the people involved better off; or
2. there is no way to make some individual better off without making someone else worse off; or
3. all of the gains from trade have been exhausted; or
4. there are no mutually advantageous trades to be made, and so on.

Indeed we have mentioned the concept of Pareto efficiency several times already in the context of a single market: we spoke of the Pareto efficient level of output in a single market as being that amount of output where the marginal willingness to buy equaled the marginal willingness to sell. At any level of output where these two numbers differed, there would be a way to make both sides of the market better off by carrying out a trade. In this chapter we will examine more deeply the idea of Pareto efficiency involving many goods and many traders.

Note the following simple geometry of Pareto efficient allocations: the indifference curves of the two agents must be tangent at any Pareto efficient allocation in the interior of the box. It is easy to see why. If the two indifference curves are not tangent at an allocation in the interior of the box, then they must cross. But if they cross, then there must be some mutually advantageous trade—so that point cannot be Pareto efficient. (It is possible to have Pareto efficient allocations on the sides of the box—where one consumer has zero consumption of some good—in which the indifference curves are not tangent. These boundary cases are not important for the current discussion.)

From the tangency condition it is easy to see that there are a lot of Pareto efficient allocations in the Edgeworth box. In fact, given any indifference curve for person A, for example, there is an easy way to find a Pareto efficient allocation. Simply move along A's indifference curve until you find the point that is the best point for B. This will be a Pareto efficient point, and thus both indifference curves must be tangent at this point.

The set of *all* Pareto efficient points in the Edgeworth box is known as the **Pareto set**, or the **contract curve**. The latter name comes from the idea that all "final contracts" for trade must lie on the Pareto set—otherwise they wouldn't be final because there would be some improvement that could be made!

In a typical case the contract curve will stretch from A's origin to B's origin across the Edgeworth box, as shown in Figure 32.2. If we start at A's origin, A has none of either good and B holds everything. This is Pareto efficient since the only way A can be made better off is to take something away from B. As we move up the contract curve A is getting more and more well-off until we finally get to B's origin.

The Pareto set describes all the possible outcomes of mutually advantageous trade from starting anywhere in the box. If we are given the starting point—the initial endowments for each consumer—we can look at the subset of the Pareto set that each consumer prefers to his initial endowment. This is simply the subset of the Pareto set that lies in the lens-shaped region depicted in Figure 32.1. The allocations in this lens-shaped region are the possible outcomes of mutual trade starting from the particular initial endowment depicted in that diagram. But the Pareto set itself doesn't depend on the initial endowment, except insofar as the endowment determines the total amounts of both goods that are available and thereby determines the dimensions of the box.

Let us examine the calculus conditions describing Pareto efficient allocations. By definition, a Pareto efficient allocation makes each agent as well-off as possible, given the utility of the other agent. So let us pick $\overline{u}$ as the utility level for agent B, say, and see how we can make agent A as well-off as possible.

The maximization problem is

$$\max_{x_A^1, x_A^2, x_B^1, x_B^2} u_A(x_A^1, x_A^2)$$

$$\text{such that } u_B(x_B^1, x_B^2) = \overline{u}$$

$$x_A^1 + x_B^1 = \omega^1$$

$$x_A^2 + x_B^2 = \omega^2.$$

Here $\omega^1 = \omega_A^1 + \omega_B^1$ is the total amount of good 1 available and $\omega^2 = \omega_A^2 + \omega_B^2$ is the total amount of good 2 available. This maximization problem asks us to find the allocation $(x_A^1, x_A^2, x_B^1, x_B^2)$ that makes person A's utility as large as possible, given a fixed level for person B's utility, and given that the total amount of each good used is equal to the amount available.

We can write the Lagrangian for this problem as

$$\begin{aligned} L = {} & u_A(x_A^1, x_A^2) - \lambda(u_B(x_B^1, x_B^2) - \overline{u}) \\ & - \mu_1(x_A^1 + x_B^1 - \omega^1) - \mu_2(x_A^2 + x_B^2 - \omega^2). \end{aligned}$$

Here λ is the Lagrange multiplier on the utility constraint, and the μ's are the Lagrange multipliers on the resource constraints. When we differentiate with respect to each of the goods, we have four first-order conditions that must hold at the optimal solution:

$$\frac{\partial L}{\partial x_A^1} = \frac{\partial u_A}{\partial x_A^1} - \mu_1 = 0$$
$$\frac{\partial L}{\partial x_A^2} = \frac{\partial u_A}{\partial x_A^2} - \mu_2 = 0$$
$$\frac{\partial L}{\partial x_B^1} = -\lambda\frac{\partial u_B}{\partial x_B^1} - \mu_1 = 0$$
$$\frac{\partial L}{\partial x_B^2} = -\lambda\frac{\partial u_B}{\partial x_B^2} - \mu_2 = 0.$$

If we divide the first equation by the second, and the third equation by the fourth, we have

$$MRS_A = \frac{\partial u_A/\partial x_A^1}{\partial u_A/\partial x_A^2} = \frac{\mu_1}{\mu_2} \tag{32.1}$$

$$MRS_B = \frac{\partial u_B/\partial x_B^1}{\partial u_B/\partial x_B^2} = \frac{\mu_1}{\mu_2}. \tag{32.2}$$

The interpretation of these conditions is: at a Pareto efficient allocation, the marginal rates of substitution between the two goods must be the same. Otherwise, there would be some trade that would make each consumer better off.

Let us recall the conditions that must hold for optimal choice by consumers. If consumer A is maximizing utility subject to her budget constraint and consumer B is maximizing utility subject to his budget constraint, and both consumers face the same prices for goods 1 and 2, we must have

$$\frac{\partial u_A/\partial x_A^1}{\partial u_A/\partial x_A^2} = \frac{p_1}{p_2} \tag{32.3}$$

$$\frac{\partial u_B/\partial x_B^1}{\partial u_B/\partial x_B^2} = \frac{p_1}{p_2}. \tag{32.4}$$

Note the similarity with the efficiency conditions. The Lagrange multipliers in the efficiency conditions, μ_1 and μ_2, are just like the prices p_1 and p_2 in the consumer choice conditions. In fact the Lagrange multipliers in this kind of problem are sometimes known as **shadow prices** or **efficiency prices**.

Every Pareto efficient allocation has to satisfy conditions like those in equations (32.1) and (32.2). The conditions describing Pareto efficiency and the conditions describing individual maximization in a market environment are virtually the same.

32.4 Market Trade

The equilibrium of the trading process described above—the set of Pareto efficient allocations—is very important, but it still leaves a lot of ambiguity about where the agents end up. The reason is that the trading process we have described is very general. Essentially we have only assumed that the two parties will move to *some* allocation where they are both made better off.

If we have a *particular* trading process, we will have a more precise description of equilibrium. Let's try to describe a trading process that mimics the outcome of a competitive market.

Suppose that we have a third party who is willing to act as an "auctioneer" for the two agents A and B. The auctioneer chooses a price for good 1 and a price for good 2 and presents these prices to the agents A and B. Each agent then sees how much his or her endowment is worth at the prices (p_1, p_2) and decides how much of each good he or she would want to buy at those prices.

One warning is in order here. If there are really only two people involved in the transaction, then it doesn't make much sense for them to behave in a competitive manner. Instead they would probably attempt to bargain over the terms of trade. One way around this difficulty is to think of the Edgeworth box as depicting the average demands in an economy with only two *types* of consumers, but with many consumers of each type. Another way to deal with this is to point out that the behavior is implausible in the two-person case, but it makes perfect sense in the many-person case, which is what we are really concerned with.

Either way, we know how to analyze the consumer-choice problem in this framework—it is just the standard consumer-choice problem we described in Chapter 5. In Figure 32.3 we illustrate the two demanded bundles of the two agents. (Note that the situation depicted in Figure 32.3 is not an equilibrium configuration since the demand by one agent is not equal to the supply of the other agent.)

As in Chapter 9 there are two relevant concepts of "demand" in this framework. The **gross demand** of agent A for good 1, say, is the total amount of good 1 that he wants at the going prices. The **net demand** of agent A for good 1 is the difference between this total demand and the initial endowment of good 1 that agent A holds. In the context of general equilibrium analysis, net demands are sometimes called **excess demands**. We will denote the excess demand of agent A for good 1 by e_A^1. By definition, if A's gross demand is x_A^1, and his endowment is ω_A^1, we have

$$e_A^1 = x_A^1 - \omega_A^1.$$

The concept of excess demand is probably more natural, but the concept of gross demand is generally more useful. We will typically use the word "demand" to mean gross demand and specifically say "net demand" or "excess demand" if that is what we mean.

tribute income? If you give a person an extra dollar to spend, then he can choose to consume more of any of the goods that he wants to consume—not necessarily just the good being subsidized.

Summary

1. General equilibrium refers to the study of how the economy can adjust to have demand equal supply in all markets at the same time.

2. The Edgeworth box is a graphical tool to examine such a general equilibrium with 2 consumers and 2 goods.

3. A Pareto efficient allocation is one in which there is no feasible reallocation of the goods that would make all consumers at least as well-off and at least one consumer strictly better off.

4. Walras' law states that the value of aggregate excess demand is zero for all prices.

5. A general equilibrium allocation is one in which each agent chooses a most preferred bundle of goods from the set of goods that he or she can afford.

6. Only relative prices are determined in a general equilibrium system.

7. If the demand for each good varies continuously as prices vary, then there will always be some set of prices where demand equals supply in every market; that is, a competitive equilibrium.

8. The First Theorem of Welfare Economics states that a competitive equilibrium is Pareto efficient.

9. The Second Theorem of Welfare Economics states that as long as preferences are convex, then every Pareto efficient allocation can be supported as a competitive equilibrium.

REVIEW QUESTIONS

1. Is it possible to have a Pareto efficient allocation where someone is worse off than he is at an allocation that is not Pareto efficient?

2. Is it possible to have a Pareto efficient allocation where everyone is worse off than they are at an allocation that is not Pareto efficient?

3. True or false? If we know the contract curve, then we know the outcome of any trading.

4. Can some individual be made better off if we are at a Pareto efficient allocation?

5. If the value of excess demand in 8 out of 10 markets is equal to zero, what must be true about the remaining two markets?

goods, we know that they will end up somewhere on this contract curve. The exact position will depend on their property rights involving smoke and money and on the precise mechanism that they use to trade.

One mechanism that they could use to trade is the price mechanism. Just as before we could imagine an auctioneer calling out prices and asking how much each agent would be willing to buy at those prices. If the initial endowment point gave A the property rights to smoke, he could consider selling some of his smoking rights to B in exchange for B's money. Similarly, if the property rights for clean air were given to B, he could sell some of his clean air to A.

When the auctioneer manages to find a set of prices where supply equals demand everything is fine: we have a nice Pareto efficient outcome. If there is a market for smoke, a competitive equilibrium will be Pareto efficient. Furthermore, the competitive prices will measure the marginal rate of substitution between the two goods, just as in the standard case.

This is just like the usual Edgeworth box analysis, but described in a slightly different framework. As long as we have well-defined property rights in the good involving the externality—no matter who holds the property rights—the agents can trade from their initial endowment to a Pareto efficient allocation. If we want to set up a market in the externality to encourage trade, that will work as well.

The only problem arises if the property rights are *not* well defined. If A believes that he has the right to smoke and B believes that he has the right to clean air, we have difficulties. *The practical problems with externalities generally arise because of poorly defined property rights.*

My neighbor may believe that he has the right to play his trumpet at 3 in the morning, and I may believe that I have the right to silence. A firm may believe that it has the right to dump pollutants into the atmosphere that I breathe, while I may believe that it doesn't. Cases where property rights are poorly defined can lead to an inefficient production of externalities—which means that there would be ways to make both parties involved better off by changing the production of externalities. If property rights are well defined, and mechanisms are in place to allow for negotiation between people, then people can trade their rights to produce externalities in the same way that they trade rights to produce and consume ordinary goods.

35.2 Quasilinear Preferences and the Coase Theorem

We argued above that as long as property rights were well defined, trade between agents would result in an efficient allocation of the externality. In general, the amount of the externality that will be generated in the efficient solution will depend on the assignment of property rights. In the case of the two roommates, the amount of smoke generated will depend on whether the smoker has the property rights or the nonsmoker has them.

But there is a special case where the outcome of the externality is independent of the assignment of property rights. If the agents' preferences are **quasilinear**, then every efficient solution must have the same amount of the externality.

This case is illustrated in Figure 35.2 for the Edgeworth box case of the smoker versus the nonsmoker. Since the indifference curves are all horizontal translates of each other, the locus of mutual tangencies—the set of Pareto efficient allocations—will be a horizontal line. This means that the amount of smoke is the same in every Pareto efficient allocation; only the dollar amounts held by the agents differ across the efficient allocations.

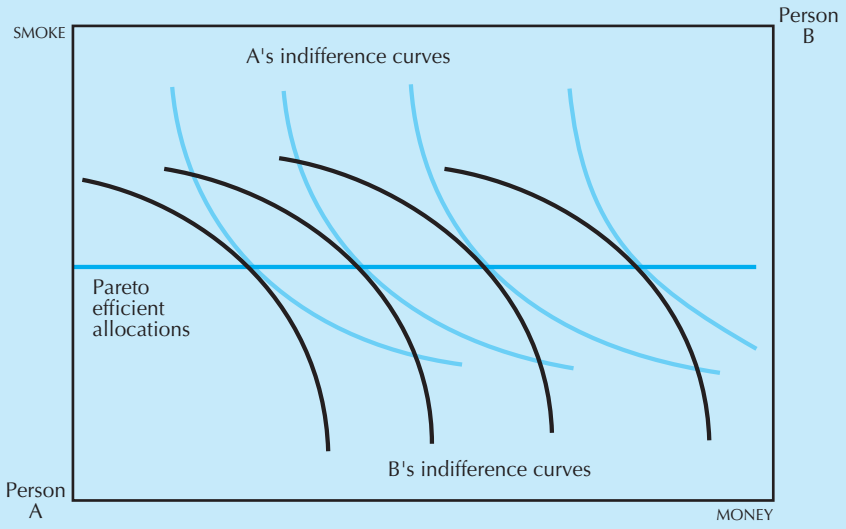


Figure 35.2 **Quasilinear preferences and the Coase theorem.** If each consumer's preferences are quasilinear, so that they are all horizontal translates of each other, the set of Pareto efficient allocations will be a horizontal line. Thus there will be a unique amount of the externality, in this case smoke, at each Pareto efficient allocation.

The result that under certain conditions the efficient amount of the good involved in the externality is independent of the distribution of property rights is sometimes known as the **Coase Theorem**. However, it should be emphasized just how special these conditions are. The quasilinear preference assumption implies that the demands for the good causing the exter-

nality doesn't depend on the distribution of income. Therefore a reallocation of endowments doesn't affect the efficient amount of the externalities. This is sometimes expressed by saying that the Coase theorem is valid if there are no "income effects."[1]

In this case, the Pareto efficient allocations will involve a unique amount of the externality being generated. The different Pareto efficient allocations will involve different amounts of money being held by the consumers; but the amount of the externality—the amount of smoke—will be independent of the distribution of wealth.

35.3 Production Externalities

Let us now consider a situation involving production externalities. Firm S produces some amount of steel, s, and also produces a certain amount of pollution, x, which it dumps into a river. Firm F, a fishery, is located downstream and is adversely affected by S's pollution.

Suppose that firm S's cost function is given by $c_s(s, x)$, where s is the amount of steel produced and x is the amount of pollution produced. Firm F's cost function is given by $c_f(f, x)$, where f indicates the production of fish and x is the amount of pollution. Note that F's costs of producing a given amount of fish depend on the amount of pollution produced by the steel firm. We will suppose that pollution increases the cost of providing fish $\partial c_f/\partial x > 0$, and that pollution *decreases* the cost of steel production, $\partial c_s/\partial x \leq 0$. This last assumption says that increasing the amount of pollution will decrease the cost of producing steel—that reducing pollution will increase the cost of steel production, at least over some range.

The steel firm's profit-maximization problem is

$$\max_{s,x} \; p_s s - c_s(s, x)$$

and the fishery's profit-maximization problem is

$$\max_{f} \; p_f f - c_f(f, x).$$

Note that the steel mill gets to choose the amount of pollution that it generates, but the fishery must take the level of pollution as outside of its control.

[1] Ronald Coase is an emeritus professor at the University of Chicago Law School. His famous paper, "The Problem of Social Costs," *The Journal of Law & Economics*, 3 (October 1960), has been given a variety of interpretations. Some authors suggest that Coase only asserted that costless bargaining over externalities achieves a Pareto efficient outcome, not that the outcome will be independent of the assignment of property rights. Coase received the 1991 Nobel Prize in Economics for this work.

be achieved by appropriate effluent fees ... and it can be achieved at a substantially lower cost than the current system of mandated standards.

Of course, there is no reason why there might not also be some federally mandated standards for the two-thirds of the vehicles that are operated in localities where pollution is not a serious problem. If it is cheaper to impose standards than to require inspections, then by all means that should be the proper choice. The appropriate method of pollution control for automobiles should depend on a rational analysis of benefits and costs—as should all social policies of this nature.

Summary

1. The First Theorem of Welfare Economics shows that a free, competitive market will provide an efficient outcome in the absence of externalities.

2. However, if externalities are present, the outcome of a competitive market is unlikely to be Pareto efficient.

3. However, in this case, the state can sometimes "mimic" the role of the market by using prices to provide correct signals about the social cost of individual actions.

4. More importantly, the legal system can ensure that property rights are well defined, so that efficiency-enhancing trades can be made.

5. If preferences are quasilinear, the efficient amount of a consumption externality will be independent of the assignment of property rights.

6. Cures for production externalities include the use of Pigouvian taxes, setting up a market for the externality, simply allowing firms to merge, or transferring property rights in other ways.

7. The tragedy of the commons refers to the tendency for common property to be overused. This is a particularly prevalent form of externality.

REVIEW QUESTIONS

1. True or false? An explicit delineation of property rights usually eliminates the problem of externalities.

2. True or false? The distributional consequences of the delineation of property rights are eliminated when preferences are quasilinear.

MWG

Andreu Mas-Colell, Michael D. Whinston, Jerry R. Green - Mic

- Finite and infinite horizon bilateral bargaining (Examples 9.AA.1–9.AA.2): Ch. 9, Appendix A (p. 296–299)
- 9.B.12, 9.B.13: Ch. 9, Exercises (p. 302–303)

in the game in Figure 9.D.3 as in that in Figure 9.D.1(b)? Does it matter for your answer whether in the game in Figure 9.D.3 a minute or a month passes between firm E's two decisions? These issues remain to be sorted out.

APPENDIX A: FINITE AND INFINITE HORIZON BILATERAL BARGAINING

In this appendix we study two models of bilateral bargaining as an economically important example of the use of the subgame perfect Nash equilibrium concept. We begin by studying a finite horizon model of bargaining and then consider its infinite horizon counterpart.

Example 9.AA.1: *Finite Horizon Bilateral Bargaining.* Two players, 1 and 2, bargain to determine the split of v dollars. The rules are as follows: The game begins in period 1; in period 1, player 1 makes an offer of a split (a real number between 0 and v) to player 2, which player 2 may then accept or reject. If she accepts, the proposed split is immediately implemented and the game ends. If she rejects, nothing happens until period 2. In period 2, the players' roles are reversed, with player 2 making an offer to player 1 and player 1 then being able to accept or reject it. Each player has a discount factor of $\delta \in (0, 1)$, so that a dollar received in period t is worth δ^{t-1} in period 1 dollars. However, after some finite number of periods T, if an agreement has not yet been reached, the bargaining is terminated and the players each receive nothing. A portion of the extensive form of this game is depicted in Figure 9.AA.1 [this model is due to Stahl (1972)].

There is a unique subgame perfect Nash equilibrium (SPNE) in this game. To see this, suppose first that T is odd, so that player 1 makes the offer in period T if no previous agreement has been reached. Now, player 2 is willing to accept *any* offer in this period because she will get zero if she refuses and the game is terminated (she is indifferent about accepting an offer of zero). Given this fact, the unique SPNE in the subgame that begins in the final period when no agreement has been previously reached has player 1 offer player 2 zero and player 2 accept.[16] Therefore, the payoffs from equilibrium play in this subgame are $(\delta^{T-1}v, 0)$.

Now consider play in the subgame starting in period $T-1$ when no previous agreement has been reached. Player 2 makes the offer in this period. In any SPNE, player 1 will accept an offer in period $T-1$ if and only if it provides her with a payoff of at least $\delta^{T-1}v$, since otherwise she will do better rejecting it and waiting to make an offer in period T (she earns $\delta^{T-1}v$ by doing so). Given this fact, in any SPNE, player 2 must make an offer in period $T-1$ that gives player 1 a payoff of exactly $\delta^{T-1}v$, and player 1 accepts this offer (note that this is player 2's best offer among all those that would be accepted, and making an offer that will be rejected is worse for player 2 because it results in her receiving a payoff of zero). The payoffs arising if the game reaches period $T-1$ must therefore be $(\delta^{T-1}v, \delta^{T-2}v - \delta^{T-1}v)$.

16. Note that if player 2 is unwilling to accept an offer of zero, then player 1 has no optimal strategy; she wants to make a strictly positive offer ever closer to zero (since player 1 will accept any strictly positive offer). If the reliance on player 1 accepting an offer over which she is indifferent bothers you, you can convince yourself that the analysis of the game in which offers must be in small increments (pennies) yields exactly the same outcome as that identified in the text as the size of these increments goes to zero.

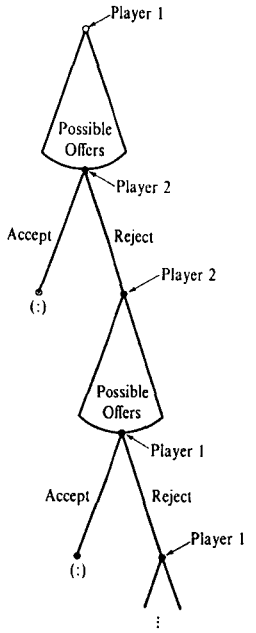


Figure 9.AA.1
The alternating-offer bilateral bargaining game.

Continuing in this fashion, we can determine that the unique SPNE when T is odd results in an agreement being reached in period 1, a payoff for player 1 of

$$v_1^*(T) = v[1 - \delta + \delta^2 - \cdots + \delta^{T-1}]$$
$$= v\left[(1-\delta)\left(\frac{1-\delta^{T-1}}{1-\delta^2}\right) + \delta^{T-1}\right],$$

and a payoff to player 2 of $v_2^*(T) = v - v_1^*(T)$.

If T is instead even, then player 1 must earn $v - \delta v_1^*(T-1)$ because in any SPNE, player 2 (who will be the first offerer in the odd-number-of-periods subgame that begins in period 2 if she rejects player 1's period 1 offer) will accept an offer in period 1 if and only if it gives her at least $\delta v_1^*(T-1)$, and player 1 will offer her exactly this amount.

Finally, note that as the number of periods grows large ($T \to \infty$), player 1's payoff converges to $v/(1+\delta)$, and player 2's payoff converges to $\delta v/(1+\delta)$. ■

In Example 9.AA.1, the application of the SPNE concept was relatively straightforward; we simply needed to start at the end of the game and work backward. We now consider the infinite horizon counterpart of this game. As we noted in Section

9.B, we can no longer solve for the SPNE in this simple manner when the game has an infinite horizon. Moreover, in many games, introduction of an infinite horizon allows a broad range of behavior to emerge as subgame perfect. Nevertheless, in the infinite horizon bargaining model, the SPNE concept is quite powerful. There is a unique SPNE in this game, and it turns out to be exactly the limiting outcome of the finite horizon model as the length of the horizon T approaches ∞.

Example 9.AA.2: *Infinite Horizon Bilateral Bargaining.* Consider an extension of the finite horizon bargaining game considered in Example 9.AA.1 in which bargaining is no longer terminated after T rounds but, rather, can potentially go on forever. If this happens, the players both earn zero. This model is due to Rubinstein (1982).

We claim that this game has a unique SPNE. In this equilibrium, the players reach an immediate agreement in period 1, with player 1 earning $v/(1+\delta)$ and player 2 earning $\delta v/(1+\delta)$.

The method of analysis we use here, following Shaked and Sutton (1984), makes heavy use of the stationarity of the game (the subgame starting in period 2 looks exactly like that in period 1, but with the players' roles reversed).

To start, let $\bar{v}_1$ denote the largest payoff that player 1 gets in *any* SPNE (i.e., there may, in principle, be multiple SPNEs in this model).[17] Given the stationarity of the model, this is also the largest amount that player 2 can expect in the subgame that begins in period 2 after her rejection of player 1's period 1 offer, a subgame in which player 2 has the role of being the first player to make an offer. As a result, player 1's payoff in any SPNE cannot be lower than the amount $\underline{v}_1 = v - \delta\bar{v}_1$ because, if it was, then player 1 could do better by making a period 1 offer that gives player 2 just slightly more than $\delta\bar{v}_1$. Player 2 is certain to accept any such offer because she will earn only $\delta\bar{v}_1$ by rejecting it (note that we are using subgame perfection here, because we are requiring that the continuation of play after rejection is an SPNE in the continuation subgame and that player 2's response will be optimal given this fact).

Next, we claim that, in any SPNE, $\bar{v}_1$ cannot be larger than $v - \delta\underline{v}_1$. To see this, note that in any SPNE, player 2 is certain to reject any offer in period 1 that gives her less than $\delta\underline{v}_1$ because she can earn at least $\delta\underline{v}_1$ by rejecting it and waiting to make an offer in period 2. Thus, player 1 can do no better than $v - \delta\underline{v}_1$ by making an offer that is accepted in period 1. What about by making an offer that is rejected in period 1? Since player 2 must earn at least $\delta\underline{v}_1$ if this happens, and since agreement cannot occur before period 2, player 1 can earn no more than $\delta v - \delta\underline{v}_1$ by doing this. Hence, we have $\bar{v}_1 \leq v - \delta\underline{v}_1$.

Next, note that these derivations imply that

$$\bar{v}_1 \leq v - \delta\underline{v}_1 = (\underline{v}_1 + \delta\bar{v}_1) - \delta\underline{v}_1,$$

so that

$$\bar{v}_1(1-\delta) \leq \underline{v}_1(1-\delta).$$

Given the definitions of $\underline{v}_1$ and $\bar{v}_1$, this implies that $\underline{v}_1 = \bar{v}_1$, and so player 1's SPNE payoff is uniquely determined. Denote this payoff by v_1°. Since $v_1^\circ = v - \delta v_1^\circ$, we find that player 1 must earn $v_1^\circ = v/(1+\delta)$ and player 2 must earn $v_2^\circ = v - v_1^\circ = \delta v/(1+\delta)$. In addition, recalling the argument in the previous paragraph, we see that an agreement will be reached in the first period (player 1 will find it worthwhile to make an offer that player 2 accepts). The SPNE strategies are as follows: A player who has just received an offer accepts it if and only if she is offered at least δv_1°, while a player whose turn it is to make an offer offers exactly δv_1° to the player receiving the offer.

Note that the equilibrium strategies, outcome, and payoffs are precisely the limit of those in the finite game in Example 9.AA.1 as $T \to \infty$. ■

17. This maximum can be shown to be well defined, but we will not do so here.

The coincidence of the infinite horizon equilibrium with the limit of the finite horizon equilibria in this model is not a general property of infinite horizon games. The discussion of infinitely repeated games in Chapter 12 provides an illustration of this point.

We should also point out that the outcomes of game-theoretic models of bargaining can be quite sensitive to the precise specification of the bargaining process and players' preferences. Exercises 9.B.7 and 9.B.13 provide an illustration.

APPENDIX B: EXTENSIVE FORM TREMBLING-HAND PERFECT NASH EQUILIBRIUM

In this appendix we extend the analysis presented in Section 9.C by discussing another equilibrium notion that strengthens the consistency conditions on beliefs in the weak PBE concept: *extensive form trembling-hand perfect Nash equilibrium* [due to Selten (1975)]. In fact, this equilibrium concept is the strongest among those discussed in Section 9.C.

The definition of an extensive form trembling-hand perfect Nash equilibrium parallels that for the normal form (see Section 8.F) but has the trembles applied not to a player's mixed strategies, but rather to the player's choice at each of her information sets. A useful way to view this idea is with what Selten (1975) calls the *agent normal form*. This is the normal form that we would derive if we pretended that the player had a set of agents in charge of moving for her at each of her information sets (a different one for each), each acting independently to try to maximize the player's payoff.

Definition 9.BB.1: Strategy profile σ in extensive form game Γ_E is an ***extensive form trembling-hand perfect Nash equilibrium*** if and only if it is a normal form trembling-hand perfect Nash equilibrium of the agent normal form derived from Γ_E.

To see why it is desirable to have the trembles occurring at each information set rather than over strategies as in the normal-form concept considered in Section 8.F, consider Figure 9.BB.1, which is taken from van Damme (1983). This game has a unique subgame perfect Nash equilibrium: $(\sigma_1, \sigma_2) = ((NR, L), \ell)$. But you can check that $((NR, L), \ell)$ is not the only normal form trembling-hand perfect Nash equilibrium: so are $((R, L), r)$ and $((R, M), r)$. The reason that these two strategy profiles are normal form trembling-hand perfect is that, in the normal form, the tremble to strategy (NR, M) by player 1 can be larger than that to (NR, L) despite the fact that the latter is a better choice for player 1 at her second decision node.

9.B.4[B] Prove that in a finite *zero-sum* game of perfect information, there are unique subgame perfect Nash equilibrium payoffs.

9.B.5[B] (E. Maskin) Consider a game with two players, player 1 and player 2, in which each player i can choose an action from a finite set M_i that contains m_i actions. Player i's payoff if the action choices are (m_1, m_2) is $\phi_i(m_1, m_2)$.

(a) Suppose, first, that the two players move simultaneously. How many strategies does each player have?

(b) Now suppose that player 1 moves first and that player 2 observes player 1's move before choosing her move. How many strategies does each player have?

(c) Suppose that the game in (b) has multiple SPNEs. Show that if this is the case, then there exist two pairs of moves (m_1, m_2) and (m_1', m_2') (where either $m_1 \neq m_1'$ or $m_2 \neq m_2'$) such that either

$$\text{(i)} \quad \phi_1(m_1, m_2) = \phi_1(m_1', m_2')$$

or

$$\text{(ii)} \quad \phi_2(m_1, m_2) = \phi_2(m_1', m_2').$$

(d) Suppose that for any two pairs of moves (m_1, m_2) and (m_1', m_2') such that $m_1 \neq m_1'$ or $m_2 \neq m_2'$, condition (ii) is violated (i.e., player 2 is never indifferent between pairs of moves). Suppose also that there exists a pure strategy Nash equilibrium in the game in (a) in which π_1 is player 1's payoff. Show that in any SPNE of the game in (b), player 1's payoff is at least π_1. Would this conclusion necessarily hold for any *Nash* equilibrium of the game in (b)?

(e) Show by example that the conclusion in (d) may fail either if condition (ii) holds for some strategy pairs (m_1, m_2), (m_1', m_2') with $m_1 \neq m_1'$ or $m_2 \neq m_2'$ or if we replace the phrase *pure strategy Nash equilibrium* with the phrase *mixed strategy Nash equilibrium*.

9.B.6[B] Solve for the mixed strategy equilibrium involving actual randomization in the post-entry subgame of the Niche Choice game in Example 9.B.4. Is there an SPNE that induces this behavior in the post-entry subgame? What are the SPNE strategies?

9.B.7[B] Consider the finite horizon bilateral bargaining game in Appendix A (Example 9.AA.1); but instead of assuming that players discount future payoffs, assume that it costs $c < v$ to make an offer. (Only the player making an offer incurs this cost, and players who have made offers incur this cost even if no agreement is ultimately reached.) What is the (unique) SPNE of this alternative model? What happens as T approaches ∞?

9.B.8[C] Prove that every (finite) game Γ_E has a mixed strategy subgame perfect Nash equilibrium.

9.B.9[B] Consider a game in which the following simultaneous-move game is played twice:

		Player 2		
		b_1	b_2	b_3
Player 1	a_1	10, 10	2, 12	0, 13
	a_2	12, 2	5, 5	0, 0
	a_3	13, 0	0, 0	1, 1

The players observe the actions chosen in the first play of the game prior to the second play. What are the pure strategy subgame perfect Nash equilibria of this game?

9.B.10[B] Reconsider the game in Example 9.B.3, but now change the post-entry game so that when both players choose "accommodate", instead of receiving the payoffs $(u_E, u_I) = (3, 1)$, the players now must play the following simultaneous-move game:

		Firm I	
		ℓ	r
Firm E	U	3, 1	0, 0
	D	0, 0	x, 3

What are the SPNEs of this game when $x \geq 0$? When $x < 0$?

9.B.11[B] Two firms, A and B, are in a market that is declining in size. The game starts in period 0, and the firms can compete in periods 0, 1, 2, 3, . . . (i.e., indefinitely) if they so choose. Duopoly profits in period t for firm A are equal to $105 - 10t$, and they are $10.5 - t$ for firm B. Monopoly profits (those if a firm is the only one left in the market) are $510 - 25t$ for firm A and $51 - 2t$ for firm B.

Suppose that at the start of each period, each firm must decide either to "stay in" or "exit" if it is still active (they do so simultaneously if both are still active). Once a firm exits, it is out of the market forever and earns zero in each period thereafter. Firms maximize their (undiscounted) sum of profits.

What is this game's subgame perfect Nash equilibrium outcome (and what are the firms' strategies in the equilibrium)?

9.B.12[C] Consider the infinite horizon bilateral bargaining model of Appendix A (Example 9.AA.2). Suppose the discount factors δ_1 and δ_2 of the two players differ. Now what is the (unique) subgame perfect Nash equilibrium?

9.B.13[B] What are the subgame perfect Nash equilibria of the infinite horizon version of Exercise 9.B.7?

9.B.14[B] At time 0, an incumbent firm (firm I) is already in the widget market, and a potential entrant (firm E) is considering entry. In order to enter, firm E must incur a cost of $K > 0$. Firm E's only opportunity to enter is at time 0. There are three production periods. In any period in which both firms are active in the market, the game in Figure 9.Ex.1 is played. Firm E moves first, deciding whether to stay in or exit the market. If it stays in, firm I decides whether to fight (the upper payoff is for firm E). Once firm E plays "out," it is out of

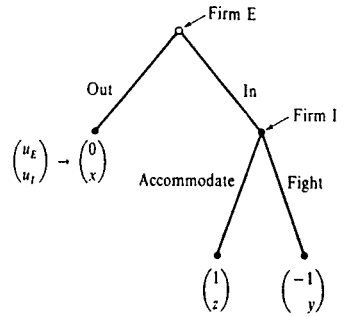


Figure 9.Ex.1